

Intangible Capital, Firm Scope, and Growth^{*}

Nicolas Crouzet^{*}

Kellogg School of Management
n-crouzet@kellogg.northwestern.edu

Janice Eberly

Kellogg School of Management and NBER
eberly@kellogg.northwestern.edu

Andrea Eisfeldt

UCLA Anderson School of Management and NBER
andrea.eisfeldt@anderson.ucla.edu

Dimitris Papanikolaou

Kellogg School of Management and NBER
d-papanikolaou@kellogg.northwestern.edu

^{*}Corresponding author. Kellogg School of Management, Northwestern University, 2211 Campus Drive, Evanston, IL 60208.

June 2026

Abstract

How do intangible assets differ from physical capital, like machines? We model intangible assets with two defining properties: they can be replicated across locations, and they are imperfectly excludable. If entrepreneurs want to increase scope, they must share proprietary knowledge with managers – who can carry it into competing ventures. The resulting hold-up problem limits scope and entry. When entrepreneurs' replication technology improves, scope rises but value shifts toward managers: profitability, concentration, and valuations rise, while entry, and possibly long-run growth, fall. Taxing intangible investment or strengthening IP protection can raise welfare. Undermeasured capital deployment means that productivity is overstated.

Keywords: Investment, Intangible capital, Technological change, Long-run growth.

JEL codes: G32, G33, O31, O32.

^{*}We thank Yudan Ying and Siqi Xue for extraordinary research assistance. We are also grateful to Philippe Aghion, Andy Atkeson, Anmol Bhandari, Maarten De Ridder, Teresa Fort, Chad Jones, Pete Klenow, Huiyu Li, Ellen McGrattan, Ben Moll, Jane Olmsted-Rumsey, Juliana Salomao, Vincent Stark, Alwyn Young, along with seminar participants at Brown, UT Austin, Columbia Business School, UT Dallas, Universite Paris Dauphine, Dartmouth Tuck, HEC, University of Houston, INSEAD, ITAM, Kellogg, LBS Economics, LSE Economics, LSE Finance, McGill, Minneapolis Fed, UCL, SAIF, SED, the ICI conference, and the Conference on the Economics of Innovation in Memory of Zvi Griliches.

What is the difference between physical capital, like machines, and intangible assets? Unlike machines, intangible assets lack a physical presence: they correspond to ideas that must be stored, say in people, processes, or software, in order to be used in production (Crouzet, Eberly, Eisfeldt, and Papanikolaou, 2022). This distinction has two implications. First, intangible assets can be *replicated*. Second, intangible assets may be difficult to *protect*. For instance, an entrepreneur may share an idea with managers to scale its use, but having done so, it may be difficult for her to prevent them from using this same idea in competing ventures.

These two fundamental properties of intangible assets create trade-offs. First, from a private perspective, entrepreneurs generating new intangible assets need to balance the returns from deploying their idea at greater scale with the increased risk of expropriation that accompanies sharing the idea more broadly. This trade-off leads entrepreneurs to limit the scale at which they implement their ideas, even in the absence of physical costs of doing so. Second, from a social perspective, this limited implementation has not only a static cost—loss of variety, smaller firms—but also a dynamic one—lower spillovers of ideas from existing to new firms.

To capture these trade-offs, we embed intangible assets in a model of endogenous growth. In the model, a given intangible asset can be deployed in multiple locations, which may represent physical establishments, product markets, or tasks. The measure of locations determines the *scope* of the intangible’s implementation, and the returns to increasing scope depend on a technology parameter that captures the current state of replication technology. What limits scope? Our model has no physical costs of implementation; rather, increasing the scope of deployment requires the entrepreneur to share trade secrets with key employees—managers who run each location. Once these employees possess the firm’s knowledge, they can potentially use it to start competing ventures. Their ability to do so is determined by the entrepreneur’s choice of scope together with the degree of exclusivity that the entrepreneur retains over the intangible asset. By threatening to compete, managers extract a fraction of the surplus associated with the creation of the firm by the entrepreneur. The entrepreneur optimally chooses scope to balance the gains from wider deployment against losing rents to managers.

This static trade-off limits the scope of the firm but also has dynamic implications. In our model, firms build on the shoulders of giants, as in Romer (1990); that is, they improve upon the public stock of knowledge. Unlike Romer (1990), the public stock of knowledge depends not only on the intensity of intangible creation, but also on the extent to which entrepreneurs’ existing knowledge has been shared with managers. Thus, growth in a decentralized equilibrium is determined both by the rate of entrepreneurship and by the scope of deployment of new ideas. While a social planner may want to increase both entry and scope, in equilibrium they are linked and limited by the private incentives of entrepreneurs, constraining the set of achievable outcomes.

We use our framework to address two questions. First, what is the impact of improvements in the technology for replicating ideas on economic outcomes? In our model, the degree of replicability is a parameter that is meant to capture the ease and fidelity of codifying and sharing ideas with managers. For instance, recent advances in information technology have allowed firms to more accurately implement the identical production processes across multiple plants or locations. To study this question, we compute the short- and long-run response of economic quantities in our model to a permanent increase in the parameter controlling the degree of replicability of intangible assets. In the short-run, a technological change of this kind increases scope, entry, and the share of capital income that accrues to managers. It also leads to an increase in growth, measured total factor productivity (TFP), and a decline in the valuation of incumbent firms relative to earnings and capital. The

increase in measured TFP is driven by the mismeasurement of effective capital services provided by intangible assets; standard approaches to measuring capital services fail to incorporate the scope at which they are deployed, which rises with the replicability of intangible assets. Valuations of incumbents fall due to increased competition with new entrants who operate at larger scope. In the long-run, however, the new balanced growth path always features lower entry. The higher return to scope induces entrepreneurs to hire more managers and operate at greater scale, choosing to keep a smaller share of a larger pie. This is optimal conditional on entry, but it depresses the returns of entrepreneurship relative to its opportunity cost, which is working in production. This leads to larger firms but lower entry rates in the new balanced growth path. The long-run impact on growth is ambiguous and depends on the relative magnitude of positive spillovers from higher scope versus the lower equilibrium entry rates. In brief, the long-run features larger firms with higher measured profit rates, more concentrated product markets, and possibly lower growth. We illustrate these forces with simple numerical examples.

Second, we ask: given that the private trade-off between entry and scope may not reflect its social counterpart, how would a planner choose to tax or subsidize intangible investment? We start by highlighting the key inefficiencies that give rise to a wedge between social and private valuations in our model. Two inefficiencies are standard in endogenous growth models. First, entrants fail to internalize that their decision to create a firm raises the stock of public knowledge for future entrants. In our model, these spillovers operate not only on the extensive margin (the number of new firms), but also on the intensive margin (the scope of new firms). Second, congestion externalities: diminishing returns in the entry technology imply that the social marginal product of entrepreneurial labor is less than its private marginal product. The first of these two inefficiencies leads to insufficient entry, while the latter leads to too much entry. Different from standard models, though, our framework features a third inefficiency: because managers must be awarded a fraction of enterprise value upon its creation, entrepreneurs do not enjoy the full surplus generated by entry. This wedge further depresses entry and is present even without any dynamic spillovers from existing to new firms. It represents an independent channel by which entry and scope might be inefficiently low in the decentralized equilibrium.

In our framework, intangible investment occurs when the firm is created, as the entrepreneur creates a new stock of knowledge to be deployed across locations with the help of managers. We consider the effects of a tax on this investment. Because of the managerial hold-up friction, this tax plays a double role. First, it reduces the private returns to intangible investment (entry) enjoyed by entrepreneurs. Second, and crucially, it also reduces the outside option of managers to start a competing venture using a fraction of the intangible asset from their entrepreneur. The latter occurs because we assume that the planner cannot distinguish between firms newly created by entrepreneurs and firms potentially created by managers when it applies the tax. We show that, when the rate of current entrepreneurship does not affect the rate at which new ideas are created, the optimal tax on intangible investment is positive, and that more generally it remains so when dynamic spillovers from entry are sufficiently small. While the higher tax rate on intangible investment reduces entry, it also increases scope conditional on entry. This higher scope conditional on entry raises growth because managers appropriate a larger share of the ideas created by the entrepreneur when scope is larger. On the other hand, the lower entry rate only has a level effect when entry rates don't directly affect trend growth. Discounting implies that the negative level effect of lower entry is always dominated by the positive growth effect of higher scope, leading to a positive optimal tax. The planner effectively trades off lower entry against the benefits of entrepreneurs deploying their ideas at broader scope.

Similar forces shape the optimal degree of IP protection in the model, that is, the welfare-maximizing value of the parameter controlling the rate at which managers can appropriate ideas created by entrepreneurs. With weaker IP protection, entrepreneurs choose lower scope conditional on entry and raise the share of entry surplus they appropriate. Absent dynamic spillovers from greater scope, the planner would prefer weaker IP protection in order to stimulate entry. With dynamic spillovers, there can be an optimal interior level of IP protection that trades off entry incentives with the dynamic benefits of greater scope.

Our results highlight how the trade-off between scaling and rent-sharing is at the core of intangible assets and distinguishes them, economically, from physical assets. This approach helps to clarify several important positive and normative questions in macroeconomics and finance. Because the relevant measure of capital services should incorporate the scope of deployment of intangible assets, capitalized investment measures of the intangible stock will be biased downward, and particularly so during periods of technical change. Persistent changes in entry, profitability, concentration, valuation ratios and growth can be driven purely by a higher degree of replicability of intangible assets, independent of changes in markups or market power in input or output markets, and in ways that are similar to investment-specific technical change. The optimal tax rate on intangible investment may be positive, in contrast with standard models in which maintaining optimal entry incentives requires a subsidy, and likewise, a higher degree of optimal IP protection is not necessarily welfare-improving, in both cases because positive spillovers across firms can occur when entrepreneurs choose larger scope and share their knowledge more broadly with key workers. Finally, the model has implications for understanding the potential impact of artificial intelligence. We return to these implications in our concluding remarks.

Our work complements the literature on endogenous growth that starts with [Romer \(1990\)](#). Our contribution is to formalize the key economic trade-offs that are involved in the creation and deployment of intangible assets, and relate them to economic outcomes. There are notable differences between our notion of intangibles and the notion of ideas in existing models of endogenous growth. First, ideas in most endogenous growth models are fully non-rival. Ideas in our model can also be non-rival, in that they can be used in multiple locations simultaneously; however we allow for the possibility of imperfect replication—or equivalently, a cost to replication.¹ Within our framework, firms choose the scope of deployment of the intangible subject to expropriation concerns; absent these expropriation concerns, entrepreneurs would choose infinite scope. Thus, our model features an explicit trade-off between the scope of deployment and the likelihood of expropriation, introducing a wedge between private and social returns to intangible creation that is conceptually distinct from monopolistic competition in [Romer \(1990\)](#) or the scarcity of complementary inputs in [Atkeson and Kehoe \(2005\)](#). Further, our notion of intangibles is broader than patents: intangibles in [Romer \(1990\)](#) represent the exclusive right to produce a particular good, or a technology vintage that lowers marginal cost as in [Aghion and Howitt \(1992\)](#), [Klette and Kortum \(2004\)](#), [De Ridder \(2022\)](#), or [Aghion, Bergeaud, Boppart, Klenow, and Li \(2023\)](#). Our notion encompasses not only product-specific ideas or production processes ([Markusen, 1984](#)), but also organization capital and managerial methods that can be deployed at broader scope. Finally, our framework provides new perspectives on taxing corporate investment and protecting intellectual property (IP) ([Boldrin and Levine, 2008a](#); [Acemoglu and Acikgit, 2012](#)). The core insight of our framework is that either taxing intangible investment or providing

¹As [Lucas and Moll \(2014\)](#) note, “knowledge is partially rival: it is ‘rival’ in the short run because people who want to access better knowledge must exert effort and have the good luck to run into the right people; but knowledge is ‘nonrival’ in the long run in the sense that it is in no way diminished when it spreads from one person to another.”

stronger IP protection can encourage entrepreneurs to deploy their ideas at greater scale, and that this might be beneficial from a welfare standpoint even though entry might decline.

Our model includes some elements that are familiar from the endogenous growth literature. Our treatment of knowledge spillovers connects to the literature on technology diffusion through imitation and spillovers (Lucas and Moll, 2014; Perla and Tonetti, 2014; Stokey, 2015; Perla, Tonetti, and Waugh, 2021; Akcigit and Ates, 2019). Relative to these papers, we model intangibles as a replicable capital input, analyze the impact of the degree of replicability on growth, and emphasize the dark side of knowledge spillovers: the same channel through which knowledge diffuses to future entrants also reduces the incentive to create new firms in the first place. Our modeling of intangible assets is similar to McGrattan and Prescott (2009, 2010a), in that the same intangible can be replicated in multiple locations (countries in their setting). A key difference is that, rather than taking the number of possible locations as exogenous, we endogenize the scope of deployment through concerns about expropriation risk. This cost of expanding scope is not a physical cost, which differentiates our model from recent work in which resource costs limit firms' scope of operations (Aghion, Bergeaud, Boppart, Klenow, and Li, 2022; Aghion et al., 2023). Along these lines, our work complements the literature that models intangibles as reducing the cost of entering new markets (Argente, Moreira, Oberfield, and Venkateswaran, 2020; Hsieh and Rossi-Hansberg, 2023a). Recent work has also explored non-rival capital goods such as data (Farboodi and Veldkamp, 2020; Jones and Tonetti, 2020). Relative to these papers, we allow for imperfect replicability, endogenize the limits on deployment through expropriation risk, and discuss the implications for growth.

The possibility of expropriation of an entrepreneur's ideas is a key feature of our model. This possibility arises because enforcement of cash-flow rights over intangibles is imperfect and contracts are potentially incomplete (Grossman and Hart, 1986; Hart and Moore, 1988, 1994; Aghion and Bolton, 1992; Hart and Holmstrom, 2010). Implementing an idea at higher scope requires the creator to communicate it to third parties—key employees or providers of financial capital—but ideas, once communicated, can be stolen (the Arrow information paradox, Arrow, 1962). Anton and Yao (2004) explores the tradeoff between scope and rent sharing in a setting where inventors have imperfect property rights and must partially disclose innovations to exploit them; they find that inventions may not be fully commercialized when property rights are weak. Kondo, Li, and Papanikolaou (2021) embed a similar tradeoff between scope and appropriable surplus and explore the extent to which sunspot equilibria can generate innovation cycles. Expropriation risk is a salient concern: Bhide (1999) reports that 71% of the founders of firms in the Inc 500 list replicated or modified ideas encountered through previous employment. Knowledge can also leak through financiers who transfer a firm's knowledge to other portfolio companies (Cabezon and Hoberg, 2024), or through mandatory disclosures that firms strategically redact to protect proprietary information (Kankanhalli, Kwan, and Merkley, 2024; Wang, 2025). In our setting, the possibility of expropriation arises because entrepreneurs need managers to increase the scope of production; in this regard, some elements of our model bear strong similarities to Lustig, Syverson, and Van Nieuwerburgh (2011). Lustig et al. (2011) solve for the optimal contract between entrepreneurs and managers; like our model, managers can appropriate a fraction of the surplus. Unlike our work, their primary focus is on the dynamics of managerial compensation and its relation to firm size and the IT revolution.

Finally, a literature in macroeconomics and finance has highlighted the implications of rising intangible capital for investment, asset prices, labor markets, and growth (McGrattan and Prescott, 2010b; Haskel and Westlake, 2017; Crouzet and Eberly, 2019, 2023). These papers model intangible capital as qualitatively similar to physical

capital: another factor of production accumulated subject to adjustment costs, differing in depreciation, price, or risk. [Eisfeldt, Falato, and Xiaolan \(2021\)](#) argue that equity-based compensation has dramatically increased since the 1980s, transforming high-skilled labor into a class of “human capitalists.” We revisit these issues using a model that emphasizes the key economic differences between physical and intangible capital: their degree of replicability, and their degree of excludability.

Section 1 discusses three macroeconomic trends consistent with the basic economic mechanism we highlight—rising firm scope, rising managerial compensation, and declining business entry—and presents cross-sectional evidence directly linking scope to both managerial pay and establishment entry rates. Section 2 introduces a simple static model of intangible capital that formalizes the replicability-excludability tradeoff and provides a microfoundation for imperfect replicability. Section 3 embeds this mechanism in a dynamic endogenous growth model. Section 4 characterizes the balanced growth path and its comparative statics with respect to replicability. Section 5 provides numerical examples and studies transition dynamics. Section 6 analyzes how the taxation of intangible investment, and IP protection, affect growth and welfare. Section 7 discusses the main implications of our analysis.

1 Motivating Facts

The introduction described a tradeoff between replication and expropriation that is central to intangible capital; intangibles can be replicated, but by sharing proprietary knowledge, the entrepreneur is exposed to expropriation risk. Before modeling this mechanism in detail, we first explore empirical trends consistent with this mechanism. Under this view, as the technology for replicating intangible capital improves over time, we would expect to see an increase in firm scope, rising managerial compensation, and a decline in business entry. In particular, if intangibles become easier to replicate, we would expect to see them implemented at a wider scope. Since doing so requires managerial talent, managers should capture a higher share of compensation. Finally, larger incumbent firms increase the opportunity cost for entrepreneurs relative to creating new firms, and entrepreneurs’ share of the new firms declines, leading to a decline in firm entry. These joint predictions are at the heart of our model and distinguish our model from existing approaches that aim to rationalize some of these trends ([Akcigit and Ates, 2021](#)).

We first summarize existing evidence documenting these trends in the data, and then show how these trends are related in the cross-section.

1.1 Three trends

Rising firm scope. Over the past three decades, U.S. firms have substantially expanded the range of markets and locations in which they operate. [Hoberg and Phillips \(2025\)](#) use text analysis of firm 10-K filings to construct measures of product-market scope and document that the average firm’s scope in related industries increased by approximately 60% between 1989 and 2017. This expansion was achieved largely through acquisitions and investment in R&D rather than through increases in the number of traditional operating segments, pointing to the deployment of intangible—rather than physical—assets across a wider set of activities. Evidence from establishment-level data tells a consistent story. [Hsieh and Rossi-Hansberg \(2023b\)](#) use the Census Bureau’s Lon-

itudinal Business Database to show that the largest service-sector firms grew primarily by adding establishments in new local markets, with geographic expansion concentrated in smaller cities. National concentration rose even as local concentration fell, a pattern consistent with top firms replicating standardized operations across an increasing number of locations. [Jiang and Rubinton \(2025\)](#) document that multi-establishment firms are more likely to adopt custom software and that, conditional on adoption, software’s share of investment falls with the number of establishments, consistent with the non-rival nature of software inputs enabling wider scope.

Rising managerial share of compensation. Concurrent with the expansion in firm scope, the share of total compensation accruing to managers and key employees has risen. [Eisfeldt et al. \(2021\)](#) document that equity-based compensation has dramatically increased since the 1980s, representing approximately 40% of total compensation to high-skilled labor in recent years. The rising managerial share is not confined to specific industries or to the largest firms. It represents a broad shift in how firms compensate the employees who are most critical to their operations, and who, by virtue of their roles, acquire detailed knowledge of the firm’s proprietary assets.

Declining entry rates. The rate of firm and establishment entry in the U.S. has declined persistently since the late 1970s. The firm startup rate (new firms as a share of all firms) has fallen from roughly 13% in the late 1970s to below 8% in the 2010s, a decline of more than a third, and the share of employment accounted for by young firms (less than five years old) has followed a similar trajectory ([Decker, Haltiwanger, Jarmin, and Miranda, 2014](#)). [Decker, Haltiwanger, Jarmin, and Miranda \(2016\)](#) show that after 2000 the decline was disproportionately concentrated among high-growth young firms: the right tail of the firm employment growth distribution thinned sharply, reducing the contribution of startups to aggregate job creation and to reallocation-driven productivity growth. [Akcigit and Ates \(2021\)](#) catalog ten interrelated facts on declining business dynamism—including the rise in industry concentration, the aging of the firm distribution, and the slowdown in productivity growth ([Fernald, 2014](#))—and argue that they are difficult to reconcile with standard endogenous growth models absent a shift in the underlying innovation technology. [Hopenhayn, Neira, and Singhania \(2022\)](#) and [Karahana, Pugsley, and Şahin \(2024\)](#) show that some, though not all, of the startup deficit can be traced to slower labor-force growth, leaving meaningful residual variation to be explained by forces that shift the economic returns to firm creation. Both fewer entrants and slower measured productivity growth are predictions of the model we develop below, linking these properties to the same underlying shift in the replicability of intangible assets.

1.2 Cross-sectional evidence

These three macro trends—in firm scope, compensation, and entry—could in principle be unrelated. We provide firm-level evidence that these trends are indeed driven by the same underlying economic forces. We begin by showing that, in firms that expanded their scope of operations, the share of equity value that accrues to managers has also increased—compared to firms that didn’t expand scope. Second, we show that industries that experienced an increase in average scope also experienced a decline in new firm creation. To conserve space we briefly summarize the results and leave the details to Appendices [A.1.2](#) and [A.2](#).

Firm-level scope and managerial compensation. The evidence on scope and compensation draws on three sources of firm-level data. Firm fundamentals (revenue, employment, and the market value of equity, `prcc_f` $\times$ `csho`) are from Compustat. Managerial compensation is from Execucomp, using the sum of `tdc1` across the firm’s top five named executives; we divide compensation by equity market value to obtain the managerial

share $\gamma_{i,t}$. Measures of firm geographic and operational scope (the number of establishments and the number of distinct core-based statistical areas (CBSAs) in which the firm operates) are constructed from the National Establishment Time Series (NETS), matched to Compustat at the headquarters level using a name-and-location fuzzy matching procedure described in Appendix A.1.2. The sample covers 1992 to 2018.

We estimate regressions of the form:

$$\Delta_5 \log Y_{i,t+5} = \beta \Delta_5 \log \gamma_{i,t+5} + \delta \mathbf{X}'_{i,t} + \alpha_{I,t} + \varepsilon_{i,t}, \quad (1)$$

where Δ_5 denotes the five-year log change, $Y_{i,t}$ is one of the firm outcomes listed above (revenue, employment, number of establishments, or number of distinct CBSAs). The term $\alpha_{I,t}$ denotes three-digit SIC industry-by-year fixed effects, included in even-numbered columns. The control vector $\mathbf{X}_{i,t}$ includes the five-year growth in the market value of equity, the lagged log level of equity market value, one lag of the dependent variable in log levels, and the lagged log level of the managerial share. Standard errors are computed using the Driscoll and Kraay (1998) procedure.

Table 1 reports the results. Across all eight specifications, the coefficient β on the managerial share is positive and highly statistically significant. The point estimates range from 0.038 to 0.152: a ten percent increase in the managerial share over five years is associated with a 1.5 percent increase in revenue, a 1.1 percent increase in employment, a 0.6 percent increase in the number of establishments, and a 0.4 percent increase in the number of markets. Adding industry-by-year fixed effects (even columns) has little effect on the magnitudes, indicating that the relationship is *within-industry* and cannot be explained by differential trends across sectors. Figure 1 provides a visual summary using binned scatter plots. Hence, in addition to the long run trends noted earlier, we document a robust positive relationship between firms' composition of compensation and their scope, revenue, and employment. Note that this relationship is not intended to be causal, but instead consistent with the mechanism we propose in which underlying technological change drives both compensation and firm scope.

Industry-level scope and establishment entry rates. We next relate changes in firm geographic scope to changes in establishment entry rates across three-digit SIC industries.² Establishment entry rates are from the BDS, accessed at the four-digit NAICS level and mapped to three-digit SIC industries using a payroll-weighted concordance through successive NAICS vintages (Schaller, DeCelles, and Patrick, 2025). For each industry-year, the establishment entry rate is defined as the number of establishments born during the year (BDS variable `estabs_entry`) divided by the lagged number of establishments in the industry; it measures the gross rate at which new business locations open. Scope is measured by industry-year averages, for multi-establishment firms with at least two employees, of the number of CBSAs, ZIP codes, and establishments per firm, all from NETS. Because the NETS scope series exhibit level breaks in 2011 and 2015 due to changes in data collection, we splice each series by extrapolating the pre-break trend in log scope and rescaling post-break levels to match. Full details are in Appendix A.2.

²Appendix A.2 reports industry-level time series of establishment entry rates and average firm geographic scope from the Business Dynamics Statistics (BDS) and the National Establishment Time Series (NETS). These time series illustrate visually the negative co-movement between entry and scope at the aggregate and one-digit industry level.

On the resulting panel of SIC-3 industries observed annually from 1990 to 2013, we estimate

$$\Delta_5 \ln(\text{scope}_{i,t}) = \beta \Delta_5 \text{entry rate}_{i,t} + \gamma \ln(\text{scope}_{i,t}) + \delta \text{entry rate}_{i,t} + \alpha_{d(i),t} + \varepsilon_{i,t}, \quad (2)$$

where i indexes SIC-3 industries, t is the base year, Δ_5 denotes the five-year forward difference (so $\Delta_5 x_{i,t} \equiv x_{i,t+5} - x_{i,t}$), $d(i)$ denotes the one-digit SIC division to which industry i belongs, and $\alpha_{d(i),t}$ are year $\times$ SIC-1 division fixed effects. The coefficient of interest is β , which captures the within-division, cross-industry association between changes in entry rates and changes in scope over five-year horizons; level controls for scope and the entry rate at the base year absorb any mechanical relationship between levels and changes. We place scope on the left-hand side because the NETS scope series is measured with error (as evidenced by the level breaks noted above), and classical measurement error in a right-hand-side regressor would attenuate β toward zero; placing scope on the left increases standard errors but keeps the point estimate unbiased. Standard errors are clustered at the SIC-3 level to address correlation induced by the overlapping five-year differences.

Table 2 shows that industries experiencing larger declines in establishment entry rates also experience larger increases in scope, whether scope is measured by establishments, ZIP codes, or CBSAs per firm; the coefficients are negative and statistically significant in all six specifications, both in the full sample and when excluding FIRE³ industries. Figure 2 displays binned scatter plots of the residualized five-year changes, after partialling out the division-by-year fixed effects. The analogous exercise using firm-level entry rates (reported in Appendix Table A-3) yields coefficients of the same sign but smaller in magnitude and mostly imprecisely estimated, consistent with firm births at the four-digit NAICS level being noisier than establishment births in the BDS. These correlations are consistent with the mechanism we develop in the model, in which technological change drives both higher firm scope and less entry.

2 A Simple Model of Intangibles

This section presents a static model that isolates the two fundamental economic properties of intangible capital: imperfect replicability and imperfect exclusivity. We define the *scope* of a firm, x , as the number of distinct locations—or, equivalently, product markets—in which the firm produces using its intangible capital. Firms can increase scope by deploying the same intangible across multiple locations; the degree of replicability is potentially imperfect and captured by the parameter ρ . Absent any cost of scaling up, the firm would produce using its intangible in every possible location and output would be infinite. The endogenous cost of increasing scope does not take the form of a resource cost; rather, it arises from a hold-up problem with the firm’s key employees, whose severity depends on the quality of property rights enforcement, captured by the parameter λ .

2.1 Setup

To fix ideas, consider a single firm in a static framework. The firm owns an intangible asset n that it can deploy in multiple locations $m \in [0, x]$, where x is the scope of the firm—the measure of locations or product markets in which the firm operates. Absent within-location decreasing returns, operating profits are additively separable

³Finance, Insurance, and Real Estate

across locations and equal the integral of intangible “services” deployed at each site:

$$\pi = \int_0^x n(m) dm. \quad (3)$$

For now, take the choice of scope as given. The firm decides how to allocate its intangible capital across locations, subject to the following budget constraint,

$$\left(\int_0^x n(m)^{\frac{1}{1-\rho}} dm \right)^{1-\rho} \leq n. \quad (4)$$

The budget constraint (4) captures our first main economic assumption: the intangible asset can be (potentially imperfectly) replicated across varieties absent a resource cost.

Assumption 1 (Replicability) *The parameter ρ satisfies $0 \leq \rho \leq 1$.*

The parameter ρ governs the degree to which intangible capital can be replicated without a resource cost. When $\rho = 0$, there is no replicability: intangible capital behaves like a machine, so deploying it in location m precludes deploying it in location m' . When ρ approaches 1, the constraint (4) becomes:

$$\lim_{\rho \rightarrow 1} \left(\int_0^x n(m)^{\frac{1}{1-\rho}} dm \right)^{1-\rho} = \max_{m \in [0, x]} n(m) \leq n. \quad (5)$$

In this limit the intangible is perfectly replicable: the firm can set $n(m) = n$ for all locations. Our model focuses on the intermediate case $0 < \rho < 1$, in which replication is possible but imperfect. To illustrate, consider a firm deploying a new inventory management system across its production facilities. Doing so requires re-allocating managers or key employees to each facility, which may degrade performance at other sites—effectively reducing the intangible input available elsewhere. Similarly, a firm may share a customer list across branches for promotion purposes, but only up to a point: branches competing for the same customers reduce the effective value of the list at each location. Section 2.2 provides a formal microfoundation for this assumption.

The firm maximizes profits (3) subject to (4). Given the firm faces a symmetric problem, the optimal allocation of intangibles to each variety m equals

$$\forall m, \quad n(m) = x^{-(1-\rho)} n. \quad (6)$$

When intangible capital is non-replicable ($\rho = 0$), the entrepreneur divides the total stock equally among locations, $n(m) = n/x$. When it is perfectly replicable ($\rho = 1$), the entire stock n can be deployed in every location, $n(m) = n$.

Given the entrepreneur’s optimal allocation of intangibles, the total profits of the firm are

$$\pi = x^\rho n. \quad (7)$$

Firm profits are increasing in both the quality of the intangible n and the scope of production x . When $\rho > 0$, profits (7) exhibit *increasing* joint returns to scope and intangible capital—the economies of scope emphasized by [Haskel and Westlake \(2017\)](#). The marginal return to scope rises with n , and more so the higher the value of ρ .

We turn to the choice of firm scope. Choosing a scope x requires the entrepreneur to give up a fraction $\gamma(x; \lambda)$ of the firm's value to managers and other key stakeholders. Section 3.2.2 microfound this tradeoff: expanding scope requires sharing the firm's intangible assets with managers or financiers, and ideas, once communicated, can be replicated or expropriated (Arrow, 1962).

Assumption 2 (Exclusivity) *The function $\gamma(\cdot; \cdot) : R_+^2 \rightarrow R_+$ is twice continuously differentiable. Moreover, for each λ , $\gamma(\cdot; \lambda)$ is strictly increasing and weakly convex in x , and satisfies boundary conditions specified in Appendix 1. Finally, for each $x \geq 1$, it is weakly increasing in λ .*

The parameter λ governs the tradeoff between broader scope and a smaller retained share. Since $\gamma(x; \cdot)$ is increasing in λ for every $x \geq 1$, higher λ captures weaker exclusivity—equivalently, weaker intellectual property rights enforcement.

The entrepreneur chooses x to maximize her retained share of operating profits after compensating managers:

$$\pi^E = \max_x (1 - \gamma(x; \lambda)) x^\rho n. \quad (8)$$

The enterprise value of the firm is log-additive in x and n , so the optimal scope is independent of the quality of the intangible. Appendix 1 gives boundary conditions on $\gamma(\cdot; \lambda)$ that ensure the optimal scope exists, is unique, and satisfies $x^* \geq 1$.

The model captures the two defining economic properties of intangible capital. First, intangibles can be replicated across locations, but the degree of replicability may be imperfect. Second, the lack of physical presence makes intangibles easier to expropriate, so expanding scope requires the entrepreneur to share rents with key employees.

2.2 Microfoundation for imperfect replicability

The replicability constraint (4) reflects the idea that intangible capital is productive knowledge that must be stored in a particular medium—people, processes, or software—in order to be shared within the firm and used in production. The technology for storing and sharing this information determines how replicable the intangible is.

Suppose the firm can produce output in a continuum of locations indexed by $m \in [0, x]$, where each location can represent a geographic site or a point in product space. The total profits of the firm equal

$$\pi(j) = \int_0^x n(m)^\omega dm, \quad (9)$$

where $n(m)$ is the quantity of the intangible stored at location m . We order locations by distance from the firm's origin, so m serves both as the location index and as the distance from the origin; transferring information or expertise is costly. Given a stock of intangible capital n , the effective quantity stored at location m is

$$n(m) \equiv n \cdot \left(\tilde{n}(m) - \frac{1}{2} \alpha(m) \tilde{n}(m)^2 \right), \quad (10)$$

where the cost of transfer is increasing in distance:

$$\alpha(m) = \frac{1}{2} m^{1-\rho}. \quad (11)$$

This distance can represent geographic separation or distance in product space—applying the same expertise to products further from the entrepreneur’s core knowledge is more costly. The parameter $\rho \in (0, 1)$ controls the steepness of this cost: when $\rho = 1$, the cost does not vary with distance, corresponding to perfect replicability.

The firm chooses $\tilde{n}(m)$ to maximize profits (9). The solution, $\tilde{n}(m) = 1/\alpha(m)$, yields total profits proportional to

$$\pi(j) \propto x^{1-\omega+\omega\rho} n^\omega, \quad (12)$$

which is the same profit function as in (7) (up to the exponent ω , which arises when the model is embedded in the general-equilibrium setting of Section 3).

Two examples illustrate how different intangible assets map to different values of ρ . First, consider data on the purchasing behavior of a firm’s customers, used across the firm’s call centers to target promotions. In the past, such data was stored on paper; today it resides in databases that can be replicated across all locations at near-zero cost. This corresponds to an intangible asset with relatively high ρ , one that has likely increased over time as digital storage and dissemination technologies have improved. Even so, $\rho < 1$ remains plausible: if local demand differs across locations, transferring data to a sufficiently different market is more costly than transferring it to a nearby one.

Second, consider a particular method for organizing production, deployed across a firm’s plants. In each plant, the method is stored in key employees—assembly-line workers and production managers. Replicating the method across plants requires supervision and training, which may be imperfect. Deploying the method to plants that are sufficiently different from the origin is increasingly costly, either because of distance or because the firm cannot hire and train a sufficient number of managers who can perfectly acquire the transmitted knowledge. If such talent is scarce, ρ is low. Advances in the technology for communicating instructions and supervising remote production will increase ρ .

The degree of replicability is an intrinsic feature of the storage technology used to disseminate the intangible. Since each firm’s intangible is fixed at entry, ρ is also constant within the firm. Section 4 studies the transitional dynamics when ρ changes: newly created firms take advantage of improved replication technology, while incumbents remain at the old level.

The next section embeds this mechanism in a dynamic general equilibrium model of endogenous growth.

3 The Full Model

The model embeds the key elements from the simple model—imperfect replicability and imperfect exclusivity of intangible assets—in an otherwise standard endogenous growth framework (Romer, 1990): the final good is produced by combining intermediate varieties, new varieties are endogenously created, and their creation generates positive knowledge spillovers. These features primarily aid tractability and illustrate the connection with the literature; the model remains qualitatively similar if all varieties are perfect substitutes and price equals

marginal cost, since firms still earn ex-post profits due to local decreasing returns to scale.

3.1 Setup

We begin by discussing the model setup.

3.1.1 Final good producers and households

Final good producer. The final good is produced using a CES production technology from a measure X_t of intermediate varieties,

$$Y_t = \left(\int_0^{X_t} y_t(m)^\chi dm \right)^{\frac{1}{\chi}}. \quad (13)$$

Here Y_t is total final output, X_t is the measure of intermediate varieties available at date t , and $\chi \in [0, 1]$ governs the elasticity of substitution between varieties. Each variety m may represent a different good or a different location where services are offered.

There is a perfectly competitive final good producer that purchases and aggregates intermediate varieties. Their cost-minimization problem yields the familiar demand system for each variety and the exact price index for the aggregator,

$$y_t(m) = \left(\frac{p_t(m)}{P_{Y,t}} \right)^{-\frac{1}{1-\chi}} Y_t, \quad P_{Y,t} \equiv \left(\int_0^{X_t} p_t(m)^{-\frac{\chi}{1-\chi}} dm \right)^{-\frac{1-\chi}{\chi}} = 1. \quad (14)$$

Here $y_t(m)$ is quantity demanded of variety m , $p_t(m)$ is its price, and $P_{Y,t}$ is the ideal price index associated with (13). Without loss of generality, we normalize the price of the final consumption good to one.

Households. There is a representative household with preferences over aggregate consumption given by:

$$U_t = \int_{s \geq t} e^{-\eta s} \log(C_{t+s}) ds, \quad (15)$$

where U_t is the household's expected discounted utility from calendar date t forward, C_{t+s} is aggregate consumption, and $\eta > 0$ is the subjective discount rate.

Within the household, there are two groups of agents, each with unit mass. Agents in the first group supply labor that can be freely allocated toward entrepreneurship $L_{E,t}$ or production $L_{Y,t}$. Their total time is normalized to one,

$$L_{Y,t} + L_{E,t} \leq 1. \quad (16)$$

Entrepreneurial labor is compensated through ownership stakes in firms, while production labor earns the wage rate W_t . Because of perfect substitutability between the two, in equilibrium the compensation to a unit of entrepreneurial labor will equal the wage rate. The second group of agents consists of managers who inelastically supply a fixed unit of managerial time,

$$L_{M,t} \leq 1. \quad (17)$$

In equilibrium this constraint binds with $L_{M,t} = 1$. Managers are instrumental in the creation of new firms, as

we discuss below. Managerial labor is also compensated through ownership stakes in firms, which managers co-own alongside entrepreneurs.

The household pools income from all these agents and provides full consumption insurance against idiosyncratic shocks. The household's flow budget constraint equates the change in the value of equity holdings to wage and dividend income net of consumption:

$$V_t dS_t = (W_t L_{Y,t} + D_t S_t - C_t) dt, \quad (18)$$

where V_t is the price of a unit claim to aggregate equity, D_t is the dividend rate per unit, and S_t is the number of such claims held by the household (normalized to $S_t = 1$ in equilibrium). Under log utility, the Euler equation for aggregate consumption is

$$\frac{dC_t}{C_t} = (r_t - \eta)dt, \quad (19)$$

where r_t is the riskless interest rate, equal in equilibrium to the total return on equity, $r_t \equiv \frac{D_t}{V_t} + \frac{dV_t}{V_t}$.

3.1.2 Intermediate good producers

We index intermediate goods producers by j , and refer to them as “firms”. Let $t_e(j)$ be the date at which firm j enters. This firm is created by a coalition of entrepreneurs and managers. At $t = t_e(j)$, firm j is endowed with a stock of intangible capital $n(j)$. At that point, the firm's entrepreneurs make two irreversible choices: the scope of the firm—that is, the number of locations or product markets in which it produces— $x(j)$; and how to allocate the firm's total intangible stock, $n(j)$, across the production of these varieties. After these choices have been made ($t > t_e(j)$), for each variety within their scope, entrepreneurs make all remaining production and pricing decisions, including hiring production labor. Each firm is the monopolistic producer of each variety within its scope. Each firm exits with Poisson intensity $\delta > 0$.

Output of variety m combines local intangible capital and production labor under constant returns:

$$y_t(m) = n(m)^{1-\zeta} l_t(m)^\zeta, \quad (20)$$

where $\zeta \in (0, 1)$ is the labor elasticity in this Cobb–Douglas technology, $l_t(m)$ is production labor allocated to variety m , and $n(m)$ is the intangible capital committed to that variety at entry. The latter decision is irreversible and is made when the firm is created.

Figure 3 shows the different stages of each firm's problem. We solve it backward: first the monopoly pricing problem for each variety, then the allocation of intangible capital across varieties at $t = t_e(j)$, and finally the choice of scope at $t = t_e(j)$. The determination of the initial intangible stock $n(j)$ and the size of the coalition of entrepreneurs and managers is deferred to Section 3.1.3. To simplify notation, we omit the firm index j ; variables not indexed by time t are fixed at entry.

Monopoly pricing problem for each variety. Given the allocation of intangible capital across varieties, the profit maximization problem for each variety is standard:

$$\begin{aligned} \pi_t(m) &= \max_{l_t(m), p_t(m)} p_t(m) y_t(m) - W_t l_t(m) \\ \text{s.t. } y_t(m) &\leq n_t(m)^{1-\zeta} l_t(m)^\zeta \quad [mc_t(m)] \end{aligned} \quad (21)$$

where the firm internalizes its choice of quantity on the equilibrium price (14). With a continuum of varieties, the firm's choice of scale for variety m has no effect on the price of other varieties m' within its scope. The firm therefore chooses a constant markup over marginal cost,

$$p_t(m) = \frac{1}{\chi} mc_t(m). \quad (22)$$

Define:

$$\omega \equiv \frac{(1-\zeta)\chi}{1-\zeta\chi} \in [0, 1] \quad (23)$$

$$A_t \equiv (1-\zeta\chi) \left(\frac{Y_t^{1-\chi} (\zeta\chi)^{\zeta\chi}}{W_t^{\zeta\chi}} \right)^{\frac{1}{1-\zeta\chi}} \quad (24)$$

Then the resulting labor demand and flow profits from each variety can be written as:

$$l_t(m) = \frac{\zeta\chi}{1-\zeta\chi} \frac{A_t}{W_t} n(m)^\omega \quad (25)$$

$$\pi_t(m) = A_t n(m)^\omega \quad (26)$$

The variable A_t governs the marginal revenue product of capital invested in any variety. The reduced-form parameter ω captures two sources of decreasing returns to that capital: diminishing returns in production within each variety, and, whenever $\chi < 1$, the downward-sloping demand curve.

Allocating intangible capital across varieties. Let x denote the scope of the firm, that is, the measure of varieties it produces. Taking x as given for now, the enterprise value of the firm at any time $t \geq t_e(j)$ equals the present discounted value of the profits of all varieties it produces

$$\tilde{v}_t(x, \{n(m)\}_{m \in [0, x]}) \equiv \int_{s \geq 0} e^{-\delta s - \int_0^s r_{t+u} du} \left\{ \int_0^x \pi_t(m) dm \right\} ds = \nu_t A_t \int_0^x n(m)^\omega dm \quad (27)$$

where

$$\nu_t \equiv \int_0^{+\infty} e^{-\delta s - \int_0^s r_{t+u} du} \frac{A_{t+s}}{A_t} ds \quad (28)$$

is the price-to-earnings ratio of the firm at time t .

When the firm is created at $t = t_e(j)$, it makes an irreversible decision of how much of the firm's total intangible capital n it will allocate across the varieties it is producing in $m \in [0, x]$. The firm will choose the

intangible allocation to maximize its enterprise value,

$$v_t(n, x) = \max_{\{n(m)\}_{m \in [0, x]}} \tilde{v}_t \left(x, \{n(m)\}_{m \in [0, x]} \right), \quad (29)$$

subject to the constraint (4) that captures the imperfect replicability of intangible assets.

The firm maximizes its enterprise value (27), internalizing how profits relate to the intangible capital invested in each variety (26) and subject to (4). Given the firm faces a symmetric problem, the optimal allocation of intangibles to each variety m is again given by (6) as in the simple model.

Given the entrepreneur's optimal allocation of intangibles, the total profits of the firm across all the varieties it is producing equal

$$\pi(n, x) \equiv \int_0^x \pi(m) dm = A_t x^{1-\omega+\omega\rho} n^\omega, \quad (30)$$

and the enterprise value of her firm is

$$v_t(n, x) = \nu_t A_t x^{1-\omega+\omega\rho} n^\omega. \quad (31)$$

As in the simple model, firm profits (30) and enterprise value (31) feature *increasing* returns to scope and intangible capital when $\rho > 0$. The existence of the intangible asset also implies that the firm earns ex-post profits—the difference between revenue and labor costs of production—even when the elasticity of substitution across varieties is perfect ($\omega = 1$) and markups are zero.

Choosing scope. The choice of scope occurs at entry and is irreversible. As in the simple model, choosing scope x requires providing managers with at least a fraction $\gamma(x; \lambda)$ of the firm's value. The entrepreneur's problem is to jointly choose scope and managerial compensation,

$$\tilde{v}_t^{(E)}(n) = \max_{\{\tilde{v}_t^{(M)}, x\}} v_t(n, x) - \tilde{v}_t^{(M)} \quad (32)$$

$$\text{s.t. } \tilde{v}_t^{(M)} \geq \gamma(x; \lambda) v_t(n, x) \quad (33)$$

Here $\tilde{v}_t^{(E)}(n)$ is total entrepreneurial compensation and $\tilde{v}_t^{(M)}$ is total managerial compensation. The constraint (33) ensures that managers receive at least their incentive-compatible share. We adopt the functional form

$$\gamma(x; \lambda) = \kappa x^\lambda \quad \text{with } \lambda > 0 \quad \text{and} \quad \kappa < \frac{1-\omega}{1-\omega+\lambda}. \quad (34)$$

The functional form is microfounded in Section 3.2.2 as the result of a hold-up problem between managers and entrepreneurs. The solution to the entrepreneur's problem (32) is:

$$\begin{aligned} x &= \arg \max_x (1 - \gamma(x; \lambda)) v_t(x, n), \\ \tilde{v}_t^{(E)} &= (1 - \gamma(x; \lambda)) v_t(x, n), \\ \tilde{v}_t^{(M)} &= \gamma(x; \lambda) v_t(x, n). \end{aligned} \quad (35)$$

Since enterprise value is log-additive in x and n , optimal scope is given by

$$x = \left(\frac{1 - \omega(1 - \rho)}{\kappa(1 - \omega(1 - \rho) + \lambda)} \right)^{\frac{1}{\lambda}} > 1. \quad (36)$$

Optimal scope is increasing in the degree of replicability ρ , increasing in the strength of property rights (decreasing in λ), and decreasing in the level of markups (increasing in χ). Along a balanced growth path, optimal scope is constant and identical across firms. Along transition paths, however, a shift in structural parameters affecting only newly created firms generates persistent cohort-specific differences in scope.

3.1.3 The creation of new intangible assets

Both entrepreneurs and managers are needed to create a new firm. Given total entrepreneurial effort $L_{E,t}$ and total managerial effort $L_{M,t}$, the number of new firms created in the interval $[t, t + dt]$ is equal to $E_t dt$, where

$$E_t \equiv L_{E,t}^\alpha L_{M,t}^{1-\alpha}, \quad \text{with } \alpha \in (0, 1). \quad (37)$$

After the firm is created and the entrepreneur's intangible is deployed in each location, operating the firm does not require any additional unit of time, from either managers or entrepreneurs.

For simplicity, we abstract from within-cohort differences across firms: all firms born at time t are endowed with the same amount of intangible capital:

$$\forall j \text{ s.t. } t_e(j) = t, \quad n(j) = n_t = \Xi_t. \quad (38)$$

Here, Ξ_t represents a public stock of knowledge, which evolves as:

$$\frac{d\Xi_t}{\Xi_t} = \xi (\gamma(x_t; \lambda) x_t^\rho)^\mu E_t^\sigma dt. \quad (39)$$

The parameter $\xi > 0$ governs, in equilibrium, the relative size of newly created firms compared to existing ones. As in [Romer \(1990\)](#) and [Jones \(1995\)](#), the model features knowledge spillovers from current to future entrants, with $\sigma \in [0, 1]$ determining their magnitude. The key novelty relative to the standard endogenous growth framework is that the rate of knowledge accumulation depends on the scope x_t at which new firms deploy their intangibles, and on the managerial share $\gamma(x_t; \lambda)$. The product $\gamma(x_t; \lambda) \cdot x_t^\rho$ captures the total knowledge acquired by managers across the firm's locations. The parameter $\mu \in [0, 1]$ governs the degree of diminishing returns in the conversion of acquired knowledge into public knowledge: not all acquired knowledge is equally novel, so doubling the total acquired knowledge less than doubles the spillover. The case $\mu = 0$ nests the benchmark in which the spillover term $(\gamma x^\rho)^\mu$ is identically one, so knowledge accumulation does not depend on scope or the managerial share; the quantitative analysis in [Section 5](#) includes this baseline. [Section 3.2.3](#) provides a detailed microfoundation.

We refer to $\xi (\gamma(x; \lambda) x^\rho)^\mu$ as the *spillover rate*. Since the optimal scope x depends only on structural parameters $(\rho, \lambda, \kappa, \omega)$, the spillover rate is constant along any balanced growth path. During transitions in which ρ changes over time, x_t and hence the spillover rate are time-varying.

Given the process of firm formation, the compensation to entrepreneurs and managers, per unit of time devoted, is given by

$$v_t^{(E)} \equiv \frac{E_t}{L_{E,t}} \tilde{v}_t^{(E)} = \frac{L_{E,t}^\alpha L_{M,t}^{1-\alpha}}{L_{E,t}} (1 - \gamma(x_t; \lambda)) v_t(x_t, n_t), \quad (40)$$

$$v_t^{(M)} \equiv \frac{E_t}{L_{M,t}} \tilde{v}_t^M = \frac{L_{E,t}^\alpha L_{M,t}^{1-\alpha}}{L_{M,t}} \gamma(x_t; \lambda) v_t(x_t, n_t). \quad (41)$$

Agents in the first group can freely choose to become entrepreneurs or production workers. As long as entry is positive, they must be indifferent between the two:

$$L_{E,t} > 0 \quad \text{and} \quad v_t^{(E)} = W_t. \quad (42)$$

Entry is strictly positive because the marginal benefit of entry in (37) is infinite at $L_{E,t} = 0$.

Last, the total measure of distinct intermediate varieties actively supplied at date t sums, across all incumbent firms $j \in [0, J_t]$, the scope of each firm surviving from its entry date $t_e(j)$:

$$X_t = \int_0^{J_t} e^{-\delta(t-t_e(j))} x_{t_e(j)} dj. \quad (43)$$

New varieties are therefore created together with new firms, in proportion to those firms' scopes; the law of motion for J_t links net firm creation to entry and exit. The mass of active firms evolves according to

$$\begin{aligned} dJ_t &= dE_t - \delta J_t dt \\ &= (L_{E,t}^\alpha L_{M,t}^{1-\alpha} - \delta J_t) dt. \end{aligned} \quad (44)$$

The parameter δ is the exogenous exit rate.

3.1.4 Equilibrium

The model is closed by assuming that the final good market and the market for production labor must clear:

$$Y_t = C_t, \quad L_{Y,t} + L_{E,t} = 1, \quad L_{M,t} = 1, \quad \int_0^{X_t} l_t(m) dm = L_{Y,t}, \quad (45)$$

where $l_t(m)$ is labor demand from the firm producing variety m at time t , defined in equation (25).

An equilibrium of this economy is an allocation, prices, and a number of firms such that: the conditions in Equation (45) hold; the stock market clears, $S_t = 1$; the free-entry condition (42) holds; and for each new firm, the initial stock of intangible capital must be given by Equation (38), the optimal scope of the firm by Equation (35), and the optimal allocation of intangibles across varieties by Equation (29).

3.2 Discussion of the key assumptions

This subsection provides explicit microfoundations for the hold-up function $\gamma(x; \lambda)$ in the manager's incentive-compatibility constraint (33) and for the knowledge spillover term in (39). Section 2.2 provides the microfoundation for the replicability constraint (4); for convenience, we summarize the key result here.

3.2.1 Imperfect replicability

The microfoundation in Section 2.2 models the intangible as information that must be transferred across locations at a cost that rises with distance. The parameter ρ governs the steepness of this transfer cost. When the firm optimally allocates its intangible stock across locations, total profits are proportional to $x^{1-\omega+\omega\rho} n^\omega$, delivering the profit function (30). The stored intangible at location m is $n(m) \propto m^{\rho-1} n$, a fact we use below to derive the hold-up payoff and the knowledge spillover.

3.2.2 Imperfect exclusivity

To microfound the function $\gamma(x; \lambda)$ in the incentive-compatibility constraint (33), we need to specify the manager's outside option. Managers are needed to deploy intangibles across locations. However, once information is transferred to a location m , the manager can operate that location without the entrepreneur, since the intangible is now stored and codified there. This creates an ex-post hold-up problem: the manager can attempt to use the codified knowledge and keep the value of the output.

We assume that a defecting manager captures the full output of each location they appropriate. This is supported by the following reasoning: the entrepreneur could hire a replacement manager, but if the original manager found it profitable to expropriate, a replacement would face the same incentive. The entrepreneur therefore ends up with zero payoff regardless of whether she tries to compete.

If the manager attempts to hold up the entrepreneur, the entrepreneur can pursue legal action to prevent this from happening. The success of doing so is uncertain and depends on the legal environment. In particular, property rights are harder to enforce in locations that are further from the origin. Hence, the probability that the entrepreneur *prevails* in location m —that is, that legal enforcement blocks the manager from appropriating output using the firm's intangible—is given by

$$\kappa (1 + \lambda) m^\lambda. \quad (46)$$

Here κ indexes baseline enforcement strength, while λ governs how sharply enforcement deteriorates with distance. Securing property rights is harder in locations that are geographically remote or in product markets far from the entrepreneur's core expertise; (κ, λ) jointly index the quality of the legal environment. If the manager finds it profitable to defect in one location, they will defect in every location.

As a result, integrating the per-location deviation payoff over all x locations, the manager's total deviation value is:

$$v_t^{(d)} = \int_0^{x_t} \kappa (1 + \lambda) m^\lambda \cdot \pi_t(m) dm. \quad (47)$$

From the intangible allocation problem (Section 3.2.1), the optimal intangible stored at location m is $n(m) \propto$

$m^{\rho-1}n_t(j)$, so the per-location profit from (9) satisfies $\pi_t(m) \propto n(m)^\omega \propto m^{\omega(\rho-1)}n_t(j)^\omega$. Substituting:

$$v_t^{(d)} \propto \kappa(1+\lambda)n_t(j)^\omega \int_0^{x_t} m^{\lambda+\omega(\rho-1)} dm = \kappa n_t(j)^\omega x_t^{\lambda+1-\omega(1-\rho)}. \quad (48)$$

Dividing by the total enterprise value $v_t \propto x_t^{1-\omega(1-\rho)}n_t(j)^\omega$ from (31), the factor $x_t^{1-\omega(1-\rho)}$ cancels exactly:

$$\gamma(x; \lambda) \equiv \frac{v_t^{(d)}}{v_t} = \kappa x^\lambda, \quad (49)$$

which gives the functional form in equation (34). Notably, γ does not depend on ω : markups affect the per-location profit and the total enterprise value in the same proportion, so they cancel in the ratio. More generally, different assumptions for the quality of property right enforcement (46) would lead to different functional forms for $\gamma(x; \lambda)$.

The hold-up friction causes enterprise value to be shared between the entrepreneur and the managers. The entrepreneur must compensate managers enough to prevent them from starting competing firms using the knowledge she shares with them—the Arrow information paradox (Arrow, 1962). This friction is empirically salient. In the late 1980s, software maker Peoplesoft and its founder David Duffield were sued by Integral Systems, which claimed that its software was based on code stolen while Duffield worked there. More broadly, Bhide (1999) reports that 71% of the founders of firms on the Inc 500 list replicated or modified ideas encountered through previous employment.

The managers in the model should be interpreted broadly as all agents beyond the entrepreneur who are necessary to deploy the firm’s intangible, including financiers. Financiers can appropriate rents by diluting the innovator’s stake, which is often feasible given the contractual features of venture capital arrangements (see, e.g., Kaplan and Stromberg, 2004). Atanasov, Ivanov, and Litvak (2012) describe how the founder of Pogo.com, an e-gaming company, saw his stake reduced from 13% to 0.1% through derivative securities issued by the VCs on the board. Financiers can also transfer a firm’s knowledge to other portfolio companies (Cabezon and Hoberg, 2024).

3.2.3 Knowledge spillovers

The law of motion for public knowledge (39) features the product $\gamma(x_t; \lambda) \cdot x_t^\rho$ multiplying the standard entry-driven spillover term E_t^σ . This specification is microfounded by combining the two primitives just described—imperfect replicability (Section 3.2.1) and imperfect exclusivity (Section 3.2.2)—and recognizing that they jointly determine how much knowledge managers acquire from the firm.

The connection between the two primitives is that the enforcement probability $p(m; \lambda) = \kappa(1+\lambda)m^\lambda$ from Section 3.2.2 measures not only the manager’s ability to *expropriate* but also the degree to which the manager has *acquired* the firm’s knowledge at location m . A manager can only threaten to independently use knowledge that they actually understand. At locations close to the origin, the intangible remains close to the entrepreneur’s personal expertise and need not be extensively codified; the manager has limited independent understanding. At more distant locations, the intangible must be more explicitly codified—documented in procedures, embedded in transferable systems—for the manager to operate effectively. This codification makes the knowledge both more

useful in production and more portable.

Given the stored intangible $n(m) = n \cdot m^{\rho-1}$ from Section 3.2.1 and the acquisition fraction $p(m; \lambda)$, the knowledge acquired by the manager at location m is:

$$a(m) = p(m; \lambda) \cdot n(m) = \kappa(1 + \lambda) n m^{\lambda+\rho-1}. \quad (50)$$

The total knowledge acquired by all managers across the firm's x locations is:

$$\int_0^x a(m) dm = \kappa(1 + \lambda) n \frac{x^{\lambda+\rho}}{\lambda + \rho} \propto \gamma(x; \lambda) \cdot x^\rho \cdot n. \quad (51)$$

This acquired knowledge enters the public domain when firms exit. Firms die at Poisson rate δ ; upon exit, the firm's managers are released into the labor pool, carrying with them the knowledge they acquired. In balanced growth, the flow of dying firms equals the flow of new firms, so the total knowledge released per unit of time is proportional to $E_t \cdot \gamma(x; \lambda) \cdot x^\rho \cdot n_t$.

Not all of this knowledge is equally novel or useful for future innovation: much of what managers acquire is firm-specific or redundant with existing public knowledge. The transfer of acquired managerial knowledge into the public stock therefore has decreasing returns ($\mu < 1$): the spillover is proportional to $(\gamma(x; \lambda) \cdot x^\rho)^\mu$ rather than to $\gamma(x; \lambda) \cdot x^\rho$ directly, with $\mu \in [0, 1]$. The polar case $\mu = 0$ shuts this channel down, so knowledge spillovers no longer depend on scope or the managerial share. Each new firm is endowed with $n_t = \Xi_t$; combining this with diminishing returns to the quantity of new firms through σ yields the law of motion (39).

The spillover rate $\xi (\gamma(x; \lambda) \cdot x^\rho)^\mu$ combines three factors: the non-exclusivity share $\gamma(x; \lambda) = \kappa x^\lambda$ (the aggregate probability that managers have acquired the intangible), the knowledge footprint x^ρ (total effective intangible stored across locations), and the baseline spillover intensity ξ . The concavity parameter μ governs how quickly additional acquired knowledge translates into public knowledge; when $\mu < 1$, there are diminishing returns to knowledge diffusion.

3.2.4 Increasing scope versus new entry

The model features two dimensions of aggregate growth: the creation of new intangibles (the standard mechanism in endogenous growth models) and the endogenous choice of scope at which each firm deploys its intangible. These two dimensions differ in important ways, and it is useful to distinguish which of our assumptions are fundamental to the economics versus those made for tractability.

A key feature is that increasing firm scope *does* generate positive knowledge spillovers. As derived in Section 3.2.3, the rate of public knowledge accumulation depends on $\gamma(x; \lambda) \cdot x^\rho$, which is increasing in x . This creates a fundamental tension: stronger IP protection (lower λ) raises scope and the knowledge spillover per firm, but depresses new firm entry; weaker IP protection does the opposite. Section 4 shows that an interior optimal level of IP protection can maximize long-run growth.

The two dimensions of growth also differ in their timing and whether they operate within or across firms. Intangible capital is fixed within a firm and cannot be accumulated after entry, in line with putty-clay models of technology adoption (Gilchrist and Williams, 2000). Aggregate intangible capital still grows over time through the

creation of new firm cohorts, so the model speaks to the behavior of aggregate investment. The assumption that scope is chosen immediately at entry while innovation takes time is a simplification; the essential requirement is that an optimal level of scope exists for each firm, so that once it is reached, further growth requires new intangibles.

Finally, we have taken a strong stance on what constitutes a firm. Firm boundaries are not fundamental in the model: a firm could be redefined as an arbitrary collection of projects, as in [Kogan, Papanikolaou, and Stoffman \(2020\)](#). With ex-ante heterogeneity in the likelihood that firms acquire new projects, shifts in ρ would have a heterogeneous impact on firms, much as vintage-specific capital productivity shocks in [Kogan et al. \(2020\)](#) displace large incumbents in favor of younger, fast-growing firms.

3.3 Equilibrium characterization

We now characterize the equilibrium. We first aggregate firm-level decisions into an aggregate production function. We then collect the key equilibrium conditions into a dynamic system.

3.3.1 Aggregation

Aggregate demand for production labor can be written as

$$L_{Y,t} = \chi \frac{\zeta Y_t}{W_t}. \quad (52)$$

This follows from aggregating firms' labor demand for each variety (25) and clearing the labor market. The markup $1/\chi$ distorts labor demand downward. Aggregate output satisfies

$$Y_t = N_t^{1-\zeta} L_{Y,t}^\zeta, \quad (53)$$

where N_t is the quantity index of capital services formed from the intangibles stored across varieties; for brevity we refer to it below as the *capital index*:

$$N_t \equiv \left(\int_0^{X_t} n(m)^\omega dm \right)^{\frac{1}{\omega}}. \quad (54)$$

The derivation of (52)–(54) is in Appendix A.4. Section 5 discusses how N_t relates to empirical measures of the intangible capital stock.

3.3.2 Global dynamics

Optimal scope is pinned down by structural parameters alone, and not by endogenous variables describing the state of the economy. Recall from (35) that the entrepreneur chooses scope to maximize her retained share of enterprise value. Enterprise value $v_t(\tilde{x}, n)$ is proportional to $\tilde{x}^{1-\omega+\omega\rho} n^\omega$, from (31), so that the intangible stock n enters multiplicatively and drops out of the maximization. We therefore write x rather than x_t throughout.⁴

⁴Given the functional form $\gamma(\tilde{x}; \lambda) = \kappa \tilde{x}^\lambda$ from (34), the solution is given by (36). When we consider transitional dynamics following a change in ρ in Section 5, the same equations hold with ρ indexed by t , and therefore with scope written as x_t .

The equilibrium is then characterized by three objects: entrepreneurial effort $L_{E,t}$, the price-to-earnings ratio ν_t , and the sufficient state variable θ_t . Using optimal scope (36), the aggregation relations above, the labor market clearing condition (16), and the free-entry condition (42), entrepreneurial effort satisfies

$$\frac{L_{E,t}^{1-\alpha}}{1-L_{E,t}} = \frac{1-\zeta\chi}{\zeta\chi} (1-\gamma(x;\lambda)) x^{1-\omega+\rho\omega} \nu_t e^{\omega\theta_t}, \quad (55)$$

where

$$\theta_t \equiv \log\left(\frac{n_t}{N_t}\right) = \log\left(\frac{\Xi_t}{N_t}\right). \quad (56)$$

The variable θ_t is a sufficient state for this economy: all endogenous variables can be expressed as functions of θ_t . The law of motion for θ_t combines the growth rates of public knowledge Ξ_t from (39) and of the capital index N_t :

$$d\theta_t = \frac{1}{\omega} \left(\delta + \omega \xi (\gamma(x;\lambda) x^\rho)^\mu L_{E,t}^{\alpha\sigma} - x^{1-\omega+\rho\omega} L_{E,t}^\alpha e^{\omega\theta_t} \right) dt. \quad (57)$$

The price-to-earnings ratio $\nu_t = \nu(\theta_t)$ solves the ordinary differential equation

$$\eta \nu(\theta_t) = 1 - x^{1-\omega+\rho\omega} L_{E,t}^\alpha e^{\omega\theta_t} \nu(\theta_t) + \frac{1}{\omega} \left(\delta + \omega \xi (\gamma(x;\lambda) x^\rho)^\mu L_{E,t}^{\alpha\sigma} - x^{1-\omega+\rho\omega} L_{E,t}^\alpha e^{\omega\theta_t} \right) \nu'(\theta_t). \quad (58)$$

Appendix A.4 derives (55)–(58). Together with optimal scope (36), these equations characterize the global dynamics of the economy starting from any value of θ_t .

4 Comparative Statics

This section compares balanced growth paths corresponding to different values of ρ , the replicability parameter. We highlight two main results. First, higher replicability does not necessarily enhance the long-run growth potential of an economy. When knowledge spillovers are sufficiently small, higher replicability may harm growth. Second, higher replicability generally increases the level of measured TFP, the pure profit share of value added, firm valuations relative to the replacement cost of intangibles, and concentration.

4.1 Balanced growth path

A balanced growth path (BGP) is defined as an equilibrium in which the state variable θ_t is constant.

Result 1 (Balanced growth path) *Suppose that $\gamma(x;\lambda)$ is as in Equation (34). For any $(\rho, \lambda) \in [0, 1] \times [1, +\infty)$, there exists a unique BGP. Entrepreneurial effort L_E , the optimal scope of new firms x , and the price-to-earnings ratio ν are all constant. The size of new firms grows at rate g_N .*

The proof is in Appendix A.5. On the BGP, output growth equals

$$g = (1-\zeta) g_N = (1-\zeta) \xi (\gamma(x;\lambda) x^\rho)^\mu L_E^{\alpha\sigma}. \quad (59)$$

The growth rate is the product of two endogenous components: the spillover rate $\xi (\gamma(x;\lambda) x^\rho)^\mu$, which is

increasing in scope and in the managerial share; and the entry rate $L_E^{\alpha\sigma}$, which depends on the incentives to create new firms. We now study how these two components respond to changes in ρ .

4.2 Firm scope, entry, and growth

We characterize the comparative statics of the model across BGPs as replicability ρ varies from 0 to 1, holding λ fixed. Figure 4 illustrates the resulting BGP growth rate at four values of the knowledge-diffusion concavity μ .

Higher replicability always raises firm scope. From (36), a higher ρ increases the returns to deploying intangibles across locations. Entrepreneurs respond by choosing a wider scope x . The managerial share $\gamma = \kappa x^\lambda$ rises in lockstep, as wider scope requires more delegation.

Higher replicability always depresses entry. The key mechanism runs through the managerial share γ . To isolate its role, consider a hypothetical economy in which γ is held fixed while scope and wages adjust. Higher ρ raises scope, which makes each new firm more valuable. But larger firms also hire more production workers, which bids up the equilibrium wage. To see that these two effects exactly cancel, note that on the BGP, entrepreneurial effort satisfies

$$\frac{\zeta}{1-\zeta} \frac{L_E}{1-L_E} = \frac{1-\gamma}{\omega} \frac{\delta + \omega \bar{\xi} L_E^{\alpha\sigma}}{\eta + \delta + \omega \bar{\xi} L_E^{\alpha\sigma}}, \quad (60)$$

where $\bar{\xi} \equiv \xi(\gamma x^\rho)^\mu$ is the spillover rate. The right-hand side depends on ρ only through γ and $\bar{\xi}$. When γ is held fixed, neither side changes with ρ , so L_E is unchanged.

In the actual model, γ is not fixed. Higher scope raises the managerial share, which means entrepreneurs must give up a larger fraction of enterprise value to expand. With the adjustment in the managerial share, the increase in enterprise value and the increase in the wage rate do not offset one another; instead, the return to entrepreneurship falls relative to wages, and entry declines. Formally, Appendix A.5 shows that $\partial L_E / \partial \rho < 0$.

Higher replicability may or may not depress growth. The effect of ρ on growth depends on the strength of knowledge spillovers. Recall from (59) that the BGP growth rate has two components: the spillover rate $\xi(\gamma x^\rho)^\mu$ and the entry rate $L_E^{\alpha\sigma}$. Differentiating with respect to ρ :

$$\frac{d \log g}{d\rho} = \underbrace{\mu \frac{d}{d\rho} \log(\gamma(x; \lambda) x^\rho)}_{\geq 0 \text{ (spillover channel)}} + \underbrace{\alpha\sigma \frac{d \log L_E}{d\rho}}_{\leq 0 \text{ (entry channel)}}. \quad (61)$$

The first term captures knowledge spillovers: higher ρ raises both x^ρ and optimal scope x , increasing the spillover rate. The second term captures the entry disincentive described above. The parameter μ governs the balance between the two forces. Appendix A.5.2 establishes formally the following result.

Result 2 (Growth and replicability) *Under appropriate conditions on μ and other parameters, $d \log g / d\rho|_{\rho=0} > 0$ and $d \log g / d\rho|_{\rho=1} < 0$, so $g(\rho)$ has an interior maximum at some $\hat{\rho} \in (0, 1)$.*

Figure 4 illustrates how greater values of the spillover intensity μ can turn a decreasing relationship between growth and replicability into an increasing one. When $\mu = 0$, the spillover channel is absent and higher ρ unambiguously reduces g through the entry disincentive. As μ rises, the spillover channel strengthens, and at the calibration of Section 5 growth becomes monotonically increasing in ρ for $\mu \in \{0.10, 0.25, 0.50\}$.

The replicability of intangible assets need not enhance the long-run growth potential of an economy. To reap the scope economies that intangibles afford, entrepreneurs must rely more heavily on managers and share the firm's core knowledge with them. This creates an incentive problem that depresses entry. Knowledge spillovers provide a countervailing force: larger firms diffuse more knowledge to managers. The net effect on growth depends on which force dominates.⁵

4.3 Other BGP comparative statics

We now study how higher replicability affects measured total factor productivity, the profit share, firm valuations, and concentration. Unlike its ambiguous effect on growth, the effect of higher ρ on each of these quantities is unambiguously positive when markups are sufficiently small.

Result 3 (Other BGP comparative statics) *When markups are sufficiently small (ω close to 1), the following hold across balanced growth paths:*

- (i) *measured TFP Z is increasing in ρ ;*
- (ii) *the pure profit share Π and aggregate Tobin's Q are increasing in ρ ;*
- (iii) *the sales Herfindahl–Hirschman index H is increasing in ρ .*

The remainder of this subsection explains the underlying economics. The proofs are in Appendix A.5.

The stock of intangible capital. In the model, aggregate output satisfies $Y_t = N_t^{1-\zeta} L_{Y,t}^\zeta$, where N_t is the capital index from (54). Existing approaches (Corrado, Hulten, and Sichel, 2009; Eisfeldt and Papanikolaou, 2013) measure the intangible capital stock K_t by capitalizing past investment expenditures. In the model, this approach works as follows. Spending on intangible creation, and the replacement cost of each unit of non-deployed intangible capital, are:

$$I_t = W_t L_{E,t} \quad \text{and} \quad P_{N,t} = W_t L_{E,t}^{1-\alpha} / n_t. \quad (62)$$

The latter follows from the fact that $L_{E,t}$ units of entrepreneurial labor create $L_{E,t}^\alpha n_t$ new units of intangible capital. The perpetual inventory method then implies:

$$\frac{dK_t}{K_t} = \left(\frac{I_t / P_{N,t}}{K_t} - \delta \right) dt = \left(\frac{n_t}{K_t} L_{E,t}^\alpha - \delta \right) dt. \quad (63)$$

This equation capitalizes the real value of past expenditures on R&D into an aggregate stock, K_t .

By contrast, the capital index N_t evolves as

$$\frac{dN_t}{N_t} = \frac{1}{\omega} \left(\left(\frac{n_t}{N_t} \right)^\omega x_t^{1-\omega+\rho\omega} L_{E,t}^\alpha - \delta \right) dt. \quad (64)$$

The growth of N_t depends on scope x_t , but K_t does not, because K_t measures the cost of creating intangibles while N_t measures the productive value of deploying them at scale. When $\rho > 0$, N_t grows faster than K_t , and the wedge between the two widens with ρ .

⁵Section 6 applies the analogous decomposition to IP enforcement λ and derives optimal policy.

To illustrate the implications of our model for measuring the stock of intangible capital, consider the case without markups ($\omega = 1$). In this case, existing measures of the intangible stock (K_t) omit the scope of intangibles in production. In practice, one could modify the existing approaches to account for (possibly) time-varying scope after taking a stance on the estimate of ρ . In particular, one could recover the appropriate measure of the intangible capital stock (N_t) by multiplying the observed expenditures on intangibles by x_t^ρ , which represents the scope at which intangible expenditures are effectively deployed.

Measured TFP. We define measured TFP as the Solow residual:

$$Z_t \equiv \frac{Y_t}{K_t^{1-\zeta} L_{Y,t}^\zeta} = \left(\frac{N_t}{K_t} \right)^{1-\zeta}. \quad (65)$$

Appendix A.5 shows that along any balanced growth path, measured TFP is given by

$$Z = x^{\frac{(1-\zeta)(1-\omega+\rho\omega)}{\omega}} \left(\frac{\delta + g_N}{\delta + \omega g_N} \right)^{1-\zeta} \left(\frac{L_E^\alpha}{\delta + \omega g_N} \right)^{\frac{1-\omega}{\omega}(1-\zeta)}. \quad (66)$$

When $\omega = 1$, the expression simplifies to $Z = x^{(1-\zeta)\rho}$, which is increasing in ρ because both x and ρ increase, establishing part (i) of Result 3 in the competitive limit. When $\omega < 1$, lower entry works against higher TFP through the last factor, but the scope effect dominates when markups are small. Section 5 examines TFP dynamics along transition paths.

Profit share and valuation ratios. On a balanced growth path, total output is divided into labor income $\zeta\chi Y_t$ and capital income $(1 - \zeta\chi) Y_t$. Capital income in turn decomposes into competitive returns and rents; Appendix A.5 provides details. In balanced growth, the pure profit share is the rent term relative to output, which is given by:

$$\Pi = (1 - \zeta\chi) \left[\gamma + (1 - \gamma) \frac{\eta(1 - \omega) g_N}{(g_N + \delta)(\eta + \delta + \omega g_N)} \right]. \quad (67)$$

Rents accrue partly to managers (the γ term) and partly to entrepreneurs (the markup term, which vanishes when $\omega = 1$). Higher ρ raises both channels, shifting income from competitive returns toward rents.

Aggregate Tobin's Q equals total enterprise value divided by the replacement cost of capital. Enterprise value is split between entrepreneurs (share $1 - \gamma$) and managers (share γ). In balanced growth, Appendix A.5 shows that:

$$Q = \frac{1}{1 - \gamma} \cdot \frac{g_N + \delta}{\omega g_N + \delta}. \quad (68)$$

The first factor in (68) reflects the hold-up friction: managers capture a share γ of enterprise value, inflating Q . Since γ is increasing in ρ , this factor is unambiguously increasing. The second factor reflects markups: when $\omega < 1$, entrepreneurs earn rents above competitive returns. When $\omega = 1$, the second factor equals one and $Q = 1/(1 - \gamma)$. When ω is sufficiently close to 1, entrepreneurs don't earn monopoly rents, and Q is close to $1/(1 - \gamma)$ for the same reason. Any additional effects arising because of the effect of replicability on growth are muted. The hold-up channel dominates, and higher ρ leads to higher profit shares and higher Tobin's Q , establishing part (ii) of Result 3.

Concentration. On the BGP, each firm's sales share declines over time as newer, larger firms enter. Appendix A.5.3.3 shows that the Herfindahl–Hirschman index of firm sales is:

$$H = \frac{(\delta + \omega g_N)^2}{L_E^\alpha (\delta + 2\omega g_N)}. \quad (69)$$

Higher ρ reduces entry, so fewer firms compete for market share, but also raises the spillover rate $\bar{\xi}$ (when $\mu > 0$), which means new firms are born at a larger size relative to incumbents. Both channels combine to yield part (iii) of Result 3 when markups are sufficiently small.

The comparative statics of Section 4.2 show that higher replicability raises firm scope, the managerial share, and depresses entry. These predictions match qualitatively the motivating facts of Section 1. Result 3 shows that higher replicability also raises the profit share, firm valuations, and concentration. These predictions are consistent with several broader trends in the U.S. economy: the rise in the profit share of value added (Barkai, 2020; De Loecker, Eeckhout, and Unger, 2020), the increase in firm valuations relative to measured capital (Hall, 2001; Crouzet and Eberly, 2023), and the rise in sales concentration (Autor, Dorn, Katz, Patterson, and Reenen, 2020). Section 5 discusses whether the model can also match these patterns quantitatively.

These comparative statics also raise a normative question. The analysis of Section 4.2 highlighted precisely how entrepreneurs must share entry rents with managers, and what the effects on growth were. This rent-sharing could potentially lead the private return to entry to be depressed relative to its social value, raising the question of whether the decentralized equilibrium is efficient, and what role policy could play. Section 6 tackles this question.

5 Numerical illustration

This section illustrates the quantitative implications of the model. We study two BGPs, indexed by 1990 and 2020, with replicability ρ as the only changing parameter. We choose BGP parameters to match empirical targets for the managerial share γ_t , average firm scope x_t , and 1990 output growth. We then study the model's predictions at the two BGPs and along the unanticipated transition between them.

5.1 Parameter choices

We assign parameter values to two balanced growth paths, indexed by 1990 and 2020, holding all parameters fixed across the two BGPs except for ρ , the degree of replicability. Externally fixed parameters are the time discount rate $\eta = 0.02$, the markup $1/\chi = 1.05$, and the labor-share elasticity $\zeta = 0.70$. The exit rate is $\delta = 0.10$, matching the BDS gross firm exit rate. The entry-technology parameters are set to $(\alpha, \sigma) = (1/2, 1)$, the symmetric Cobb–Douglas–Romer benchmark.⁶ The remaining parameters $\{\rho_{1990}, \rho_{2020}, \lambda, \kappa, \xi\}$ are pinned down by cross-BGP movements in the managerial share γ_t and average firm scope x_t jointly with output growth in 1990, as we explain below. The auxiliary parameter μ governing the strength of the manager-mediated knowledge spillover is not identified separately from ξ from BGP moments, so we set its value to $\mu = 0$ as a benchmark in what follows, and explore the sensitivity of model predictions with respect to μ in Section 5.2.

⁶These choices imply a reduced-form parameter $\omega = 0.857$. The comparative statics in Result 3 are established analytically only when ω is close to 1; Section 5.2 confirms that they hold numerically at the calibrated value.

We obtain estimates of the managerial equity share γ_{1990} and γ_{2020} as follows. The model has only two equity-holding groups, managers and entrepreneurs. Letting m denote the managerial equity stake and e denote the entrepreneurial equity stake, the managerial share is:

$$\gamma = m / (m + e). \quad (70)$$

The closest direct measure in the data is the equity-based compensation series of [Eisfeldt et al. \(2021\)](#), who report the dollar value of reserved shares for employee equity-based compensation scaled by aggregate market capitalization. Letting o denote outsider equity, this ratio can be written as:

$$\underline{\gamma} = m / (m + e + o). \quad (71)$$

Assuming all incentive-based equity in the data series of [Eisfeldt et al. \(2021\)](#) accrues to managers, the relationship between γ and $\underline{\gamma}$ is:

$$\gamma = \frac{\underline{\gamma}}{s^{\text{insider}}}, \quad \text{where } s^{\text{insider}} \equiv \frac{m + e}{m + e + o}. \quad (72)$$

We proxy s^{insider} using the post-IPO insider equity share, which we collect from SEC IPO filings. Combining decade averages of $\bar{\gamma}$ with decade averages of post-IPO insider share s^{insider} yields the targets $\gamma_{1990} = 0.15$ and $\gamma_{2020} = 0.30$. Appendix [A.1.3](#) provides the details.

We then combine these empirical targets with the average number of establishments per firm in the NETS data matched to Compustat, and data on output growth for the non-financial corporate business sector (NFCB) to pin down the remaining parameters. The empirical 1990–2020 change in scope is:

$$d \equiv \log(x_{2020} / x_{1990}) \approx 0.40. \quad (73)$$

Along with the managerial equity shares, these empirical values pin down $(\rho_{1990}, \rho_{2020}, \lambda, \kappa)$ as follows:

$$\hat{\lambda} = \log\left(\frac{\gamma_{2020}}{\gamma_{1990}}\right) / d, \quad \hat{\rho}_t = 1 - \omega^{-1}(1 - \hat{\lambda}\gamma_t / (1 - \gamma_t)), \quad t \in \{1990, 2020\}, \quad \hat{\kappa} = \gamma_{1990} x_{1990}^{-\hat{\lambda}}. \quad (74)$$

Given these parameter values, $\hat{\xi}$ is set so that BGP growth rate fits the observed growth rate $g_{1990} = 0.027$ of output in the NFCB sector, obtained from BEA data. Implementation details are in Appendix [A.6.1](#). Panels A and B of Table [3](#) report the implied parameter choices.

5.2 Implications across balanced growth paths

Panel C of Table [3](#) compares model-implied moments at the 1990 and 2020 BGPs. The 1990 column is independent of μ by construction. The four 2020 columns trace the model's predictions across $\mu \in \{0, 0.10, 0.25, 0.50\}$. We discuss the $\mu = 0$ benchmark first, and then turn to the role of μ in shaping 2020 predictions.

Implications when $\mu = 0$. In 1990, the model fits g_{1990} exactly by construction. It also delivers untargeted moments that are broadly consistent with the data: aggregate Tobin's Q of 1.26 (versus approximately 1.6 in the data), a price-to-output ratio V/Y of 1.69 (versus approximately 1.7 in the data), and a profit share of 5.2%

(versus a profit share of approximately 7.5% in the data).⁷ Moving from the 1990 to the 2020 BGP, the increase in replicability ρ generates qualitative movements in the right direction for every untargeted moment in Panel B of Table 3. Finally, output growth falls in both the model and the data, but the model decline is much smaller in magnitude: the empirical decline in NFCB output growth is about 1.6 percentage points, while the model predicts a decline of only 0.2 percentage points, from 2.7% to 2.5%.

It is worth clarifying why, as growth falls, the price-to-output ratio V/Y rises across BGPs. In this model, the valuation of the stock market reflects the valuation of incumbents, whose capital stock and set of varieties are set at birth. This is by contrast with other models, in which valuations may reflect future investment opportunities that might accrue to incumbents. Forces that depress growth by reducing entry also improve expected future cash-flows of incumbents, and therefore their valuations, by reducing the competition they face in output and labor markets. More precisely, in the model, an incumbent's market share of output is proportional to $(n_\tau / N_t)^\omega$, where n_τ is the stock of intangible capital the incumbent was born with, and N_t is the current level of the aggregate stock of intangible capital. When the aggregate stock N_t grows more slowly, incumbents are effectively displaced at a lower rate. In this sense, a transition to a higher ρ is akin to a form of investment-specific technical change that affects the wedge between the productivity of existing and new firms.

The role of μ . The auxiliary parameter μ controls how strongly knowledge spills across the manager-mediated channel as scope and the managerial share rise. The four 2020 columns of Panel C of Table 3 show how its value reshapes 2020 predictions, holding the 1990 BGP fixed. Output growth is by far the moment that is most sensitive to μ : it rises from 2.51% at $\mu = 0$ to 6.25% at $\mu = 0.50$. The price-to-output ratio V/Y is the second most sensitive, but declines instead of increasing. Under log utility, the growth in the intangible stock shrinks the present-value factor in this calibration. Profit share, concentration, and Tobin's Q exhibit only moderate sensitivity, and the level of TFP is approximately invariant to μ . Because μ is identified from BGP moments separately from other factors affecting the equilibrium growth rate, we treat it as a sensitivity parameter throughout the rest of the paper.

5.3 Transitional dynamics

We trace the response of the economy to an unanticipated, deterministic increase in replicability from ρ_{1990} to ρ_{2020} , taking place smoothly over a ten-year transition window, with all other parameters held at the values reported in Table 3 and $\mu = 0$. The economy is on the 1990 BGP for $t \leq 0$. At $t = 0$ agents learn the entire path of ρ_t and reoptimize under perfect foresight; we then solve the model forward using the algorithm described in Appendix A.6.2, which uses the 2020 BGP as the terminal condition. Figure 5 reports the resulting trajectories. Panels (b) and (c) display the cross-sectional *averages* of firm scope, $\hat{x}_t$, and managerial share, $\hat{\gamma}_t$, across all incumbent firms.⁸ The remaining four panels show how stocks, flows, and prices respond.

Output growth, productivity, and concentration. Output growth (panel d) is non-monotone: a small upward spike on impact, a brief dip below the 1990 level, a sharp acceleration during the transition, and a slow decline

⁷The empirical counterpart for aggregate Tobin's Q is shown in Figure A-2, panel (c). The empirical counterpart for the price-to-output ratio is constructed as the aggregate Compustat enterprise value (the numerator of Tobin's Q in Figure A-2, panel (c)) divided by NFCB nominal output (NIPA Table 1.14). The empirical counterpart for the profit share is shown in Figure A-2, panel (d).

⁸The average scope and average managerial share are survival-weighted mean of past entrants' scopes, which causes them to lag behind the marginal entrant.

thereafter. On impact, the announcement compresses the price-to-earnings multiple of existing firms (faster expected intangible accumulation accelerates the dilution of any vintage of intangibles), so entry contracts; the labor freed from entry flows to production and produces a brief upward growth spike. The brief decline in the entry rate then translates into less variety creation, producing a brief dip below the 1990 BGP.

The growth boom that follows reflects two reinforcing forces. First, new cohorts enter with markedly higher scope and contribute many more varieties (and much more intangible capital) than the firms they replace. Second, the cross-section is still dominated by low- ρ incumbents, so labor demand is low and wages have not yet caught up: the marginal product of labor at new high- ρ entrants is well above wages, the entry rate temporarily rises above its 1990 BGP value, and contributes to the boom in growth and TFP. As wages catch up, entry becomes less attractive and growth declines toward the lower 2020 BGP value. Measured TFP growth (panel e) inherits the same pattern. Sales concentration (panel f), predetermined at the 1990 value on impact, is hump-shaped along the transition: as high-scope cohorts replace small 1990 incumbents, the Herfindahl index rises and overshoots the long-run level, then mean-reverts as continued variety creation dilutes incumbent shares.

Two points are worth highlighting. First, these transitional dynamics assume $\mu = 0$: Section 4.2 established that, in this case, long-run growth necessarily declines with replicability. The short- and medium-run dynamics in panel (d) are therefore qualitatively very different from the long-run implications of the model, with a sustained boom in growth through roughly year 15 set against a long-run decline from 2.7% to 2.5%. Second, the boom-bust profile of growth, productivity, and concentration is qualitatively reminiscent of the U.S. experience in the 1990s and early 2000s, though the model overstates the magnitudes: peak output and TFP growth substantially exceed the empirical peaks visible in Appendix Figure A-2, panels (a) and (e).

Asset-pricing ratios. The model's valuation ratios also respond non-monotonically, governed in the short run by a replacement effect. As ρ rises, new cohorts operate at higher scope and generate more revenue and intangible capital per firm than the low- ρ incumbents: older firms are therefore effectively less productive than newer ones, and the intangible capital they have already accumulated is worth relatively less. Agents anticipate this and discount existing firms' future cash flows more aggressively. The price-to-output ratio (panel h) jumps down sharply on impact, falls further as more productive entrants continue to erode the relative position of older firms, troughs midway through the transition, and only recovers toward the 2020 BGP value once the long-run earnings benefit of higher replicability dominates the replacement effect. Tobin's Q (panel g) follows a qualitatively similar pattern, but does not jump on impact.⁹ These dynamics, however, get the cyclical timing of the U.S. experience wrong: in the data, Q and V/Y rose sharply through the late 1990s before collapsing in the early 2000s, whereas the model produces an on-impact bust with no preceding boom. The model also continues to understate the long-run rise in valuations between the two BGPs, as indicated in Appendix Figure A-2, panel (c).

6 Optimal Policy

Having explored numerically the implications of the model across balanced growth paths and during the transition, we now ask whether the decentralized equilibrium (DE) is efficient and study second-best optimal policy, with

⁹The continuity of Q_t at the start of the transition is exact: Q_t admits a closed-form expression in terms of the predetermined states (N_t, K_t, θ_t) and continuous functions of ρ_t , with the value multiple ν_t and the entry rate $L_{E,t}$ eliminated. Since this expression is a continuous function of ρ_t , Q_t inherits the continuity of ρ_t directly. See equation (A99) in Appendix A.5.3 for the precise statement.

numerical examples tied to the parameter choices of Section 5. This section identifies the sources of inefficiency, with a focus on the wedges that arise from the hold-up friction. We then characterize optimal taxes on the income from intangible capital creation and the optimal degree of intellectual property protection.

6.1 Sources of inefficiency

Consider a social planner who maximizes the household's utility (15) by choosing entrepreneurial effort $L_{E,t}$ and the scope of new firms x_t , subject to the same production technology and laws of motion as in Section 3. Unlike private agents, the planner internalizes how entry and scope affect the public knowledge stock Ξ_t . The planner also faces no hold-up friction: she can assign managers to locations without giving them bargaining power.¹⁰

First-best benchmark. In the DE, the entrepreneur cannot write contracts that prevent managers from using the knowledge they acquire on the job. This incomplete contracting friction gives managers bargaining power and forces the entrepreneur to share enterprise value. Under perfect contractibility, the planner would not face this constraint: she could assign managers to locations and retain the full value of each firm. In that case, scope would be set to $x \rightarrow \infty$, because the spillover rate is increasing in x and there is no incentive constraint to limit it. The first-best therefore entails unbounded growth and serves only as a theoretical benchmark.¹¹

Three wedges. To isolate sources of inefficiency, we compare the DE to a planning allocation in which scope is held at the same value x as in the DE, but the planner otherwise chooses entrepreneurial effort optimally.¹² Both allocations share the same spillover rate, so any difference in entry rates reflects frictions in the allocation of labor to entrepreneurship, not in the scope margin. This is a useful decomposition because it separates the entry distortions (which are standard in growth models) from the scope distortion (which is specific to this framework).

For a given scope x , Appendix A.7.2 shows that the first-best BGP entry rate satisfies

$$\frac{\zeta}{1-\zeta} \frac{L_E^{FB}}{1-L_E^{FB}} = \frac{1}{\omega} \frac{\delta + \omega \bar{\xi} L_E^{\alpha\sigma}}{\eta + \delta + \omega \bar{\xi} L_E^{\alpha\sigma}} \left(1 + \frac{\sigma}{\eta} \omega \bar{\xi} L_E^{\alpha\sigma} \right) \alpha. \quad (75)$$

Compare this to the decentralized BGP entry condition (60):

$$\frac{\zeta}{1-\zeta} \frac{L_E}{1-L_E} = \frac{1}{\omega} \frac{\delta + \omega \bar{\xi} L_E^{\alpha\sigma}}{\eta + \delta + \omega \bar{\xi} L_E^{\alpha\sigma}} (1 - \gamma). \quad (76)$$

Three differences separate the two conditions.

Result 4 (Sources of inefficiency) *The decentralized allocation differs from the first-best through three wedges:*

- (i) **Dynamic spillovers** ($\sigma > 0$): entrants do not internalize that current entry raises Ξ_t for future entrants through (39). This wedge generates insufficient entry in the DE.

¹⁰The planner's problem reduces to the choice of $(L_{E,t}, x_t)$ because the static allocation of labor and intangibles across varieties is the same in the first-best and the decentralized equilibrium. The markup that firms charge in the DE is uniform across varieties: it lowers wages but raises profits by the same proportion, leaving the real allocation and aggregate output unchanged. With a representative household and inelastic total labor supply, this redistribution does not affect welfare. The planner therefore inherits the same aggregate production function and the same laws of motion. See Appendix A.7.1 for the formal argument.

¹¹Appendix A.7.2 derives the first-best planning problem and shows that the FOC for scope is never satisfied at finite x .

¹²The fixed-scope planning optimum is purely a decomposition device: as noted in the previous paragraph, if the planner can ignore the incentive compatibility constraint of managers, the first-best involves infinite scope.

- (ii) **Congestion** ($\alpha < 1$): the social marginal product of entry labor is α times the private marginal product, because of diminishing returns in the entry technology (37). This wedge generates excessive entry in the DE.
- (iii) **Hold-up friction**: the entrepreneur retains only a fraction $1 - \gamma$ of enterprise value. This wedge generates insufficient entry in the DE. It also constrains scope to a finite level, limiting the spillover rate $\bar{\xi}$ below what the planner could achieve.

Wedges (i) and (ii) are standard in endogenous growth models (e.g., Jones, 1995). Wedge (iii) is specific to our framework. It operates through two channels. First, it depresses entry: the entrepreneur captures only a share $1 - \gamma$ of enterprise value, so the private return to entry falls below its social value. Second, it constrains scope. In this model, the same process that generates knowledge spillovers also creates the hold-up problem. The entrepreneur delegates knowledge to managers, who carry it into the public stock Ξ_t . That delegation also gives managers bargaining power. The two are intertwined: policies that affect the hold-up friction also affect the rate of knowledge diffusion. A planner unconstrained by the IC would set scope to infinity. In the decentralized equilibrium, the IC binds and scope is finite, limiting $\bar{\xi}$ below what the planner could achieve.

When $\sigma = 0$, wedges (i) and (ii) become irrelevant for growth: the BGP growth rate $g = (1 - \zeta)\bar{\xi}$ depends only on scope, not on entry. In this case, wedge (iii) is the sole source of growth inefficiency, and we can isolate its effect cleanly. We use this case as our benchmark in the policy analysis below.

6.2 Optimal tax on intangible investment

6.2.1 Policy instrument

Suppose the government levies a proportional tax τ on the returns to intangible investment. In this model, intangible investment is the act of creating a new firm. Both entrepreneurs and defecting managers engage in firm creation, so both face the tax. This uniformity is natural: the government cannot distinguish an innovator from someone who replicates an existing idea, so it taxes the proceeds from firm creation at the same rate.

Under the tax, the manager's incentive constraint (33) becomes

$$\tilde{v}_t^{(M)} \geq (1 - \tau) \gamma(x; \lambda) v_t(x, n), \quad (77)$$

since a defecting manager would face the same tax when starting a competing firm. With this constraint binding, the entrepreneur retains a fraction $1 - (1 - \tau)\gamma(\hat{x}(\tau); \lambda)$ of enterprise value, where $\hat{x}(\tau)$ is the tax-adjusted optimal scope.¹³ The entrepreneur's per-unit-time payoff from entry is then taxed in turn, so the taxed free-entry condition is

$$(1 - \tau) \frac{E_t}{L_{E,t}} \left(1 - (1 - \tau)\gamma(\hat{x}(\tau); \lambda) \right) v_t(\hat{x}(\tau), n_t) = W_t. \quad (78)$$

This single instrument affects two margins simultaneously. On the scope margin, a higher τ reduces the manager's outside option, relaxing the incentive constraint and allowing the entrepreneur to choose wider scope. On the

¹³The tax on intangible investment differs from a corporate profit tax. A corporate profit tax would reduce the firm's total income before it is split between entrepreneur and manager, scaling both the manager's inside compensation and outside option by the same factor. The relative bargaining position would be unchanged and scope would remain at its DE level. The tax on intangible investment, by contrast, targets the act of creating new capital. The manager's outside option (starting a competing firm) falls, but the ongoing firm's internal income is not directly reduced. This wedge is what relaxes the incentive constraint and allows wider scope.

entry margin, τ enters twice: directly, through the tax on the entrepreneur's per-unit-time entry payoff, and indirectly, through the adjusted incentive constraint of managers. The tax revenue or subsidy cost is financed by a lump-sum transfer to or from the representative household.¹⁴ Appendix A.7.4 provides a formal statement of the decentralized equilibrium with the tax, derives the optimal scope $\hat{x}(\tau)$ as a function of the tax rate, and states the planner's problem.

6.2.2 The case of no dynamic externalities

When $\sigma = 0$, the BGP growth rate depends only on scope:

$$g = (1 - \zeta) \bar{\xi}, \quad \bar{\xi} = \xi(\gamma x^\rho)^\mu. \quad (79)$$

Entry does not contribute to growth. The dynamic spillover wedge (i) and the congestion wedge (ii) are absent. The only source of growth inefficiency is the hold-up friction, which constrains scope below the first-best level and reduces the spillovers that generate growth.

The tax corrects this distortion. A higher τ relaxes the incentive constraint, allowing wider scope $\hat{x}(\tau) > x^{DE}$ and raising $\bar{\xi}$. The cost of the tax is lower entry: fewer new firms are created at each date. But when $\sigma = 0$, this cost does not reduce growth. It only reduces the level of the capital stock N_t relative to the no-tax path, lowering output at each date by a fixed proportion. The tax thus trades off a permanent level loss against a permanent increase in the growth rate.

Result 5 (Optimal tax on intangible investment when $\sigma = 0$) *Suppose $\sigma = 0$. On the BGP, the optimal tax on intangible investment is strictly positive:*

$$\tau^{SB} > 0. \quad (80)$$

The planner taxes intangible investment to expand scope, raising the growth rate above its DE level.

The intuition is straightforward. Starting from the DE allocation ($\tau = 0$), a small positive tax has a first-order effect on growth, because wider scope raises the spillover rate. But it only has a second-order effect on the level of consumption. To see this, note that BGP welfare can be written as

$$W(\tau) = \frac{\zeta}{\eta} \log(1 - L_E(\tau)) + \frac{g(\tau)}{\eta^2}, \quad (81)$$

where η is the household's time discount rate. The first term captures static consumption; the second captures the growth dividend. Close to $\tau = 0$, growth is increasing with respect to the tax rate, because scope (and therefore the spillover rate) rises with the tax. The level effect on consumption is negative because entry falls with the tax rate. But the welfare gain from faster growth is proportional to $1/\eta^2$, while the consumption loss is proportional to $1/\eta$. The growth effect therefore dominates. Appendix A.7.4 provides the formal proof.

¹⁴The lump-sum transfer is non-distortionary in this model. Total labor supply is inelastic, there is no physical capital, and all output is consumed. The transfer therefore redistributes income within the representative household without affecting any margin. The tax changes welfare only through its effect on entry and on scope.

6.2.3 The general case

When $\sigma > 0$, entry contributes to growth through the knowledge spillover (39). A higher entry rate accelerates the accumulation of public knowledge, Ξ_t . The tax on intangible investment now acts on two margins. On the one hand, a higher τ widens scope and raises the spillover rate. On the other hand, the tax reduces entry, which slows knowledge accumulation. The sign of the optimal tax depends on which channel dominates.

BGP welfare (81) can be differentiated as

$$W'(\tau) = -\frac{\zeta}{\eta(1-L_E)} L'_E(\tau) + \frac{g'(\tau)}{\eta^2}. \quad (82)$$

The growth rate is

$$g(\tau) = (1-\zeta) \bar{\xi}(\tau) L_E(\tau)^{\alpha\sigma}, \quad (83)$$

so its derivative decomposes into a scope channel and an entry channel:

$$g'(\tau) = (1-\zeta) L_E^{\alpha\sigma} \bar{\xi}'(\tau) + (1-\zeta) \alpha\sigma L_E^{\alpha\sigma-1} \bar{\xi}(\tau) L'_E(\tau). \quad (84)$$

The first term is positive: higher scope increases spillovers. When $\sigma > 0$, the second term is generically negative, as the tax reduces entry. When σ is sufficiently large, the second term dominates and a tax reduces growth.

Result 6 (Optimal tax on intangible investment, general case) *On the BGP, the sign of the optimal tax on intangible investment depends on σ . There exists a threshold σ^* such that:*

- (i) *For $\sigma < \sigma^*$, the optimal tax is strictly positive ($\tau^{SB} > 0$). The scope channel dominates: the planner taxes intangible investment to expand scope and benefit from higher managerial spillovers;*
- (ii) *For $\sigma > \sigma^*$, the optimal tax is strictly negative ($\tau^{SB} < 0$). The entry channel dominates: the planner subsidizes intangible investment to raise entry and accelerate knowledge accumulation.*

Appendix A.7.4 gives details. The key economic force is that σ governs the social value of entry for growth. When σ is small, entry contributes little to knowledge accumulation, and the planner prefers to redirect resources toward wider scope. When σ is large, each additional unit of entry labor generates substantial knowledge spillovers, and the planner prefers to subsidize entry even at the cost of narrower scope. Result 5 is the special case $\sigma = 0$, where the entry channel is absent and the optimal tax is unambiguously positive.¹⁵

6.2.4 Illustration

Figure 6 illustrates Results 5 and 6 with a numerical example. Panel (a) plots the optimal tax against the spillover elasticity σ , holding replicability ρ constant. Recall that a positive tax reduces the manager's outside option,

¹⁵The tax operates by relaxing a contractual friction, the hold-up problem between managers and entrepreneurs. The welfare gain comes from the fact that the hold-up makes firms too small in the decentralized equilibrium. Taxing intangible investment relaxes the hold-up problem, widens scope and, through spillovers, raises the growth rate, despite the fact that it lowers entry. This tax can turn into a subsidy when direct spillovers from higher entry rates are high enough, that is, when σ is sufficiently high. The positive optimal tax on intangible investment is reminiscent of results in public finance showing that the optimal capital income tax can be positive when the intertemporal elasticity of substitution is below one (Chamley, 1986; Judd, 1985; Straub and Werning, 2020), but it rests on an entirely different economic foundation, as there is no intertemporal consumption-smoothing margin in our model.

which lowers managerial compensation and allows the entrepreneur to choose wider scope. At low σ , the growth benefit from wider scope dominates, and the optimal policy is a tax. As σ increases, entry becomes more valuable for knowledge accumulation, and the planner shifts toward subsidizing intangible investment. When $\mu = 0$, the spillover rate does not depend on scope, so the scope channel is absent and the tax turns negative at a moderate value of σ . When $\mu > 0$, wider scope raises the spillover rate, giving the planner an additional reason to tax. A larger dynamic spillover is therefore needed to overturn the case for taxing.

Panel (b) plots the optimal tax against replicability ρ , holding the spillover elasticity σ constant. When $\mu = 0$, the tax declines with ρ and eventually becomes a subsidy. When $\mu > 0$ the tax remains positive in this numerical example. Higher replicability amplifies the scope channel, because wider scope translates into a larger spillover rate.

Panels (c) and (d) compare the DE and SB allocations as ρ varies, when $\mu = 0$ and $\sigma > 0$ are kept constant. Under the optimal policy, growth and entry are exactly flat with respect to ρ . The reason is that when $\mu = 0$, growth in the FB does not depend on replicability. The planner uses the tax only to set entry at the level that optimally trades off consumption against growth; that level is independent of ρ . In the DE, by contrast, entry falls as ρ rises because the hold-up friction worsens. At low ρ , the DE features too much entry and too much growth relative to the SB. At high ρ , the DE features too little of both. As ρ increases, the DE thus moves from a situation where the congestion wedge dominates to one where the hold-up friction dominates.

6.3 Optimal intellectual property protection

The IP enforcement parameter λ governs the degree to which the legal system protects the firm's claims over its intangible assets. Unlike the tax τ , which is a fiscal instrument, λ is a structural parameter of the property-rights regime. It affects the economy through the incentive constraint (33).

Weaker IP protection—which in this model corresponds to higher λ —has a counterintuitive effect on entry. Recall that $\gamma(x; \lambda) = \kappa x^\lambda$. As a result, a higher λ makes the managerial share rise more steeply with scope. At any given scope $x > 1$, managers can extract a larger share of entry surplus from the entrepreneur. The entrepreneur responds by choosing narrower scope to limit the managerial claim. At the resulting, narrower equilibrium scope, the managerial share therefore declines, and the entrepreneur retains a larger fraction of enterprise value. This raises returns to entrepreneurship and therefore the entry rate. Thus, weaker IP protection leads to narrower firms, a lower managerial share, and more entry.

The effect on growth depends on the strength of knowledge spillovers. Differentiating the BGP growth rate (59) with respect to λ :

$$\frac{d \log g}{d \lambda} = \underbrace{\mu \frac{d}{d \lambda} \log(\gamma x^\rho)}_{\leq 0 \text{ (scope channel)}} + \underbrace{\alpha \sigma \frac{d \log L_E}{d \lambda}}_{\geq 0 \text{ (entry channel)}}. \quad (85)$$

Narrower scope reduces the spillover rate (through the first term in the equation above), but higher entry raises it (through the second term). The parameter μ governs which channel dominates. The welfare-maximizing value of λ solves:

$$\max_{\lambda} W(\lambda) = \frac{\zeta}{\eta} \log(1 - L_E(\lambda)) + \frac{g(\lambda)}{\eta^2}, \quad (86)$$

where $L_E(\lambda)$ and $g(\lambda)$ are the BGP entry and growth rates at $\tau = 0$.

Result 7 (Optimal IP protection) *The welfare-maximizing degree of IP protection depends on μ .*

- (i) *When $\mu = 0$, $W(\lambda)$ is increasing in λ . The planner prefers weaker IP protection, since the only effect is to raise entry toward the social optimum.*
- (ii) *When $\mu > 0$ and sufficiently large, there exists an interior welfare maximum λ^W . The welfare-maximizing λ^W is strictly below the growth-maximizing $\hat{\lambda}$. The planner accepts lower growth to protect consumption from the entry effects of weaker IP protection.*

The intuition for why the welfare-maximizing value of λ is below its growth-maximizing value in case (ii) is straightforward. At the growth-maximizing value of λ , the welfare derivative (A205) reduces to

$$\left. \frac{dW}{d\lambda} \right|_{\hat{\lambda}} = -\frac{\zeta}{\eta(1-L_E)} \frac{dL_E}{d\lambda} < 0, \quad (87)$$

where inequality holds because weaker IP protection (a higher value of λ) raises entry. Therefore, welfare is already declining at the growth-maximizing value of λ . The planner prefers stronger IP protection (that is, a smaller value of λ) to avoid the consumption cost of excessive entry. Appendix A.7.4 provides the full derivation.

Figure 7 illustrates Result 7. At $\mu = 0$, welfare rises monotonically with λ : weaker IP protection raises entry with no knowledge-diffusion cost. At intermediate $\mu = 0.033$, welfare has an interior peak: relaxing IP protection initially raises welfare through higher entry, but eventually destroys the scope-spillover channel. At $\mu = 0.08$, welfare falls throughout: the planner prefers strong IP protection to sustain large firms that diffuse knowledge broadly.

7 Discussion and Conclusion

We conclude by highlighting the novel implications of our framework.

Measurement of Intangible Capital. Our model has implications for the measurement of the stock of intangible capital. Existing approaches measure the stock of intangibles by capitalizing real expenditures in intangible investments. Our model implies that this approach misses a key feature of intangible assets: they can be deployed in multiple locations. Hence, the appropriate measure of the services from intangible capital used in generating revenue accumulates real investment expenditures and also adjusts for the scope of implementation and the current technology for replicating intangible assets.

To the extent to which the scope of implementation has increased and the technology for replicating intangibles has improved over the last few decades, our model would therefore imply that the existing approaches underestimate the stock of intangibles and their services. This affects both the measurement of output—when intangibles are produced—and the measurement of inputs—when intangibles are used. Hence, an additional implication is that Solow residual measures of Total Factor Productivity will generally be misstated. This occurs both in absolute levels and, during periods when the scope of implementation of technology increases, in growth rates.

Mapping to investment-specific technical change. The last few decades have seen significant increases in firm valuation combined with lower economic growth. This pattern can represent a puzzle through the lens of standard models in which the valuation of the stock market corresponds to the present discounted value of

capital income. Our model is qualitatively consistent with this pattern. Unlike standard models, the benefits of future growth do not accrue to existing firms: they accrue to firms that are yet to be created. As a result, the valuation of the stock market corresponds to the valuation of incumbent firms. Forces that lead to lower economic growth in the long run will generate both higher concentration and higher valuations for incumbents, because they lower the degree of competition with new firms in the market for products and workers, even when markups remain constant. In this respect, changes in the degree of replicability, ρ , have effects that are similar to investment-specific technical change, in that they amplify the wedge between the effective productivity of existing and new firms. This prediction is in line with the evidence in [Greenwood and Jovanovic \(1999\)](#) on the IT revolution of the 1990s.

Taxation of Intangible Investment. [Jones and Williams \(1998\)](#) and [Jones and Williams \(2000\)](#) study whether decentralized economies invest too much or too little in R&D. They balance several distortions, including dynamic knowledge spillovers and congestion externalities. Their conclusion is that underinvestment is the typical case, and the standard prescription is to subsidize R&D. Our model features a related set of forces. Dynamic spillovers push toward too little entry, while congestion pushes toward too much. The hold-up friction adds a third force that depresses entry and reduces scope relative to the planner's problem. When dynamic spillovers arising directly from the rate of entrepreneurship are small (that is, when σ is small), the planner taxes intangible investment to widen scope, accepting lower entry. This overturns the standard prescription of subsidizing R&D and reflects the fact that the planner can improve welfare by shifting resources from higher entry to higher scope.

Optimal degree of IP protection. The tradeoff between innovation incentives and the static costs of monopoly goes back to [Nordhaus \(1969\)](#). A central lesson of this literature is that strong IP protection is not always optimal: when innovation is cumulative, strong patents can block follow-on research and slow down knowledge diffusion, as [Acemoglu and Akgigit \(2012\)](#) formalize in a step-by-step innovation model. A more radical strand, associated with [Boldrin and Levine \(2008a,b, 2013\)](#), goes further and argues that IP protection is neither necessary nor sufficient for innovation; in practice, patents and copyright function as government-granted monopolies that stifle entry. Both strands share a common premise: IP protection operates by granting innovators a monopoly right, and the debate is over whether the resulting appropriable rents are necessary for innovation and worth the costs of blocked diffusion and deterred entry.

Our framework provides a different perspective on this debate. IP protection in our model governs neither the duration nor the breadth of a patent, and it does not grant a monopoly—firms are monopolists regardless of the degree of IP protection. Instead, λ determines the extent to which entrepreneurs can prevent managers from carrying their knowledge into competing ventures. This difference inverts the standard tradeoff. Stronger IP protection induces entrepreneurs to choose broader scope, and can thus raise knowledge diffusion instead of reducing it; its cost is that it depresses entry, because broader scope forces entrepreneurs to share more rents with managers. When $\mu = 0$, the knowledge that managers carry has no impact on future growth, and our conclusions align with the radical critique: weaker IP protection is unambiguously welfare-improving because it decreases scope but increases entry. When $\mu > 0$, some IP protection is optimal but for reasons that are subtly different from earlier work: stronger protection allows entrepreneurs to deploy their knowledge over broader scope, hence potentially increasing the diffusion of knowledge into the public stock Ξ_t , even if entry declines.

Rise of Artificial Intelligence. Our model can be used to understand the impact of the recent rise of Artificial Intelligence (AI) on economic outcomes. The key feature of AI is that it allows for the codification of existing knowledge that was previously embodied in people. Through the lens of our model, the rise of AI can therefore be interpreted as an increase in the value of ρ , that is, the ease through which the entrepreneur’s intangible capital can be replicated and deployed across locations. However, the key difference with our model is that the deployment of AI may not require managerial talent as a key input. This distinction has implications for understanding how the rents of new intangibles are appropriated by the entrepreneur and other parties. At first glance, this simply reduces the power of the managers to extract rents and allows the entrepreneur to scale more freely. However, even if managers are no longer required in order to scale up the entrepreneur’s intangible, other parties—such as the AI providers or other outsiders that can *now easily use the same knowledge*—may appropriate a non-trivial fraction of rents, leading to similar implications about incentives to create new firms.

References

- Acemoglu, D. and U. Akcigit (2012). Intellectual property rights policy, competition and innovation. *Journal of the European Economic Association* 10(1), 1–42.
- Aghion, P., A. Bergeaud, T. Boppart, P. Klenow, and H. Li (2022). Good rents versus bad rents: R&d misallocation and growth. Technical report, Working Paper.
- Aghion, P., A. Bergeaud, T. Boppart, P. J. Klenow, and H. Li (2023). A theory of falling growth and rising rents. *Review of Economic Studies* 90(6), 2675–2702.
- Aghion, P. and P. Bolton (1992, 07). An Incomplete Contracts Approach to Financial Contracting. *The Review of Economic Studies* 59(3), 473–494.
- Aghion, P. and P. Howitt (1992). A Model of Growth Through Creative Destruction. *Econometrica* 60(2), 323–351.
- Akcigit, U. and S. T. Ates (2019). What Happened to U.S. Business Dynamism? NBER Working Paper 25756, National Bureau of Economic Research.
- Akcigit, U. and S. T. Ates (2021). Ten facts on declining business dynamism and lessons from endogenous growth theory. *American Economic Journal: Macroeconomics* 13(1), 257–98.
- Anton, J. J. and D. A. Yao (2004). Little patents and big secrets: Managing intellectual property. *RAND Journal of Economics* 35(1), 1–22.
- Argente, D., S. Moreira, E. Oberfield, and V. Venkateswaran (2020). Scalable Expertise.
- Arrow, K. (1962, October). Economic welfare and the allocation of resources for invention. In *The Rate and Direction of Inventive Activity: Economic and Social Factors*, NBER Chapters, pp. 609–626. National Bureau of Economic Research, Inc.
- Atanasov, V., V. Ivanov, and K. Litvak (2012). Does reputation limit opportunistic behavior in the VC industry? Evidence from litigation against VCs. *The Journal of Finance* 67(6), 2215–2246.
- Atkeson, A. and P. J. Kehoe (2005). Modeling and Measuring Organization Capital. *Journal of Political Economy* 113(5), 1026–1053.

- Autor, D., D. Dorn, L. F. Katz, C. Patterson, and J. V. Reenen (2020). The fall of the labor share and the rise of superstar firms. *The Quarterly Journal of Economics* 135(2), 645–709.
- Barkai, S. (2020). Declining labor and capital shares. *The Journal of Finance* 75(5), 2421–2463.
- Bhide, A. (1999). *The Origin and Evolution of New Businesses*. Oxford University Press, USA.
- Boldrin, M. and D. K. Levine (2008a). *Against Intellectual Monopoly*. Cambridge: Cambridge University Press.
- Boldrin, M. and D. K. Levine (2008b). Perfectly competitive innovation. *Journal of Monetary Economics* 55(3), 435–453.
- Boldrin, M. and D. K. Levine (2013). The case against patents. *Journal of Economic Perspectives* 27(1), 3–22.
- Cabezón, F. and G. Hoberg (2024). Leaky director networks and innovation herding. Working paper, available at SSRN 4158977.
- Chamley, C. (1986). Optimal taxation of capital income in general equilibrium with infinite lives. *Econometrica: Journal of the Econometric Society* 54(3), 607–622.
- Corrado, C., C. Hulten, and D. Sichel (2009). Intangible capital and us economic growth. *Review of income and wealth* 55(3), 661–685.
- Crouzet, N. and J. Eberly (2019). Understanding Weak Capital Investment: the Role of Market Power and Intangibles. Proceedings of the Jackson Hole Economic Policy Symposium, Federal Reserve Bank of Kansas City.
- Crouzet, N. and J. Eberly (2023). Rents and Intangible Capital: A Q+ Framework. *The Journal of Finance* 78(4), 1873–1916.
- Crouzet, N., J. C. Eberly, A. L. Eisfeldt, and D. Papanikolaou (2022). The Economics of Intangible Capital. *Journal of Economic Perspectives* 36(3), 29–52.
- De Loecker, J., J. Eeckhout, and G. Unger (2020). The rise of market power and the macroeconomic implications. *The Quarterly Journal of Economics* 135(2), 561–644.
- De Ridder, M. (2022). Market Power and Innovation in the Intangible Economy. Working paper, London School of Economics and Political Science.
- Decker, R., J. Haltiwanger, R. Jarmin, and J. Miranda (2014, September). The role of entrepreneurship in us job creation and economic dynamism. *Journal of Economic Perspectives* 28(3), 3–24.
- Decker, R. A., J. Haltiwanger, R. S. Jarmin, and J. Miranda (2016). Where has all the skewness gone? the decline in high-growth (young) firms in the u.s. *European Economic Review* 86, 4–23. The Economics of Entrepreneurship.
- Driscoll, J. C. and A. C. Kraay (1998). Consistent covariance matrix estimation with spatially dependent panel data. *The Review of Economics and Statistics* 80(4), 549–560.
- Eisfeldt, A. L., A. Falato, and M. Z. Xiaolan (2021, May). Human Capitalists. Working Paper 28815, National Bureau of Economic Research.
- Eisfeldt, A. L. and D. Papanikolaou (2013). Organization Capital and the Cross-Section of Expected Returns. *The Journal of Finance* 68(4), 1365–1406.

- Farboodi, M. and L. Veldkamp (2020). Long-run Growth of Financial Data Technology. *American Economic Review* 110(8), 2485–2523.
- Fernald, J. (2014). A quarterly, utilization-adjusted series on total factor productivity. Working paper, Federal Reserve Bank of San Francisco.
- Gilchrist, S. and J. C. Williams (2000). Putty-clay and investment: A business cycle analysis. *Journal of Political Economy* 108(5), 928–960.
- Greenwood, J. and B. Jovanovic (1999). The information-technology revolution and the stock market. *American Economic Review* 89(2), 116–122.
- Grossman, S. J. and O. D. Hart (1986, August). The Costs and Benefits of Ownership: A Theory of Vertical and Lateral Integration. *Journal of Political Economy* 94(4), 691–719.
- Hall, R. E. (2001). The stock market and capital accumulation. *American Economic Review* 91(5), 1185–1202.
- Hart, O. and B. Holmstrom (2010, 05). A Theory of Firm Scope*. *The Quarterly Journal of Economics* 125(2), 483–513.
- Hart, O. and J. Moore (1988). Incomplete contracts and renegotiation. *Econometrica* 56(4), 755–785.
- Hart, O. and J. Moore (1994). A theory of debt based on the inalienability of human capital. *The Quarterly Journal of Economics* 109(4), 841–879.
- Haskel, J. and S. Westlake (2017). *Capitalism Without Capital*. Princeton University Press.
- Hoberg, G. and G. M. Phillips (2025). Scope, scale, and concentration: The 21st-century firm. *The Journal of Finance* 80(1), 415–466.
- Hopenhayn, H., J. Neira, and R. Singhania (2022). From population growth to firm demographics: Implications for concentration, entrepreneurship and the labor share. *Econometrica* 90(4), 1879–1914.
- Hsieh, C.-T. and E. Rossi-Hansberg (2023a). The Industrial Revolution in Services. *Journal of Political Economy Macroeconomics* 1(1), 3–42.
- Hsieh, C.-T. and E. Rossi-Hansberg (2023b). The Industrial Revolution in Services. *Journal of Political Economy Macroeconomics* 1(1), 3–42.
- Jiang, X. and H. Rubinton (2025). The adoption of non-rival inputs and firm scope. Working paper.
- Jones, C. I. (1995). R&D-based Models of Economic Growth. *Journal of Political Economy* 103(4), 759–784.
- Jones, C. I. and C. Tonetti (2020). Nonrivalry and the Economics of Data. *The American Economic Review* 110(9), 2819–2858.
- Jones, C. I. and J. C. Williams (1998). Measuring the social return to r&d. *The Quarterly Journal of Economics* 113(4), 1119–1135.
- Jones, C. I. and J. C. Williams (2000). Too much of a good thing? the economics of investment in r&d. *Journal of economic growth* 5(1), 65–85.
- Judd, K. L. (1985). Redistributive taxation in a simple perfect foresight model. *Journal of public Economics* 28(1), 59–83.

- Kankanhalli, G., A. Kwan, and K. Merkley (2024). The Paradox of Innovation Nondisclosure: Evidence from Licensing Contracts. *American Economic Journal: Applied Economics* 16(4), 220–256.
- Kaplan, S. N. and P. Stromberg (2004). Characteristics, contracts, and actions: Evidence from venture capitalist analyses. *Journal of Finance* 59(5), 2177–2210.
- Karahan, F., B. Pugsley, and A. Şahin (2024). Demographic origins of the startup deficit. *American Economic Review* 114(7), 1986–2023.
- Klette, T. and S. Kortum (2004). Innovating Firms and Aggregate Innovation. *Journal of Political Economy* 112(5), 986–1018.
- Kogan, L., D. Papanikolaou, and N. Stoffman (2020). Left Behind: Creative Destruction, Inequality, and the Stock Market. *Journal of Political Economy* 128(3), 855–906.
- Kondo, J., D. Li, and D. Papanikolaou (2021). Trust, Collaboration, and Economic Growth. *Management Science* 67(3), 1825–1850.
- Lucas, R. E. and B. Moll (2014). Knowledge Growth and the Allocation of Time. *Journal of Political Economy* 122(1), 1–51.
- Lustig, H., C. Syverson, and S. Van Nieuwerburgh (2011, March). Technological change and the growing inequality in managerial compensation. *Journal of Financial Economics* 99(3), 601–627.
- Markusen, J. R. (1984). Multinationals, multi-plant economies, and the gains from trade. *Journal of International Economics* 16(3–4), 205–226.
- McGrattan, E. R. and E. C. Prescott (2009). Openness, Technology Capital, and Development. *Journal of Economic Theory* 144(6), 2454–2476.
- McGrattan, E. R. and E. C. Prescott (2010a). Technology Capital and the US Current Account. *American Economic Review* 100(4), 1493–1522.
- McGrattan, E. R. and E. C. Prescott (2010b). Unmeasured Investment and the Puzzling US Boom in the 1990s. *American Economic Journal: Macroeconomics* 2(4), 88–123.
- Nordhaus, W. D. (1969). *Invention, growth and welfare: A Theoretical Treatment of Technological Change*. MIT Press.
- Perla, J. and C. Tonetti (2014). Equilibrium Imitation and Growth. *Journal of Political Economy* 122(1), 52–76.
- Perla, J., C. Tonetti, and M. E. Waugh (2021, January). Equilibrium Technology Diffusion, Trade, and Growth. *American Economic Review* 111(1), 73–128.
- Peters, R. H. and L. A. Taylor (2017). Intangible capital and the investment-q relation. *Journal of Financial Economics* 123(2), 251–272.
- Ritter, J. R. (2020). Initial public offerings: Dual Class Structure of IPOs Through 2020. Working paper, University of Florida.
- Ritter, J. R. and I. Welch (2002). A review of ipo activity, pricing, and allocations. *The Journal of Finance* 57(4), 1795–1828.
- Romer, P. M. (1990). Endogenous Technological Change. *Journal of Political Economy* 98(5, Part 2), S71–S102.

- Schaller, Z., P. DeCelles, and H. Patrick (2025). Weighted crosswalks for NAICS and SIC industry codes. Version 3.
- Stokey, N. L. (2015). Catching up and falling behind. *Journal of Economic Growth* 20(1), 1–36.
- Straub, L. and I. Werning (2020). Positive long-run capital taxation: Chamley-judd revisited. *American Economic Review* 110(1), 86–119.
- Wang, X. (2025). Patenting and information disclosure. Working paper.

	Growth in							
	Firm Revenue		Employment		# Establishment		# Markets (CBSAs)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Growth in Managerial Share (γ), 5 years	0.152*** (13.64)	0.135*** (14.65)	0.105*** (7.34)	0.092*** (7.39)	0.055*** (3.83)	0.060*** (4.63)	0.044*** (4.24)	0.038*** (3.29)
Observations	40,348	38,909	40,158	38,718	24,688	23,499	24,638	23,458
R^2	0.448	0.601	0.275	0.471	0.125	0.382	0.125	0.368
Fixed effects								
Year $\times$ SIC-3 industry		Y		Y		Y		Y

Table 1: Regression of five-year growth in firm outcomes on five-year growth in the managerial share $\gamma_{i,t}$. The managerial share is defined as the ratio of total compensation of the firm's top 5 executives (the sum of Execucomp tdc1) to the market value of equity ($\text{prcc_f} \times \text{csho}$ in Compustat). All specifications include the following controls: growth in the market value of equity; the lagged (log) level of equity market value; one lag of the dependent variable in (log) levels; log of the lagged managerial share. Depending on the specification we also include the interaction of industry times year fixed effects where industries are defined at the SIC3 level. Standard errors are computed using the [Driscoll and Kraay \(1998\)](#) procedure. The number of establishments per firm and number of CBSAs are from the NETS data merged to Compustat (see Appendix [A.1.2](#)) for details. Sample period is 1992 to 2018.

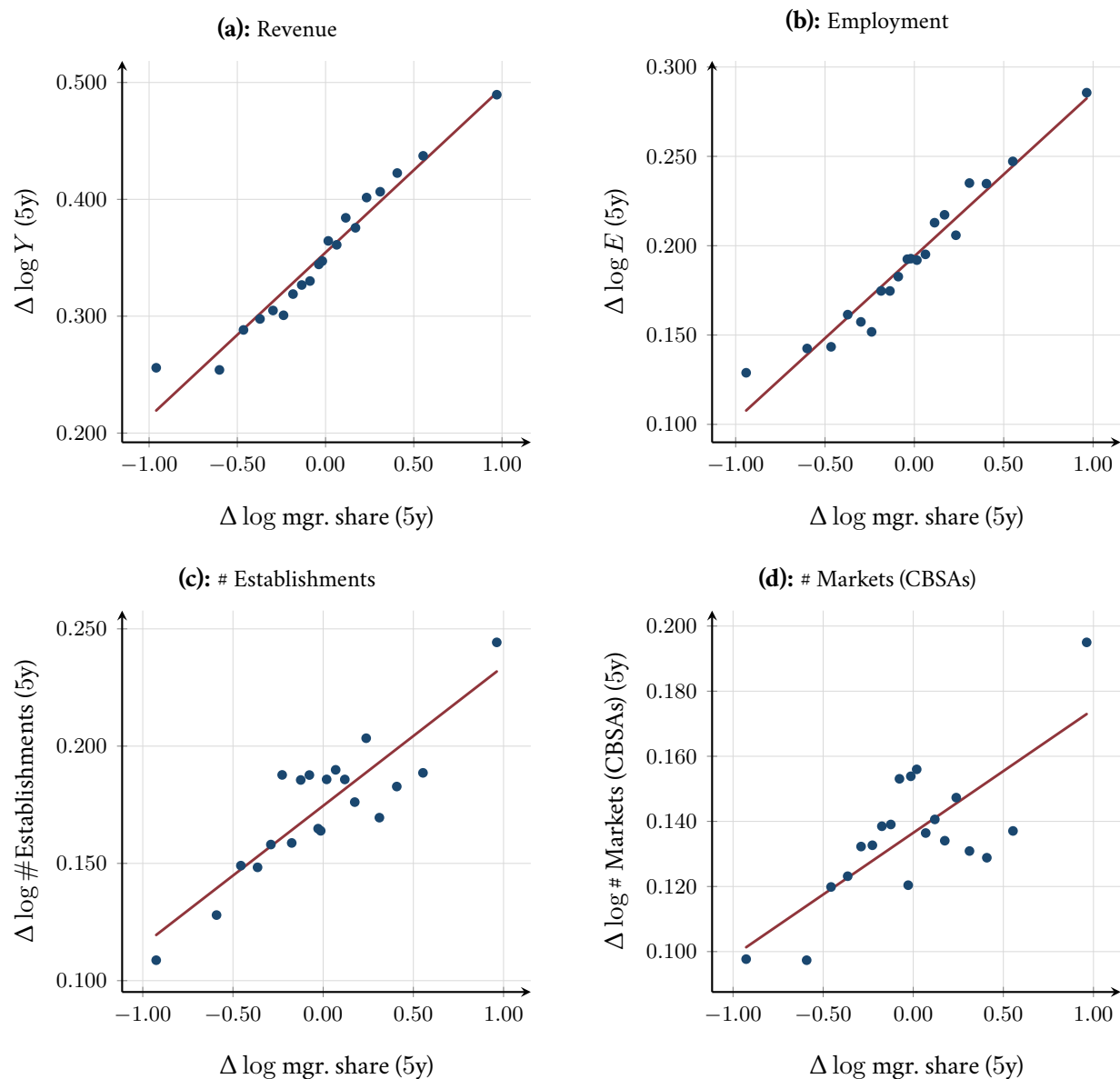


Figure 1: Top four panels show the relation between five-year changes in the managerial share of compensation (horizontal axis, each panel) and four firm outcomes: revenue, number of establishments, and two market-scope measures (# Establishments or # CBSAs per firm).

	Five-year change in log scope					
	CBSA / firm		ZIP / firm		Establishment / firm	
	(1)	(2)	(3)	(4)	(5)	(6)
Δ_5 establishment entry rate	−0.289*** (0.082)	−0.483*** (0.159)	−0.383*** (0.096)	−0.464** (0.198)	−0.423*** (0.104)	−0.549*** (0.206)
Sample	All	Ex-FIRE	All	Ex-FIRE	All	Ex-FIRE
Observations	8,545	7,855	8,547	7,857	8,547	7,857
R^2	0.099	0.091	0.105	0.097	0.086	0.081
Fixed effects						
Year $\times$ SIC-1 division	Y	Y	Y	Y	Y	Y

Table 2: Five-year changes in industry-average firm geographic scope and establishment entry rates. The dependent variable in each column is the change in log scope (CBSAs, ZIP codes, or establishments per firm, for multi-establishment firms with at least two employees in NETS, with structural breaks spliced). The key regressor is the five-year change in the establishment entry rate from the Business Dynamics Statistics. All specifications include controls for log scope and the entry rate in levels at the base year (not reported). “Ex-FIRE” excludes SIC codes 600–679. Standard errors, clustered at the SIC3 level (358 clusters), are in parentheses. See Appendix A.2 for sample construction and Appendix Table A-3 for analogous results using firm entry rates. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

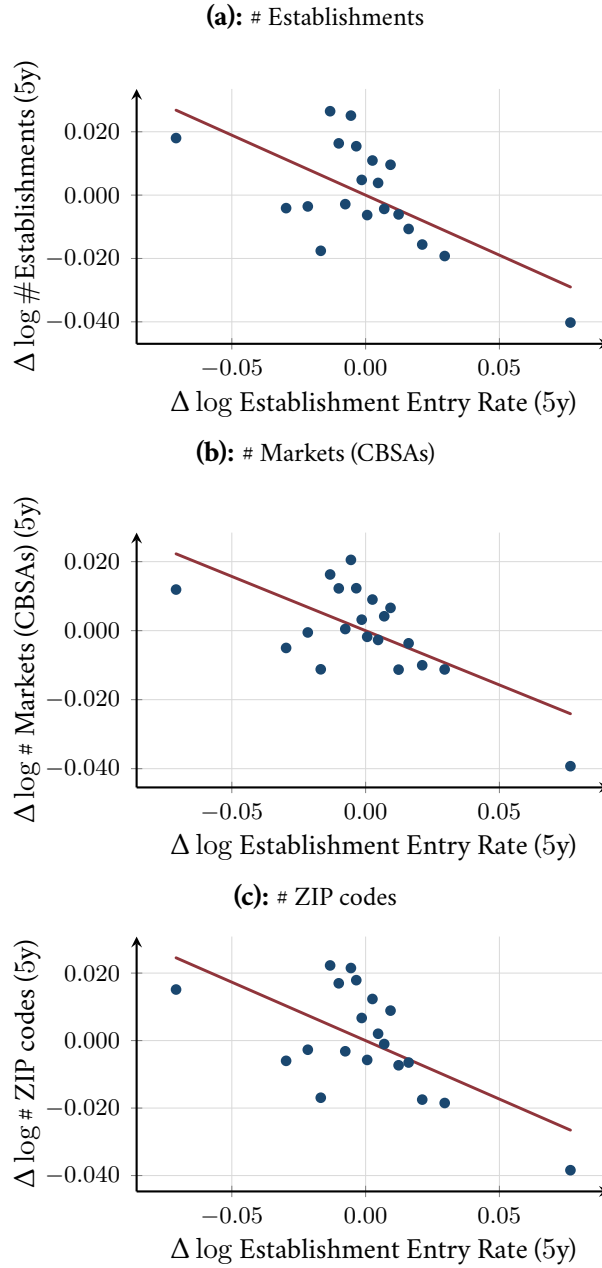


Figure 2: Binned scatter plots of residualized five-year changes in log firm geographic scope (vertical axis) against residualized five-year changes in the establishment entry rate (horizontal axis), after partialling out year $\times$ one-digit SIC division fixed effects. Scope measures are industry averages from NETS for multi-establishment firms with at least two employees (adjusted for structural breaks in the NETS series); establishment entry rates are from the BDS mapped to SIC3 (see Appendix A.2). Each panel bins observations into twenty equal-sized groups; markers are bin means and the line is an OLS fit through the underlying residuals.

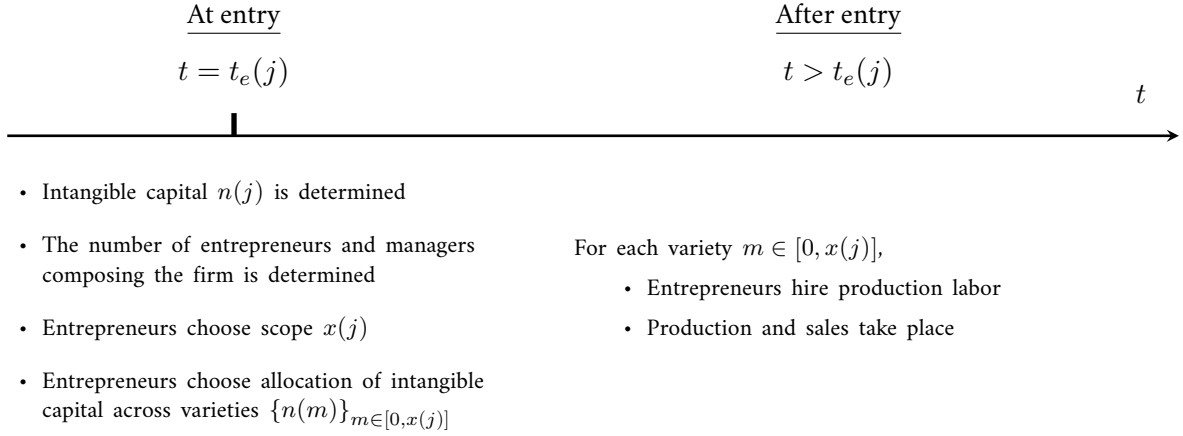


Figure 3: The timing of firm j 's problem.

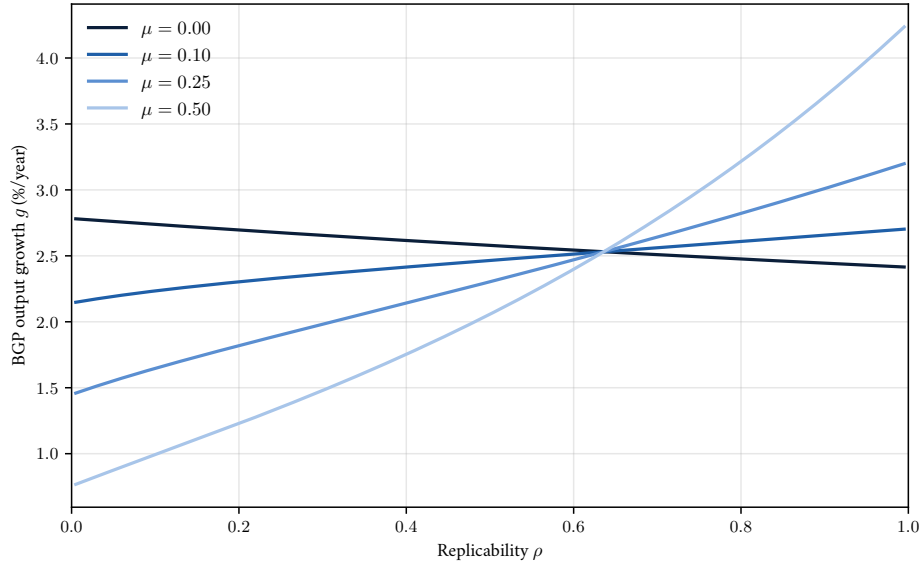


Figure 4: BGP output growth rate g as a function of replicability ρ , at four values of the knowledge-diffusion concavity $\mu \in \{0, 0.10, 0.25, 0.50\}$. All other parameters are held at the calibrated values from Panels A and B of Table 3. The spillover scale ξ is held fixed at its $\mu = 0$ calibrated value, so the spillover $\bar{\xi}(\rho, \mu) = \xi(\gamma(\rho) x(\rho)^\rho)^\mu$ varies endogenously with ρ and μ .

Panel A. Parameters set externally

Parameter	Description	Value	Source
η	Time discount rate	0.020	Annual frequency
δ	Firm exit rate	0.100	BDS gross firm exit rate, 1990
$1/\chi$	Markup	1.050	Crouzet and Eberly (2023)
ζ	Cobb-Douglas labor elasticity	0.700	Crouzet and Eberly (2023)
α	Entry-tech weight on L_E	0.500	Symmetric in (L_E, L_M)
σ	Spillover-to-entry elasticity	1.000	Romer (1990) benchmark
ω	$\equiv (1 - \zeta)\chi / (1 - \zeta\chi)$	0.857	Derived

Panel B. Parameters set to match data targets, $\mu = 0$

Parameter	Description	Value	Data target	Value
λ	Slope of $\gamma(x; \lambda)$	1.733	$\Delta \log \gamma / d$	1.733
ρ_{1990}	Replicability, 1990	0.190	γ_{1990}	0.150
ρ_{2020}	Replicability, 2020	0.700	γ_{2020}	0.300
κ	Level of $\gamma(x; \lambda)$ at x_{1990}	0.009	x_{1990}	5.000
ξ	Spillover scale	0.171	g_{1990}	0.027

Panel C. Balanced growth path comparison, 1990 versus 2020

	1990 BGP	2020 BGP, model prediction at...			
		$\mu = 0$	$\mu = 0.10$	$\mu = 0.25$	$\mu = 0.50$
Output growth, g	2.70%	2.51%	3.01%	3.96%	6.25%
Relative TFP level, Z_t / Z_{1990}	1.000	1.415	1.412	1.408	1.397
Price-to-output ratio, V/Y	1.691	1.739	1.618	1.430	1.117
Tobin's Q	1.262	1.528	1.539	1.555	1.581
Profit share, Π	5.2%	10.2%	10.2%	10.2%	10.2%
Concentration, HHI	0.235	0.248	0.260	0.283	0.342

Table 3: Calibration and balanced growth path comparison. Panel A reports parameters fixed externally. Panel B reports parameters set to match the data targets shown in the rightmost column, holding $\mu = 0$. The mapping from data targets to parameters is $\hat{\lambda} = \log(\gamma_{2020}/\gamma_{1990})/d$ where $d \equiv \log(x_{2020}/x_{1990})$; $\hat{\rho}_t = 1 - \omega^{-1}(1 - \hat{\lambda}\gamma_t/(1 - \gamma_t))$, $t \in \{1990, 2020\}$; $\hat{\kappa} = \gamma_{1990} x_{1990}^{-\hat{\lambda}}$; and $\hat{\xi}$ is pinned down by the growth rate g_{1990} . Data sources: γ_t from [Eisfeldt et al. \(2021\)](#) rescaled by the post-IPO insider equity share (Appendix A.1.3); x_{1990} and d from NETS establishments per firm matched to Compustat; g_{1990} from BEA NFCB output growth (Appendix A.1.4). See Appendix A.6.1 for further details on the procedure. Panel C reports model-implied moments at the 1990 and 2020 BGPs at the calibrated parameters of Panels A and B; 1990 values are independent of μ by construction (Appendix A.6.1.6). Except for the 1990 growth rate, the moments reported in Panel C are not targeted by our parameter choices. The aggregate TFP level Z is the BGP closed-form derived in Appendix A.5.3; price-to-output, Tobin's Q , profit share, and HHI are evaluated at the BGP closed-forms in Appendix A.5.

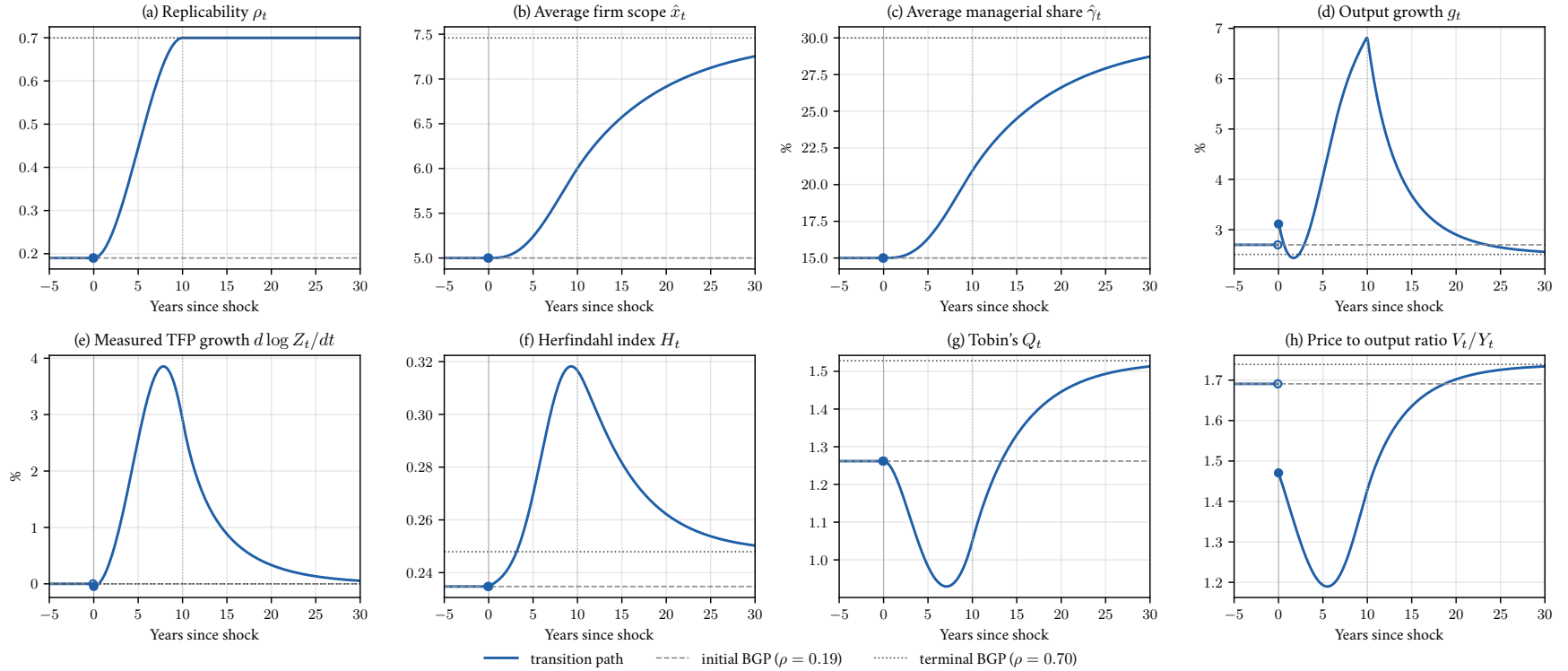


Figure 5: Transitional dynamics following an unanticipated, deterministic increase in replicability from $\rho_{1990} = 0.190$ (initial BGP) to $\rho_{2020} = 0.700$ (terminal BGP) over a window of ten years. All other parameters are held fixed at their calibrated values, reported in Panels A and B of Table 3; we set $\mu = 0$. The path of ρ_t follows a cubic smoothstep, described in Appendix A.6.3. The economy is at the initial BGP for $t \leq 0$. The vertical solid line marks the start of the unanticipated increase in ρ_t ($t = 0$); the vertical dotted line marks the end of the smoothstep window. In each panel the dashed (resp. dotted) gray line marks the value of the corresponding moment at the initial (resp. terminal) BGP. Open and filled circles at $t = 0$ indicate the pre- and post-shock values when they differ on impact. The numerical procedure for solving the transition path is described in Appendix A.6.2; the cohort-accounting equations used to construct each aggregate—average scope $\hat{x}_t$, average managerial share $\hat{\gamma}_t$, measured intangible capital K_t and replacement cost $P_{N,t}$, the firm-sales Herfindahl index H_t , Tobin's Q_t , and the price-to-output ratio V_t/Y_t —are derived in Appendix A.6.3. The closed-form expressions for the same quantities at the two endpoint BGPs (used to draw the dashed and dotted reference lines) are derived in Appendix A.5.

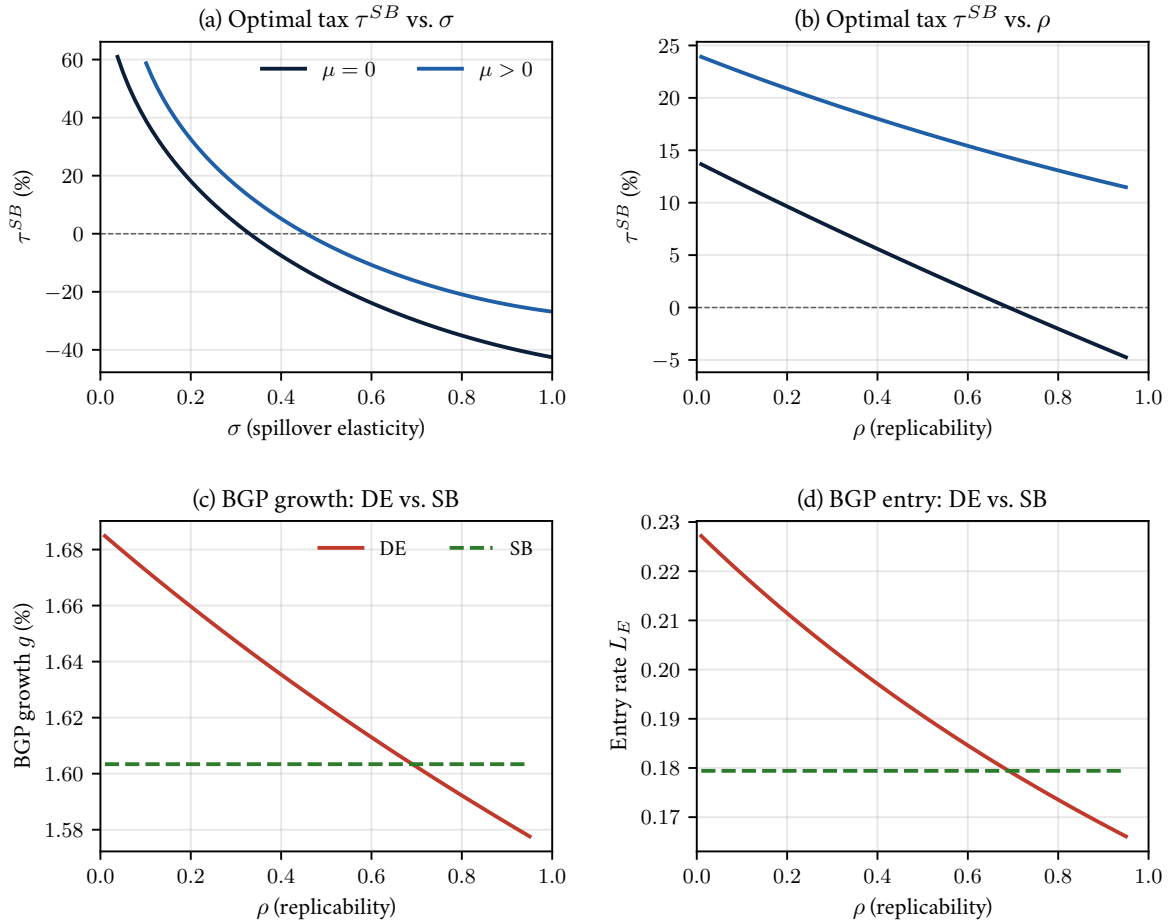


Figure 6: Numerical illustration of the optimal tax on intangible investment (Results 5 and 6). Parameters: $\alpha = 0.70$, $\chi = 1/1.10$, $\zeta = 0.77$, $\kappa = 0.005$, $\xi = 0.10$, $\mu \in \{0, 0.05\}$; not calibrated to data. Dashed horizontal lines mark the tax/subsidy boundary ($\tau = 0$); regions where $\gamma(x^{SB}) \geq 1$ are omitted. Panel (a): optimal tax τ^{SB} as a function of the spillover elasticity σ , with $\rho = 0.5$; the darker curve is $\mu = 0$, the lighter curve is $\mu = 0.05$. Panel (b): optimal tax τ^{SB} as a function of replicability ρ , with $\sigma = 0.3$; same colour convention as panel (a). Panels (c) and (d): BGP growth rate g and entry rate L_E under the decentralized equilibrium (DE, solid) and the second-best planner (SB, dashed), as functions of ρ at $\mu = 0$ and $\sigma = 0.3$.

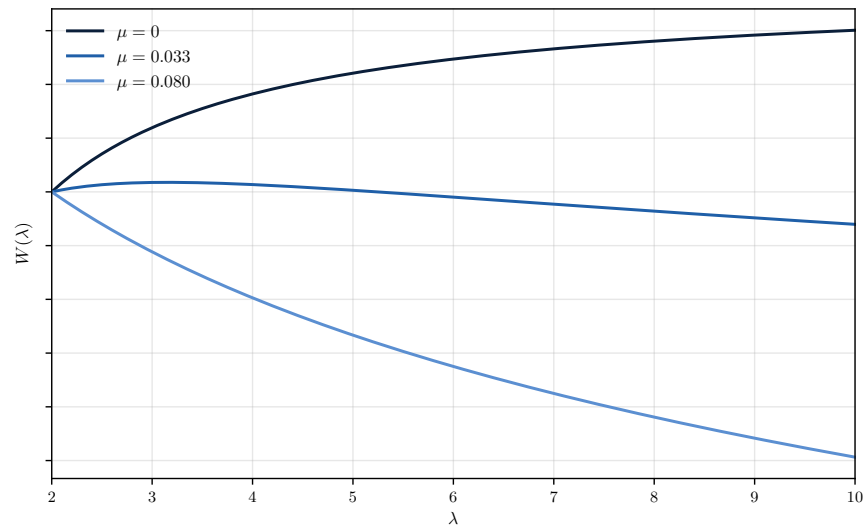


Figure 7: Welfare $W(\lambda)$ as a function of IP enforcement λ , for three values of the knowledge-diffusion concavity $\mu \in \{0, 0.033, 0.08\}$. Parameters: $\alpha = 0.5, \sigma = 1, \chi = 1/1.10, \zeta = 0.77, \kappa = 0.005, \xi = 0.50, \rho = 0.25$; chosen to illustrate, not calibrated. The colour ramp runs from darkest (lowest μ) to lightest (highest μ). Welfare is identified only up to a vertical shift, so tick labels on the vertical axis are suppressed; each curve is shifted so that $W(\lambda = 2) = 0$.

Appendix for “Intangible Capital, Firm Scope, and Growth”

Nicolas Crouzet, Janice Eberly, Andrea Eisfeldt, and Dimitris Papanikolaou

June 2026

A.1 Data sources and construction

This appendix collects the construction of all the data series used in the paper: the NETS establishment-level data and its match to Compustat (used in the empirical motivation of Section 1 and in the calibration of Section 5); the post-IPO insider equity share constructed from SEC IPO filings (used to rescale the Eisfeldt–Falato–Xiaolan equity-compensation series in the calibration); and the macroeconomic time series from NIPA, BDS, Compustat, and Fernald (2014) that are either targets in or untargeted comparators for the calibrated model.

A.1.1 DNB/NETS data construction

A.1.1.1 Raw Data and Sample Restrictions

We use the Dun & Bradstreet (DNB) National Establishment Time-Series (NETS) database, which provides annual establishment-level records for the universe of U.S. business establishments from 1990 to 2018.¹⁶ Each record contains the establishment’s DUNS number, company name, physical address (including ZIP code), four-digit SIC code, site-level employment, firm-wide employment, sales, and linkage fields to parent and headquarters DUNS numbers.

We apply two filters at the establishment level before any aggregation:

1. **Positive employment:** We drop establishments reporting zero employees ($\text{EMPLOYEEESTHISITE} = 0$).
2. **Industry exclusions:** We drop establishments in educational services (SIC 82), social services (SIC 83), museums and gardens (SIC 84), membership organizations including churches and nonprofits (SIC 86), all public administration and government (SIC 91–97), and Federal Reserve banks (SIC 6011).

A.1.1.2 Firm Identification

A central challenge in the NETS data is linking individual establishments to their parent firms. We implement a five-pass hierarchical linkage pipeline:

1. **NETS headquarters lookup.** We use the Walls & Associates NETS headquarters linkage file, which maps each establishment’s DUNS number to its headquarters DUNS. This file provides 824,000–2.1 million branch-to-headquarters linkages per year and is available for 1990–2017 (we apply the 2017 lookup to 2018).
2. **Ultimate DUNS resolution.** For establishments reporting an ULTDUNSNO (ultimate parent DUNS number), we build a lookup mapping each establishment to its ultimate parent.
3. **Headquarters DUNS resolution.** Similarly, we build a lookup from HQDUNSNO (headquarters DUNS number) where available.

¹⁶The raw data files (DMI CSV format) also cover 1969–1989, but coverage is substantially thinner pre-1990 and the NETS headquarters linkage file is unavailable before 1990. We therefore restrict our analysis to the 1990–2018 period.

4. **Transitive resolution.** For each establishment, we resolve the ultimate parent using the following priority chain: (i) ULTDUNSNO, (ii) PARENTDUNSNO resolved through the ultimate DUNS lookup, (iii) HQDUNSNO resolved through the ultimate DUNS lookup, (iv) NETS headquarters DUNS from the Walls & Associates file, (v) the establishment’s own DUNS number (standalone). We then apply one additional hop through the headquarters DUNS lookup.
5. **Name-based grouping.** Remaining standalone establishments are grouped by normalized company name and one-digit SIC code. If a (name, SIC-digit) pair appears in three or more distinct ZIP codes, we treat all matching establishments as belonging to the same firm. This captures chains and franchises that lack DUNS-based parent linkages. These firms receive a synthetic identifier of the form NAME_{name}_{sic_digit}.

After linkage, each firm’s characteristics are aggregated across its establishments. The modal four-digit SIC code (most common across establishments) serves as the firm’s industry. Geographic footprint is measured by the number of distinct ZIP codes and CBSAs, and employment and sales are summed.

A.1.1.3 CBSA Crosswalk

To measure geographic scope at the metropolitan level, we map each establishment’s five-digit ZIP code to a Core Based Statistical Area (CBSA) using a two-step crosswalk:

1. We use the Census Bureau’s 2010 ZCTA–County Relationship File to map each ZIP Code Tabulation Area to its primary county (the county with the largest population overlap).
2. We then use the Office of Management and Budget’s 2013 CBSA delineation file to map each county FIPS code to its CBSA code.

This yields a crosswalk covering 24,269 ZIP codes mapped to CBSAs. Each establishment’s ZIP code is mapped through this crosswalk, and the firm-level variable `n_cbsas` counts the number of distinct CBSAs in which the firm has at least one establishment. We apply fixed 2013 CBSA definitions throughout to ensure consistent geographic boundaries across years.

A.1.1.4 Summary Statistics

The resulting firm-level panel contains approximately 5.1 million firms in 1990, rising to 20.4 million in 2018. Of these, roughly 4–5% are multi-establishment firms in any given year. Each firm-year observation records the firm’s establishment count, number of distinct ZIP codes, number of distinct CBSAs, total employment, and total sales.

A.1.2 Matching Compustat to DNB/NETS

We construct a crosswalk between Compustat and the DNB/NETS database at the headquarters level for the period 1990–2018. The match links each Compustat firm (identified by `GVKEY`) to a DNB firm identifier, enabling us to attach establishment-level geographic information—including the number of distinct ZIP codes, CBSAs, and total establishment counts—to publicly traded firms.

A.1.2.1 Data Preparation

Compustat. We obtain historical company names, SIC codes, and headquarters ZIP codes from Compustat’s historical name snapshot file, which provides date-stamped records of each firm’s name, SIC, and address. For each GVKEY-year, we select the name and attributes that were effective during that fiscal year. When the snapshot is missing a field, we fall back to the standard Compustat annual file (such for historical SIC, addzip for ZIP code). This yields 871,229 GVKEY-year observations covering 47,119 unique GVKEYs.

DNB/NETS. For each year, we construct a headquarters roster from the raw DMI establishment files. We first identify firms using a five-pass linkage pipeline: (i) NETS headquarters lookup, (ii) ultimate DUNS resolution, (iii) headquarters DUNS resolution, (iv) transitive resolution across parent and headquarters linkages, and (v) name-based grouping of standalone establishments appearing in three or more ZIP codes. We then extract the headquarters establishment’s company name and ZIP code for each firm, along with the firm’s modal four-digit SIC code (computed across all establishments) and total employment. To keep the matching computationally tractable, we restrict the DNB roster to firms with at least 100 employees, yielding approximately 67,000–101,000 candidate firms per year.

A.1.2.2 Name Normalization

Both Compustat and DNB company names undergo a multi-step normalization procedure:

1. Remove Compustat-specific artifacts (e.g., “-CL A,” “-OLD,” “-REDH”).
2. Strip legal entity suffixes using the `cleanco` library (e.g., “Inc.,” “Corp.,” “LLC”).
3. Remove additional corporate descriptors (e.g., “Company,” “Holdings,” “International”).
4. Expand common abbreviations (e.g., “INTL” → “International,” “MFG” → “Manufacturing,” “SVCS” → “Services”).
5. Remove punctuation, collapse whitespace, and strip leading/trailing “The.”
6. Apply targeted concatenation fixes (e.g., “JPMORGAN” → “JP MORGAN”).

A.1.2.3 Blocking and Fuzzy Matching

For each year, we match Compustat firms to the DNB roster using a four-pass blocking strategy with progressively relaxed constraints. This approach balances precision against recall while keeping the candidate set manageable for fuzzy string comparison.

1. **Pass 1 (ZIP + 2-digit SIC):** Candidates must share both the five-digit headquarters ZIP code and the two-digit SIC code. Minimum employment: 100. Name score threshold: 70.
2. **Pass 2 (ZIP only):** Candidates must share the ZIP code but may differ in SIC. Minimum employment: 100. Threshold: 75.

3. **Pass 3 (2-digit SIC only):** Candidates must share the two-digit SIC but may differ in ZIP. Minimum employment: 500. Threshold: 80.
4. **Pass 4 (No blocking):** All firms with at least 10,000 employees are considered. Threshold: 85.

Within each block, we compute a composite match score. The name similarity score combines `token_sort_ratio` and `token_set_ratio` from the `rapidfuzz` library:

$$\text{name_score} = \max(\text{token_sort_ratio}(a, b), 0.8 \times \text{token_set_ratio}(a, b)). \quad (\text{A1})$$

The composite score adds bonuses for exact matches on ZIP code (+15 points) and four-digit SIC code (+5 points):

$$\text{composite} = \text{name_score} + 15 \cdot \mathbf{1}[\text{ZIP match}] + 5 \cdot \mathbf{1}[\text{SIC4 match}]. \quad (\text{A2})$$

For each Compustat firm, we retain the DNB candidate with the highest composite score, provided the name score exceeds the pass-specific threshold. Firms matched in earlier (tighter) passes are excluded from subsequent passes.

A.1.2.4 Cross-Year Consolidation

Because the same Compustat firm may match to different DNB identifiers across years (due to name changes, mergers, or noise), we consolidate matches by selecting the modal DNB firm identifier for each GVKEY across all matched years. We assign confidence tiers based on match quality:

- **High confidence:** Best name score ≥ 95 with ≥ 3 years matched, or best name score ≥ 85 with ≥ 2 years matched.
- **Medium confidence:** Best name score ≥ 80 .
- **Low confidence:** All remaining matches.

A.1.2.5 Match Results

Table A-1 summarizes the matching outcomes. Of the 47,119 unique GVKEYs in Compustat over 1990–2018, we match 20,379 (43.2%) to a DNB firm. The unmatched firms are predominantly non-U.S. companies, investment vehicles, and very small firms that do not appear in DNB with 100 or more employees.

The resulting firm-year panel contains 207,185 observations across 20,379 GVKEYs and 29 years, with each observation recording the firm’s establishment count, number of distinct ZIP codes, number of distinct CBSAs, total employment, and total sales from the DNB/NETS data.

A.1.3 Insider equity share at IPO

This subsection describes the construction of the insider equity share variable used in Section 5 as a proxy for s^{insider} , the share of public-firm equity held collectively by managers and entrepreneurs in the model. The variable is constructed from initial public offering (IPO) prospectuses for U.S. public firms over 1996–2023.

A.1.3.1 Sample and sources

The sample consists of U.S. public firms that completed an IPO on Nasdaq or the NYSE between 1996 and 2023. Following [Ritter and Welch \(2002\)](#), we exclude American depositary receipts, natural resource limited partnerships and trusts, closed-end funds, real estate investment trusts (REITs), special-purpose acquisition companies (SPACs), banks and savings and loans (SIC codes 6011–6099), unit offers, penny stocks (offer price below \$5), and spinoffs.

The 1996–2016 portion of the panel reuses an earlier dataset constructed by Andrea Eisfeldt as part of related work on firm life-cycle ownership dynamics.¹⁷ The 2017–2023 portion is a new extension constructed for this paper. Both portions follow the same sample-selection criteria. Identifiers are merged across sources using firm CUSIP. IPO dates are taken from Compustat fundamentals annual; missing dates are filled with offering dates from the IPO-age dataset of [Ritter \(2020\)](#). SPACs and spinoffs are identified primarily through SDC Platinum and verified manually by inspection of the IPO prospectus.

For each IPO firm, we obtain ownership information directly from the firm’s IPO prospectus, filed as Form 424B or Form S-1 with the U.S. Securities and Exchange Commission (SEC). For the 2017–2023 extension, we scrape these filings from SEC EDGAR using the firm’s central index key (CIK) and parse the beneficial-ownership tables they contain. The 1996–2016 portion of the panel uses the analogous tables compiled in the earlier dataset.

A.1.3.2 Construction of the insider share variable

The IPO prospectus reports, for a list of named individuals and entities, the number of common shares beneficially owned both before and after the IPO. We aggregate these holdings into three categories:

- **Insiders:** the firm’s executive officers and directors, identified by the table heading “all executive officers and directors as a group” or analogous language.
- **5% shareholders:** external holders that own at least 5% of common stock and are not executive officers or directors.
- **Other holders:** smaller external holders, including selling stockholders not in either of the prior categories.

Identification is implemented by keyword matching on the table headings and verified manually for ambiguous cases. When ownership is reported by class of stock, we use the common stock percentages where available; when only Class A and Class B shares are reported, we use the share of Class A common stock. Pre-IPO shares are computed assuming none of the offered shares have been distributed; post-IPO shares incorporate the dilution from the IPO offering itself.

The insider share variable used in Section 5 is the post-IPO share of equity held by the insiders group:

$$s_{j,t}^{\text{insider}} \equiv \frac{\text{shares held by insiders, after IPO}}{\text{total shares outstanding, after IPO}}, \quad (\text{A3})$$

¹⁷We are grateful to Andrea for sharing this resource.

where j indexes IPO firms and t is the IPO year. The yearly aggregate s_t^{insider} is the equal-weighted mean of $s_{j,t}^{\text{insider}}$ across firms going public in year t .

A.1.3.3 Time series

Figure A-1 plots the resulting time series, both for the pre-IPO snapshot and for the post-IPO snapshot used in Section 5. The post-IPO insider share has trended down over the period, from approximately 48% in the late 1990s to approximately 30% in the early 2020s.

A.1.4 Aggregate time series: NIPA, BDS, Compustat, and other sources

This subsection describes the construction of the aggregate time series used in the numerical illustration of Section 5, both as direct data targets for the parameter choices (g_{1990} and δ) and as untargeted moments to which we compare the model's predictions (Q , the profit share, total factor productivity growth, and concentration). The corresponding time series are plotted in Figure A-2.

NFCB real value added growth. The data on the real growth rate of value added in the non-financial corporate business (NFCB) sector are drawn from the National Income and Product Accounts (NIPA), Table 1.14 (chain-type quantity index for gross value added of nonfinancial corporate business). The corresponding FRED series is B455RX1Q027SBEA. We aggregate quarterly observations to annual frequency and compute the annualized log growth rate of real value added. The 1990 calibration target $g_{1990} = 0.027$ is the average annualized log growth rate over the five-year window 1988–1992.

NFCB profit share. The data on profits as a percentage of value added in the NFCB sector are drawn from NIPA Table 1.15; the corresponding FRED series is A463RD3Q052SBEA. The ratio is defined as corporate profits with inventory valuation and capital consumption adjustments, divided by total gross value added. Corporate profits are computed as domestic corporate gross value added, minus compensation of employees, minus taxes on production and imports less subsidies, minus consumption of fixed capital (depreciation), minus net interest payments, minus net business current transfer payments. Conceptually, this measure is close to accounting net income, except that it does not net out corporate income taxes (since the model abstracts from corporate income taxes).

Gross business entry and exit rates. The data on gross business entry and exit rates are drawn from the U.S. Census Bureau's Business Dynamics Statistics (BDS), specifically the one-way time series aggregated at the sectoral level.¹⁸ To obtain a sample approximately consistent with the broader empirical analysis, we drop the two-digit NAICS sectors corresponding to agriculture (11), utilities (22), finance and insurance (52), real estate rental and leasing (53), and public administration (92). For each year t , let N_t denote the number of firms and D_t denote the number of firm deaths. The gross entry rate is defined as $e_t \equiv (N_t - N_{t-1} + D_t) / N_{t-1}$, and the gross exit rate as $o_t \equiv D_t / N_{t-1}$. The 1990 calibration target $\delta = 0.10$ is the average gross firm exit rate over the five-year window 1988–1992. The exit rate is broadly stable across the full sample period, justifying the assumption that δ is constant across the two BGPs in Section 5.

¹⁸The data are available at <https://www2.census.gov/programs-surveys/bds/tables/time-series/>.

Aggregate Tobin’s Q . The aggregate Tobin’s Q series is constructed from Compustat data. For each year, aggregate enterprise value V_t is the sum, across all firms in the sample, of the market value of equity (CRSP variables `prc` and `shrout`) plus the book value of debt outstanding (Compustat variables `dlc` and `dltt`). Aggregate productive capital K_t is the sum of the one-year-lagged value of net property, plant, and equipment (Compustat variable `ppent`), the capitalized value of R&D expenditures (variable `k_int_know` from Peters and Taylor, 2017), and the capitalized value of organizational capital expenditures (variable `k_int_org` from Peters and Taylor, 2017). The aggregate Tobin’s Q is the ratio V_t / K_t .

Total factor productivity. The data on total factor productivity are from Fernald (2014); we use the utilization-adjusted, annual time series for the business sector.

Aggregate concentration. The aggregate Herfindahl index of sales is also constructed from Compustat. For each year and each NAICS three-digit industry, we compute the Herfindahl index of sales as the sum of the squared revenue shares of firms in that industry. The aggregate index for each year is the weighted average of the within-industry Herfindahls across sectors, with weights proportional to the total revenue of each sector.

A.2 Entry rates, firm scope, and industry concordances

This appendix supports the industry-level evidence in Section 1 relating establishment entry rates from the Business Dynamics Statistics (BDS) to firm geographic scope from the National Establishment Time Series (NETS). It describes data construction, reports summary statistics, shows aggregate and one-digit-industry time series, and reports regressions using firm-level entry rates (a weaker correlate of scope in our sample).

A.2.1 Data sources

Business Dynamics Statistics (BDS). The BDS is produced by the U.S. Census Bureau from the Longitudinal Business Database (LBD). It provides annual counts of firms, establishments, employment, establishment births, establishment deaths, and firm deaths, tabulated by industry. We access the BDS through the Census Bureau API at the four-digit NAICS level (`INDLEVEL=4`), covering 288 industry groups for each year from 1978 to 2023. The BDS uses the 2017 NAICS classification for all years. There is no data suppression at this level of detail.

National Establishment Time Series (NETS). The NETS database, maintained by Walls & Associates, tracks U.S. business establishments using Dun & Bradstreet data. We use a pre-constructed industry-year panel at the three-digit SIC level (`industry_year_panel.dta`), covering 376 SIC-3 codes over 1990–2018. The panel contains firm counts and several measures of firm geographic scope: the average number of CBSAs, ZIP codes, and establishments per firm, computed for multi-establishment firms with at least two employees (variables with the `_multi2` suffix). Construction of the underlying NETS micro data and the match to Compustat is described in Appendix A.1.2 and Section A.1.1.

A.2.2 Industry crosswalk

Because the BDS uses NAICS codes and the NETS panel uses SIC codes, a crosswalk is required. We use a payroll-weighted crosswalk from four-digit NAICS 2017 to three-digit SIC 1987, constructed by chaining the

weighted concordances of [Schaller et al. \(2025\)](#) through intermediate NAICS vintages: NAICS 2017 $\rightarrow$ 2012 $\rightarrow$ 2007 $\rightarrow$ 2002 $\rightarrow$ 1997 $\rightarrow$ SIC 1987. County Business Patterns (CBP) national payroll provides level weights for aggregating from six-digit to four-digit NAICS. Since the BDS is coded in NAICS 2017 for every year, we apply a single time-invariant mapping: BDS counts at the NAICS four-digit level are split fractionally across SIC-3 codes using the crosswalk weights for year = 2017, then summed within each SIC-3 code. The Python script `01b_build_crosswalk.py` in the replication code produces `input/naics4_to_sic3_crosswalk.csv`; `02_build_entry_panel.py` then selects the 2017 rows for use with the BDS.

Adjustment for breaks in NETS scope series. The NETS scope variables exhibit sharp discontinuities around 2010–2011 and 2014–2015, likely reflecting changes in data collection. Following the procedure in the replication code, we splice these breaks by extrapolating from pre-break trends in log scope and shifting post-break levels to match the extrapolated values: for each SIC-3 industry and each scope variable, we use average annual log growth over 2008–2010 to predict a 2011 level and shift all years from 2011 onward by the gap between predicted and actual; we then repeat for the 2014–2015 break using growth rates on the already-adjusted 2012–2014 segment.

A.2.3 Variable definitions

Establishment entry rate. Number of establishments born during the year (BDS variable `estabs_entry`), divided by the lagged number of establishments.

Firm entry rate. The BDS does not report firm births directly at the four-digit NAICS level. We recover them from the accounting identity $\text{firm_births}_t = \max(0, \text{firms}_t - \text{firms}_{t-1} + \text{firm_deaths}_t)$. The firm entry rate is firm births divided by lagged firms.

Scope measures. We use three scope measures from NETS, all defined for multi-establishment firms with at least two employees: average CBSAs per firm (`cbsa_multi2`), average ZIP codes per firm (`zip_multi2`), and average establishments per firm (`est_multi2`). All enter regressions in logs.

A.2.4 Specification

The panel regressions in Section 1 take the form:

$$\Delta_5 \ln(\text{scope}_{i,t}) = \beta \Delta_5 \text{entry rate}_{i,t} + \gamma \ln(\text{scope}_{i,t}) + \delta \text{entry rate}_{i,t} + \alpha_{d(i),t} + \varepsilon_{i,t}, \quad (\text{A4})$$

where i indexes SIC-3 industries, t is the base year, $d(i)$ is the one-digit SIC division of industry i , and $\alpha_{d(i),t}$ are year $\times$ SIC-1 division fixed effects. The operator Δ_5 denotes the five-year forward difference. Standard errors are clustered at the SIC-3 level. Scope is placed on the left-hand side because measurement error in NETS scope (including the documented breaks) would attenuate coefficients if scope were on the right. Further discussion appears in the replication note `note_entry_scope.tex` bundled with the code.

A.2.5 Summary statistics

Table A-2 reports summary statistics for the regression sample (SIC-3 industries with base years 1990–2013 for which five-year forward differences are observed through 2018; FIRE industries in 1991 are excluded to match the regressions).

A.2.6 Aggregate and one-digit industry time series

Figures A-3 and A-4 plot, for the industry coverage used in the merged BDS–NETS sample, the BDS-establishment-weighted establishment entry rate together with log CBSAs per firm from NETS (adjusted series). Figure A-3 aggregates across SIC-3 industries; Figure A-4 shows the same series by one-digit SIC division.

A.2.7 Regressions using firm entry rates

Table A-3 repeats the exercise of Table 2 using the firm entry rate constructed from the accounting identity instead of the establishment entry rate. Coefficients are negative but smaller in magnitude and generally insignificant, consistent with noisier measurement of firm births at this level of aggregation.

A.3 Appendix to Section 2

Consider the problem of choosing scope for the entrepreneur as described in Section 2:

$$\sup_{x \geq 0} (1 - \gamma(x; \lambda)) x^{1-(1-\rho)\omega} \quad (\text{A5})$$

Here we allow for markups, so that $\omega \in [0, 1]$, as in the full model.

Appendix result 1 (Optimal scope) Fix $\lambda \in \mathbb{R}_+$. Suppose that $\gamma(\cdot; \lambda) : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is twice continuously differentiable, strictly increasing, and weakly convex on $\mathbb{R}_+$ and satisfies the following conditions:

$$\text{C1 } 1 - \gamma(0; \lambda) > 0$$

$$\text{C2 } \text{There exists } \bar{x} > 0 \text{ such that } 1 - \gamma(\bar{x}; \lambda) \geq 0 \text{ and } 1 - \gamma(\bar{x}; \lambda) - \bar{x}\gamma'(\bar{x}; \lambda) < 0$$

Then the supremum in problem (A5) is a maximum, and it is reached at unique $x^* \in [0, \bar{x}]$. If, moreover:

$$\text{C3 } (1 - (1 - \rho)\omega)(1 - \gamma(1; \lambda)) - \gamma'(1; \lambda) \geq 0$$

then $x^* \in [1, \bar{x}]$.

Proof of Appendix Result 1. Since $\gamma(\cdot; \lambda)$ is twice continuously differentiable, so is the objective function in Equation (A5). Its derivative is

$$\frac{d}{dx} [(1 - \gamma(x; \lambda)) x^{1-(1-\rho)\omega}] = x^{-(1-\rho)\omega} R(x; \lambda), \quad (\text{A6})$$

where

$$R(x; \lambda) \equiv (1 - (1 - \rho)\omega)(1 - \gamma(x; \lambda)) - x\gamma'(x; \lambda). \quad (\text{A7})$$

Since $x^{-(1-\rho)\omega} > 0$ for $x > 0$, the derivative of the objective has the same sign as $R(x; \lambda)$.

This function is itself continuously differentiable in x , with derivative

$$\begin{aligned} R'(x; \lambda) &= -((1 + (1 - (1 - \rho)\omega))\gamma'(x; \lambda) + x\gamma''(x; \lambda)) \\ &\leq -(\gamma'(x; \lambda) + x\gamma''(x; \lambda)) \\ &< 0, \end{aligned} \quad (\text{A8})$$

These bounds hold for $x > 0$: the second line uses $1 - (1 - \rho)\omega \geq 0$ and $\gamma'(x; \lambda) \geq 0$ (since $\gamma(\cdot; \lambda)$ is increasing), and the third uses $\gamma'(x; \lambda) > 0$ (strict monotonicity) and $\gamma''(x; \lambda) \geq 0$ (weak convexity). Thus $R(\cdot; \lambda)$ is strictly decreasing in x .

Next, because $\gamma(\cdot; \lambda)$ is twice continuously differentiable on $\mathbb{R}_+$, its derivative $\gamma'(\cdot; \lambda)$ is continuous at 0, so $x\gamma'(x; \lambda) \rightarrow 0$ as $x \rightarrow 0^+$. Thus:

$$\lim_{x \rightarrow 0^+} R(x; \lambda) = R(0; \lambda) = (1 - (1 - \rho)\omega)(1 - \gamma(0; \lambda)) > 0, \quad (\text{A9})$$

where the inequality follows from Assumption C1.

Next, we have that:

$$\begin{aligned} R(\bar{x}; \lambda) &= (1 - (1 - \rho)\omega)(1 - \gamma(\bar{x}; \lambda)) - \bar{x}\gamma'(\bar{x}; \lambda) \\ &\leq 1 - \gamma(\bar{x}; \lambda) - \bar{x}\gamma'(\bar{x}; \lambda) \\ &< 0, \end{aligned} \quad (\text{A10})$$

where the second line uses $0 \leq 1 - (1 - \rho)\omega \leq 1$ and $\gamma(\bar{x}; \lambda) \leq 1$, and the third line uses the remaining part of Assumption C2.

Since $R(0; \lambda) > 0$, $R(\bar{x}; \lambda) < 0$, and R is continuous and strictly monotone with respect to x , by the intermediate value theorem there is exactly one x^* such that $R(x^*; \lambda) = 0$. Moreover, because R is strictly decreasing with respect to x , this x^* maximizes the objective function in problem A5.

Finally, suppose condition C3 holds. Then:

$$R(1; \lambda) = (1 - (1 - \rho)\omega)(1 - \gamma(1; \lambda)) - \gamma'(1; \lambda) \geq 0, \quad (\text{A11})$$

where the equality is the definition of $R(1; \lambda)$ and the inequality is condition C3. The intermediate value theorem then implies that $x^* \geq 1$. ■

Two points are worth noting in Result 1. First, the condition that $\gamma(x; \lambda)$ is weakly convex is sufficient but not necessary. A weaker sufficient condition that can be substituted for convexity is:

$$\forall x \geq 0, \quad \gamma'(x; \lambda) + x\gamma''(x; \lambda) > 0. \quad (\text{A12})$$

Second, Condition C3 involves the value of both ρ and ω . However, it can be weakened to be independent of the value of $\rho \in [0, 1]$, so long as $\omega < 1$, as is the case in the baseline model. This is useful because later we will compute comparative statics across the entire range of $\rho \in [0, 1]$, holding the functional form for $\gamma(\cdot; \lambda)$ fixed. Specifically, suppose that:

$$C3' \quad \omega < 1 \text{ and } (1 - \omega)(1 - \gamma(1; \lambda)) - \gamma'(1; \lambda) \geq 0.$$

Then, first, note that it must be the case that:

$$1 - \gamma(1; \lambda) \geq 0. \quad (A13)$$

Therefore:

$$\begin{aligned} R(1; \lambda) &= (1 - (1 - \rho)\omega)(1 - \gamma(1; \lambda)) - \gamma'(1; \lambda) \\ &\geq (1 - \omega)(1 - \gamma(1; \lambda)) - \gamma'(1; \lambda) \\ &\geq 0, \end{aligned} \quad (A14)$$

where the second line uses $1 - (1 - \rho)\omega \geq 1 - \omega$ together with $1 - \gamma(1; \lambda) \geq 0$, and the last inequality is condition C3'. The intermediate value theorem then implies that $x^* \geq 1$.

In the full model, we use the functional form:

$$\gamma(x; \lambda) = \kappa x^\lambda, \quad \lambda \geq 1, \quad \kappa \leq \frac{1 - \omega}{1 - \omega + \lambda} \quad (A15)$$

We can check that the conditions of Result (1) hold. $\gamma(\cdot; \lambda)$ is positive, twice continuously differentiable, and strictly increasing and weakly convex because $\lambda \geq 1$. Moreover $\gamma(0; \lambda) = 0$.

Next, let:

$$\bar{x} = \kappa^{-1/\lambda}. \quad (A16)$$

Then we have:

$$\gamma(\bar{x}; \lambda) = 1, \quad 1 - \gamma(\bar{x}; \lambda) - \bar{x}\gamma'(\bar{x}; \lambda) = -\bar{x}\gamma'(\bar{x}; \lambda) < 0, \quad (A17)$$

so that condition C2 holds.

Finally, note that:

$$1 - \gamma(1; \lambda) = 1 - \kappa, \quad \gamma'(1; \lambda) = \kappa\lambda, \quad (A18)$$

so that:

$$(1 - \omega)(1 - \kappa) - \kappa\lambda = (1 - \omega) - \kappa(1 - \omega + \lambda) \geq 0, \quad (A19)$$

where the last inequality follows from Equation (A15). Thus the weaker condition C3' holds and so x^* exists, is unique, and is weakly larger than 1.

Appendix result 2 (Effects of non-exclusivity on scope) Suppose that conditions C1, C2 and C3 hold and let $x^* \geq 1$ denote optimal scope in problem A5. Moreover, suppose that $\gamma(x; \lambda) : \mathbb{R}_+^2 \rightarrow \mathbb{R}_+$, as a function of two variables (x, λ) , is twice continuously differentiable, and satisfies:

$$C4 \quad \forall x \geq 1, \partial_\lambda \gamma > 0$$

$$C5 \quad \forall x \geq 1, \partial_{x\lambda}\gamma > 0.$$

Then the optimal scope satisfies:

$$\frac{\partial x^*}{\partial \lambda} < 0. \quad (A20)$$

Proof of Appendix Result 2. Apply the implicit function theorem to the first-order condition:

$$R(x^*; \lambda) = 0 \quad (A21)$$

to obtain:

$$\frac{\partial x^*}{\partial \lambda} = -\frac{\partial_\lambda R(x^*, \lambda)}{\partial_x R(x^*, \lambda)} = -\frac{(1 - (1 - \rho)\omega)\partial_\lambda \gamma + x\partial_{x\lambda}\gamma}{(1 + (1 - (1 - \rho)\omega))\partial_x \gamma + x\partial_{xx}\gamma}. \quad (A22)$$

By conditions C4 and C5 and $x^* \geq 1$, the numerator is positive; because $\gamma(\cdot; \lambda)$ is strictly increasing and weakly convex in x , the denominator is positive. Hence $\partial x^* / \partial \lambda < 0$. ■

For the functional form used in the full model, we have:

$$\partial_\lambda \gamma(x; \lambda) = \kappa x^\lambda \log(x), \quad \partial_{x\lambda} \gamma(x; \lambda) = \kappa x^{\lambda-1} (1 + \lambda \log(x)). \quad (A23)$$

Both of these expressions are positive for any $x \geq 1$, so they satisfy conditions C4 and C5.

A.4 Appendix to Section 3

This appendix derives the equilibrium characterization of Section 3.3. Throughout, structural parameters $(\rho, \lambda, \omega, \kappa)$ are held constant. Since optimal scope depends only on these parameters, we write x for the constant value solving (36), and $\gamma \equiv \gamma(x; \lambda) = \kappa x^\lambda$.

A.4.1 Aggregation

Equation (25) implies that:

$$l_t(m) \propto n(m)^\omega. \quad (A24)$$

Thus for each m ,

$$y_t(m) \propto n(m)^{\frac{\omega}{\chi}}. \quad (A25)$$

Using Equation (13), output of each variety can therefore be written as:

$$y_t(m) = \left(\frac{n(m)}{N_t} \right)^{\frac{\omega}{\chi}} Y_t, \quad (A26)$$

where the aggregate capital stock N_t is defined as:

$$N_t \equiv \left(\int_0^{X_t} n(m)^\omega dm \right)^{\frac{1}{\omega}}. \quad (A27)$$

Additionally, the labor market clearing condition implies:

$$l_t(m) = \left(\frac{n(m)}{N_t} \right)^\omega L_{Y,t}. \quad (\text{A28})$$

Using this, we obtain the following expression for output of variety m :

$$y_t(m) = l_t(m)^\zeta n(m)^{1-\zeta} = n(m)^\omega N_t^{-\omega\zeta} L_{Y,t}^\zeta. \quad (\text{A29})$$

Comparing this to Equation (A26), we see that total output Y_t is given by:

$$Y_t = N_t^{1-\zeta} L_{Y,t}^\zeta. \quad (\text{A30})$$

Moreover, using Equation (25), we can derive aggregate labor demand as a function of output, capital, and the wage rate:

$$L_{Y,t} = (\zeta\chi)^{\frac{1}{1-\zeta\chi}} N_t^{\frac{(1-\zeta)\chi}{1-\zeta\chi}} Y_t^{\frac{1-\chi}{1-\zeta\chi}} W_t^{-\frac{1}{1-\zeta\chi}}. \quad (\text{A31})$$

Using the aggregate production function to eliminate N_t , we arrive at the following expression:

$$L_{Y,t} = \chi \frac{\zeta Y_t}{W_t}. \quad (\text{A32})$$

A.4.2 Global dynamics

We combine free entry, the definition of relative firm size θ_t , and the laws of motion for the intangible index N_t and public knowledge Ξ_t to obtain a complete dynamic system. We then derive the ordinary differential equation satisfied by the price-earnings ratio $\nu(\theta_t)$.

From the results above, we can express A_t as:

$$\frac{A_t}{W_t} = \frac{1-\zeta\chi}{\zeta\chi} L_{Y,t} N_t^{-\omega}. \quad (\text{A33})$$

Compensation of the coalition of entrepreneurs at entry, given intangibles $n_t(j) = n_t$, relative to the wage rate, is given by:

$$\frac{\tilde{v}_t^{(E)}}{W_t} = \frac{1-\zeta\chi}{\zeta\chi} (1-\gamma)x^{1-(1-\rho)\omega} \nu_t L_{Y,t} \left(\frac{n_t}{N_t} \right)^\omega. \quad (\text{A34})$$

Recall that the compensation per unit of entrepreneurial effort is:

$$v_t^{(E)} = \frac{E_t}{L_{E,t}} \tilde{v}_t^{(E)} = L_{E,t}^{-(1-\alpha)} \tilde{v}_t^{(E)}, \quad (\text{A35})$$

where we used the fact that $L_{M,t} = 1$ in equilibrium. Thus we can write Equation (A34) as:

$$\frac{v_t^{(E)}}{W_t} L_{E,t}^{1-\alpha} = \frac{1-\zeta\chi}{\zeta\chi} (1-\gamma)x^{1-(1-\rho)\omega} \nu_t L_{Y,t} \left(\frac{n_t}{N_t} \right)^\omega. \quad (\text{A36})$$

Finally, free-entry requires that $v_t^{(E)} = W_t$ and $L_{E,t} + L_{Y,t} = 1$, leading to the condition:

$$\frac{L_{E,t}^{1-\alpha}}{1-L_{E,t}} = \frac{1-\zeta\chi}{\zeta\chi} (1-\gamma)x^{1-(1-\rho)\omega} \nu_t e^{\omega\theta_t}, \quad (\text{A37})$$

where we have also substituted for:

$$\theta_t = \log\left(\frac{n_t}{N_t}\right) = \log\left(\frac{\Xi_t}{N_t}\right). \quad (\text{A38})$$

The law of motion for capital is:

$$\begin{aligned} N_t^\omega &= \int_{s \leq t} x^{1-\omega+\rho\omega} n_s^\omega E_s e^{-\delta(t-s)} ds \\ d(N_t^\omega) &= \underbrace{(E_t dt)}_{\# \text{ new firms}} \times \underbrace{x^{1-\omega+\rho\omega}}_{\text{effect of scope}} \times \underbrace{\left(\frac{\Xi_t}{N_t}\right)^\omega}_{\text{capital per firm}} - \underbrace{(\delta dt)}_{\text{exit risk}} N_t^\omega \\ \frac{d(N_t^\omega)}{N_t^\omega} &= \left(E_t x^{1-\omega+\rho\omega} \left(\frac{\Xi_t}{N_t}\right)^\omega - \delta \right) dt \\ \frac{dN_t}{N_t} &= \frac{1}{\omega} \left(L_{E,t}^\alpha x^{1-\omega+\rho\omega} \left(\frac{\Xi_t}{N_t}\right)^\omega - \delta \right) dt \\ &= \frac{1}{\omega} \left(x^{1-\omega+\rho\omega} L_{E,t}^\alpha e^{\omega\theta_t} - \delta \right) dt \end{aligned} \quad (\text{A39})$$

The law of motion for Ξ_t is:

$$\frac{d\Xi_t}{\Xi_t} = \xi (\gamma x^\rho)^\mu L_{E,t}^{\alpha\sigma} dt, \quad (\text{A40})$$

where $\xi (\gamma x^\rho)^\mu = \xi \kappa^\mu x^{\mu(\lambda+\rho)}$ is the spillover rate. Thus the law of motion for θ_t is:

$$d\theta_t = \frac{1}{\omega} \left(\delta + \omega \xi (\gamma x^\rho)^\mu L_{E,t}^{\alpha\sigma} - x^{1-\omega+\rho\omega} L_{E,t}^\alpha e^{\omega\theta_t} \right) dt. \quad (\text{A41})$$

Next, we turn to the evolution of the price-earnings ratio of new firms, ν_t . It is given by:

$$\nu_t = \int_0^{+\infty} e^{-\int_0^s (r_{t+u} + \delta) du} \frac{A_{t+s}}{A_t} ds, \quad (\text{A42})$$

where:

$$\begin{aligned} r_t dt &\equiv \frac{dC_t}{C_t} + \eta dt = \frac{dY_t}{Y_t} + \eta dt \\ A_t &\equiv (1 - \zeta\chi) Y_t N_t^{-\omega} \end{aligned} \quad (\text{A43})$$

The last expression can be obtained from Equation (A33) combined with Equation (A32). This can also be written

as:

$$\nu_t = \int_0^{+\infty} e^{-(\eta+\delta)s} \frac{A_{t+s}/Y_{t+s}}{A_t/Y_t} ds = N_t^\omega \underbrace{\int_0^{+\infty} e^{-(\eta+\delta)s} N_{t+s}^{-\omega} ds}_{\equiv h(N_t)}. \quad (\text{A44})$$

The function $h(N_t)$ is a solution to the ODE:

$$(\delta + \eta)h(N_t) = N_t^{-\omega} + \frac{dN_t}{dt}h'(N_t). \quad (\text{A45})$$

We then guess and verify that θ_t is a sufficient state for any stationary variable. In this case, we can write $\nu_t = \nu(\theta_t)$. Then $N_t^{-\omega}\nu(\theta_t) = h(N_t)$. Manipulation of the ODE satisfied by $h(N_t)$ then shows that $\nu(\theta_t)$ follows:

$$(\eta + \delta)\nu(\theta_t) = 1 - \omega \frac{dN_t}{dt}\nu(\theta_t) + \frac{d\theta_t}{dt}\nu'(\theta_t), \quad (\text{A46})$$

or after substituting the drifts and re-arranging,

$$\eta\nu(\theta_t) = 1 - x^{1-\omega+\rho\omega} L_{E,t}^\alpha e^{\omega\theta_t}\nu(\theta_t) + \frac{1}{\omega} \left(\delta + \omega \xi (\gamma x^\rho)^\mu L_{E,t}^{\alpha\sigma} - x^{1-\omega+\rho\omega} L_{E,t}^\alpha e^{\omega\theta_t} \right) \nu'(\theta_t). \quad (\text{A47})$$

The system of equations (A47) and (A37) is then sufficient to characterize all stationary variables in the model, verifying our guess that θ_t is a sufficient statistic. The existence and uniqueness of a balanced growth path of this system, in which θ_t is constant in calendar time, is established in the proof of Result 1 (Appendix A.5).

A.5 Appendix to Section 4

A.5.1 Balanced growth path

Proof of Result 1. Throughout, we write $\gamma \equiv \gamma(x; \lambda)$ and $\bar{\xi} \equiv \xi(\gamma x^\rho)^\mu$ for the spillover rate. Since x is pinned down by structural parameters via (36), both γ and $\bar{\xi}$ are constants that do not depend on the state of the economy.

Suppose a balanced growth path exists. Then the drift of θ_t in (57) must be zero:

$$\delta + \omega \bar{\xi} L_E^{\alpha\sigma} = x^{1-\omega+\rho\omega} L_E^\alpha e^{\omega\theta}. \quad (\text{A48})$$

The free-entry condition (55) must hold with constant L_E :

$$\frac{L_E^{1-\alpha}}{1 - L_E} = \frac{1 - \zeta\chi}{\zeta\chi} (1 - \gamma) x^{1-(1-\rho)\omega} \nu e^{\omega\theta}. \quad (\text{A49})$$

The price-to-earnings ratio from (58) with $\nu'(\theta) = 0$ gives:

$$\eta\nu = 1 - x^{1-\omega+\rho\omega} L_E^\alpha e^{\omega\theta} \nu. \quad (\text{A50})$$

Combining these three conditions and eliminating θ and ν , we obtain the following system:

$$\frac{\zeta}{1-\zeta} \frac{L_E}{1-L_E} = \frac{1-\gamma}{\omega} \frac{\delta + \omega \bar{\xi} L_E^{\alpha\sigma}}{\eta + \delta + \omega \bar{\xi} L_E^{\alpha\sigma}} \quad (\text{A51})$$

$$\nu = \frac{1}{\eta + \delta + \omega \bar{\xi} L_E^{\alpha\sigma}} \quad (\text{A52})$$

$$\theta = \frac{1}{\omega} \log \left(\frac{\delta + \omega \bar{\xi} L_E^{\alpha\sigma}}{x^{1-\omega+\rho\omega} L_E^{\alpha\sigma}} \right) \quad (\text{A53})$$

$$g = (1-\zeta) \bar{\xi} L_E^{\alpha\sigma}. \quad (\text{A54})$$

Since γ and $\bar{\xi}$ are independent of L_E , the BGP exists if and only if (A51) has a solution $L_E \in (0, 1)$. Define

$$l(L_E) \equiv \frac{\zeta}{1-\zeta} \frac{L_E}{1-L_E}, \quad (\text{A55})$$

the left-hand side of (A51), and let $r(L_E)$ denote its right-hand side.

The function l is strictly increasing and strictly convex, with $l(0) = 0$ and $l(L_E) \rightarrow +\infty$ as $L_E \rightarrow 1^-$.

The function r is strictly increasing, with:

$$r(0) = \frac{1-\gamma}{\omega} \frac{\delta}{\eta + \delta} > 0, \quad r(1) = \frac{1-\gamma}{\omega} \frac{\delta + \omega \bar{\xi}}{\eta + \delta + \omega \bar{\xi}} < +\infty, \quad (\text{A56})$$

where we use $\gamma \in (0, 1)$. Since $l(0) < r(0)$ and $l \rightarrow +\infty$ while r is bounded, at least one crossing exists.

For uniqueness, note that $\alpha\sigma \leq 1$ implies r is concave. Therefore $\phi(L_E) \equiv l(L_E) - r(L_E)$ is strictly convex. A strictly convex function with $\phi(0) < 0$ and $\phi(L_E) \rightarrow +\infty$ as $L_E \rightarrow 1^-$ has exactly one zero. This is the unique BGP.

Finally, if $\gamma = 1$ then $r \equiv 0$, so $L_E = 0$ and the BGP has zero growth. If $\bar{\xi} = 0$, entry is positive but growth is zero by (A54). ■

A.5.2 Comparative statics

A.5.2.1 Entry and growth in the baseline model ($\mu = 0$)

Appendix result 3 (Comparative statics with respect to ρ and λ when $\mu = 0$) Suppose that $\gamma(x; \lambda)$ is given by the functional form in Equation (34) and that $\mu = 0$, so that the spillover rate $\bar{\xi} = \xi$ does not depend on scope. Then, for any $\omega \in (0, 1]$, across BGPs,

$$\frac{\partial L_E}{\partial \rho} < 0, \quad \frac{\partial L_E}{\partial \lambda} \geq 0, \quad (\text{A57})$$

where the last inequality is strict except if $\omega = 1$ and $\rho = 0$. As a result,

$$\frac{\partial g}{\partial \rho} < 0, \quad \frac{\partial g}{\partial \lambda} \geq 0. \quad (\text{A58})$$

Proof of Appendix Result 3. When $\mu = 0$, the spillover rate $\bar{\xi} = \xi$ is independent of ρ and λ . The only

channel through which ρ and λ affect the BGP is through γ . Using the implicit function theorem on (A51):

$$\frac{\partial L_E}{\partial \gamma} = - \frac{\frac{1}{\omega} \frac{\delta + \omega \xi L_E^{\alpha\sigma}}{\eta + \delta + \omega \xi L_E^{\alpha\sigma}}}{\frac{\zeta}{1 - \zeta} \frac{1}{(1 - L_E)^2} - \frac{1 - \gamma}{\omega} \frac{\eta \alpha \sigma \omega \xi L_E^{\alpha\sigma - 1}}{(\eta + \delta + \omega \xi L_E^{\alpha\sigma})^2}}. \quad (\text{A59})$$

The denominator equals $\phi'(L_E) = l'(L_E) - r'(L_E)$. Since ϕ is strictly convex and crosses zero from below at the BGP, $\phi'(L_E) > 0$ at the BGP. The numerator is strictly positive. Therefore $\partial L_E / \partial \gamma < 0$.

When $\gamma(x; \lambda) = \kappa x^\lambda$ with optimal scope from (36):

$$\gamma = \frac{1 - \omega(1 - \rho)}{1 - \omega(1 - \rho) + \lambda}. \quad (\text{A60})$$

Thus, for any $\omega > 0$:

$$\frac{\partial \gamma}{\partial \rho} = \frac{\omega \lambda}{(1 - \omega(1 - \rho) + \lambda)^2} > 0, \quad \frac{\partial \gamma}{\partial \lambda} = - \frac{1 - \omega(1 - \rho)}{(1 - \omega(1 - \rho) + \lambda)^2} \leq 0, \quad (\text{A61})$$

where the last inequality is zero if and only if $\omega = 1$ and $\rho = 0$. The result follows by the chain rule, noting that when $\mu = 0$, $g = (1 - \zeta)\xi L_E^{\alpha\sigma}$ moves in the same direction as L_E . ■

A.5.2.2 Non-monotonic relationship between growth and replicability ($\mu > 0$)

When $\mu > 0$, the spillover rate $\bar{\xi} = \xi(\gamma x^\rho)^\mu = \xi \kappa^\mu x^{\mu(\lambda + \rho)}$ depends on ρ through both γ and x . We define $\beta \equiv \mu(\lambda + \rho)$. The BGP conditions from (A51)–(A54) continue to hold with $\bar{\xi} = \xi \kappa^\mu x^\beta$, and optimal scope and the managerial share are:

$$x = \left(\frac{1}{\kappa} \frac{1 - \omega(1 - \rho)}{1 - \omega(1 - \rho) + \lambda} \right)^{\frac{1}{\lambda}} \quad (\text{A62})$$

$$\gamma = \frac{1 - \omega(1 - \rho)}{1 - \omega(1 - \rho) + \lambda}, \quad (\text{A63})$$

where we assume $\kappa < (1 - \omega) / (1 - \omega + \lambda)$ so that $x > 1$ for all $\rho \in [0, 1]$. Since $(x, \gamma, \bar{\xi})$ remain independent of L_E , the proof of BGP existence and uniqueness is unchanged.

Appendix result 4 (Non-monotonic relationship between growth and replicability) *Under the knowledge spillover specification with $\xi \kappa^\mu x^{\mu(\lambda + \rho)}$ (corresponding to an effective $\beta = \mu(\lambda + \rho)$), suppose that $\delta / \eta > (1 - \omega + \lambda) / \omega$, and that:*

$$\frac{\zeta(\eta + \delta)\omega(1 - \omega)(1 - \omega + \lambda)}{(1 - \zeta)\lambda\delta + \zeta(\eta + \delta)\omega(1 - \omega + \lambda)} \leq \frac{\mu(\lambda + \rho)}{\alpha\sigma} \leq \frac{\zeta\omega(\lambda + 1)}{(1 - \zeta)\lambda + \zeta\omega(\lambda + 1)}. \quad (\text{A64})$$

The parameter $\mu \in (0, 1]$ governs the concavity of knowledge diffusion; lower μ makes the sufficient condition easier to

satisfy. Then growth g in the BGP is a non-monotonic function of ρ : there exists at least one $\hat{\rho} \in (0, 1)$ such that

$$\frac{\partial g}{\partial \rho}(\hat{\rho}) = 0. \quad (\text{A65})$$

Proof of Appendix Result 4. We reformulate the first two equations as:

$$\frac{\zeta}{1 - \zeta} \frac{Z^{\frac{1}{\alpha\sigma}}}{x^{\frac{\beta}{\alpha\sigma}} - Z^{\frac{1}{\alpha\sigma}}} = \frac{1 - \gamma}{\omega} (1 - \eta\nu) \quad (\text{A66})$$

$$\nu = \frac{1}{\eta + \delta + \omega\xi Z}, \quad (\text{A67})$$

using the change of variable $Z = x^\beta L_E^{\alpha\sigma}$, noting that $g = (1 - \zeta)\xi Z$. Using the implicit function theorem, we have that:

$$\frac{dZ}{d\rho} = -\frac{\partial\gamma/\partial\rho}{F_Z} \frac{\zeta}{1 - \zeta} \frac{Z^{\frac{1}{\alpha\sigma}}}{\left(x^{\frac{\beta}{\alpha\sigma}} - Z^{\frac{1}{\alpha\sigma}}\right)^2} \left(\frac{x^{\frac{\beta}{\alpha\sigma}} - Z^{\frac{1}{\alpha\sigma}}}{1 - \gamma} - \frac{\beta}{\lambda\alpha\sigma} \frac{x^{\frac{\beta}{\alpha\sigma}}}{\gamma} \right), \quad (\text{A68})$$

where:

$$F = \frac{\zeta}{1 - \zeta} \frac{Z^{\frac{1}{\alpha\sigma}}}{x^{\frac{\beta}{\alpha\sigma}} - Z^{\frac{1}{\alpha\sigma}}} - \frac{1 - \gamma}{\omega} (1 - \eta\nu), \quad (\text{A69})$$

$$F_Z = \frac{\zeta}{1 - \zeta} \frac{1}{\alpha\sigma} \frac{x^{\frac{\beta}{\alpha\sigma}} Z^{\frac{1}{\alpha\sigma}}}{\left(x^{\frac{\beta}{\alpha\sigma}} - Z^{\frac{1}{\alpha\sigma}}\right)^2} - (1 - \gamma)\eta\xi\nu^2. \quad (\text{A70})$$

For the same reason as in the baseline model, F is convex in L_E and crosses zero from below at the BGP value, so $F_Z > 0$. Since $\partial\gamma/\partial\rho > 0$ and $F_Z > 0$, the leading negative sign in the expression for $dZ/d\rho$ implies that $dZ/d\rho$ has the opposite sign of the parenthetical term. Define

$$\Delta(\rho) \equiv 1 - \left(\frac{Z}{x^\beta}\right)^{\frac{1}{\alpha\sigma}} - \frac{1 - \gamma}{\gamma} \frac{\beta}{\lambda\alpha\sigma} = 1 - L_E - \frac{1}{1 - \omega(1 - \rho)} \frac{\beta}{\alpha\sigma}. \quad (\text{A71})$$

Then $dZ/d\rho$ has the same sign as $-\Delta(\rho)$. We show that $\Delta(0) < 0$ and $\Delta(1) > 0$. This implies $dZ/d\rho > 0$ at $\rho = 0$ and $dZ/d\rho < 0$ at $\rho = 1$. By continuity, $g(\rho) = (1 - \zeta)\xi Z(\rho)$ has an interior critical point.

We establish these two sign conditions in turn. First, note that L_E can always be written as:

$$L_E = \frac{(1 - \zeta)\Lambda}{(1 - \zeta)\Lambda + \zeta}, \quad \Lambda \equiv \frac{1 - \gamma}{\omega} (1 - \eta\nu). \quad (\text{A72})$$

Moreover,

$$\frac{\delta}{\eta + \delta} < 1 - \eta\nu < 1. \quad (\text{A73})$$

Therefore,

$$\frac{(1 - \zeta)(1 - \gamma)\delta}{(1 - \zeta)(1 - \gamma)\delta + \zeta(\eta + \delta)\omega} < L_E < \frac{(1 - \zeta)(1 - \gamma)}{(1 - \zeta)(1 - \gamma) + \zeta\omega} \quad (\text{A74})$$

For $\Delta(0) < 0$, we need

$$L_E > 1 - \frac{1}{1-\omega} \frac{\beta}{\alpha\sigma} \equiv \underline{L}_E. \quad (\text{A75})$$

Since L_E is bounded below by the lower bound above, it is sufficient that:

$$\underline{L}_E \leq \frac{(1-\zeta)(1-\gamma)\delta}{(1-\zeta)(1-\gamma)\delta + \zeta(\eta+\delta)\omega}, \quad (\text{A76})$$

which is equivalent to:

$$\begin{aligned} \beta &\geq \frac{(\eta+\delta)\zeta\omega(1-\omega)}{(1-\zeta)(1-\gamma)\delta + \zeta(\eta+\delta)\omega} \alpha\sigma \\ &= \frac{\zeta(\eta+\delta)\omega(1-\omega)(1-\omega+\lambda)}{(1-\zeta)\lambda\delta + \zeta(\eta+\delta)\omega(1-\omega+\lambda)} \alpha\sigma \equiv \bar{\beta} \end{aligned} \quad (\text{A77})$$

Second, we need to show that when $\rho = 1$, $\Delta(1) > 0$, i.e.,

$$L_E < 1 - \frac{\beta}{\alpha\sigma} \equiv \bar{L}_E. \quad (\text{A78})$$

Since L_E is bounded above by $(1-\zeta)(1-\gamma)/((1-\zeta)(1-\gamma) + \zeta\omega)$ from the inequality above, it is sufficient that:

$$\frac{(1-\zeta)(1-\gamma)}{(1-\zeta)(1-\gamma) + \zeta\omega} \leq \bar{L}_E, \quad (\text{A79})$$

which is equivalent to:

$$\begin{aligned} \beta &\leq \frac{\zeta\omega}{(1-\zeta)(1-\gamma) + \zeta\omega} \alpha\sigma \\ &= \frac{\zeta\omega(\lambda+1)}{(1-\zeta)\lambda + \zeta\omega(\lambda+1)} \alpha\sigma \equiv \underline{\beta}. \end{aligned} \quad (\text{A80})$$

A sufficient condition for $\underline{\beta} < \bar{\beta}$ is:

$$\frac{\delta}{\eta} > \frac{1-\omega+\lambda}{\omega}, \quad (\text{A81})$$

establishing the result. ■

Proof of Result 2. Result 2 states that under appropriate conditions on μ and other parameters, $d \log g / d\rho|_{\rho=0} > 0$ and $d \log g / d\rho|_{\rho=1} < 0$, so that $g(\rho)$ has an interior maximum. This follows directly from Appendix Result 4. The sufficient conditions stated there ensure that the function $\Delta(\rho)$ defined in the proof above satisfies $\Delta(0) < 0$ and $\Delta(1) > 0$. Since $dZ/d\rho$ has the same sign as $-\Delta(\rho)$, and $g = (1-\zeta)\xi Z$, growth is increasing at $\rho = 0$ and decreasing at $\rho = 1$. By the intermediate value theorem, there exists at least one $\hat{\rho} \in (0, 1)$ at which $dg/d\rho = 0$. ■

A.5.3 Key economic quantities in the model

A.5.3.1 Capital and measured total factor productivity

Proof of Result 3, part (i). The law of motion for capital K_t measured using the perpetual inventory method is:

$$\frac{dK_t}{K_t} = \left(\frac{n_t}{K_t} L_{E,t}^\alpha - \delta \right) dt. \quad (\text{A82})$$

The law of motion for the model-consistent index of capital services N_t is:

$$\frac{dN_t}{N_t} = \frac{1}{\omega} \left(\left(\frac{n_t}{N_t} \right)^\omega x_t^{1-\omega+\rho\omega} L_{E,t}^\alpha - \delta \right) dt. \quad (\text{A83})$$

Using $Z_t = (N_t / K_t)^{1-\zeta}$, the law of motion for measured TFP is:

$$\frac{dZ_t}{Z_t} = (1-\zeta) \left(\frac{1}{\omega} \left(e^{\omega\theta_t} x_t^{1-\omega+\rho\omega} L_{E,t}^\alpha - \delta \right) - \left(e^{\theta_t} L_{E,t}^\alpha Z_t^{\frac{1}{1-\zeta}} - \delta \right) \right) dt. \quad (\text{A84})$$

On a BGP with $d\theta_t/dt = 0$, this becomes:

$$\frac{1}{1-\zeta} \frac{dZ_t}{Z_t} = \left(\overbrace{\frac{1}{\omega} \left(e^{\omega\theta} x^{1-\omega+\rho\omega} L_E^\alpha - \delta \right)}^{=g_N = \bar{\xi} L_E^{\alpha\sigma}} - \left(e^\theta L_E^\alpha Z_t^{\frac{1}{1-\zeta}} - \delta \right) \right) dt. \quad (\text{A85})$$

Because $\delta + g_N > 0$, Z_t converges monotonically from any initial condition to

$$Z = \left(\frac{\delta + g_N}{e^\theta L_E^\alpha} \right)^{1-\zeta}. \quad (\text{A86})$$

On the BGP, (A53) implies

$$e^{\omega\theta} = \frac{\delta + \omega g_N}{x^{1-\omega+\rho\omega} L_E^\alpha}, \quad (\text{A87})$$

hence

$$e^\theta = \left(\frac{\delta + \omega g_N}{x^{1-\omega+\rho\omega} L_E^\alpha} \right)^{1/\omega}. \quad (\text{A88})$$

Substituting into (A86) yields

$$Z = x^{\frac{(1-\zeta)(1-\omega+\rho\omega)}{\omega}} \left(\frac{\delta + g_N}{\delta + \omega g_N} \right)^{1-\zeta} \left(\frac{L_E^\alpha}{\delta + \omega g_N} \right)^{\frac{1-\omega}{\omega}(1-\zeta)}, \quad (\text{A89})$$

which is (66).

When $\omega = 1$, $Z = x^{(1-\zeta)\rho}$. Taking the log and differentiating:

$$\frac{d \log Z}{d\rho} = (1-\zeta) \left(\rho \frac{d \log x}{d\rho} + \log x \right) > 0, \quad (\text{A90})$$

since $x > 1$ and x is increasing in ρ .

For $\omega < 1$, the BGP level of TFP can be written compactly as:

$$\frac{\log Z}{1 - \zeta} = \frac{1 - \omega + \rho\omega}{\omega} \log x + \log(\delta + g_N) - \frac{1}{\omega} \log(\delta + \omega g_N) - \frac{\alpha(1 - \omega)}{\omega} \log L_E. \quad (\text{A91})$$

The first term is strictly increasing in ρ (both the coefficient $(1 - \omega + \rho\omega)/\omega$ and $\log x$ rise). The second and third terms combine to $\log[(\delta + g_N)/(\delta + \omega g_N)^{1/\omega}]$, which is bounded and varies with g_N . The fourth term is negative and decreasing in ρ . A sufficient condition for Z to be increasing in ρ is that the first term dominates. This holds when markups are small (ω close to 1), because the last two terms vanish as $\omega \rightarrow 1$ while the first term remains strictly positive. By continuity, Z is increasing in ρ for all ω in a neighborhood of 1. ■

A.5.3.2 Profit share, Tobin's Q , and comparative statics

This subsection defines the pure profit share and Tobin's Q as functions of the state of the economy, and derives their closed-form expressions, both as state-space identities (off and on the BGP) and on a balanced growth path. The economic state at t is summarized by (N_t, K_t, θ_t) together with the parameter ρ_t , with optimal scope and the managerial share given in closed form by $x_t = \mathbf{x}(\rho_t)$ and $\gamma_t = \kappa x_t^\lambda$, and the value multiple $\nu_t = \nu(\theta_t, t)$ and entry rate $L_{E,t} = L_E(\theta_t, t)$ given by the dynamic system in Appendix A.6.2.1. Throughout, $g_{N,t} \equiv d \log N_t / dt$ denotes the growth rate of the productive capital index, $g_t \equiv d \log Y_t / dt$ output growth, and $r_t \equiv \eta + g_t$ the real interest rate.

Definitions. The replacement cost per unit of intangible capital is

$$P_{N,t} = \frac{W_t L_{E,t}^{1-\alpha}}{n_t}, \quad W_t = \chi \zeta \left(\frac{N_t}{1 - L_{E,t}} \right)^{1-\zeta} = \frac{\zeta \chi}{L_{Y,t}} Y_t, \quad (\text{A92})$$

where $n_t = e^{\theta_t} N_t$ is the intangible stock of the marginal entrant. The measured capital stock is

$$K_t = \int_{s \leq t} e^{-\delta(t-s)} L_{E,s}^\alpha n_s ds. \quad (\text{A93})$$

The user cost of capital is

$$R_t \equiv r_t + \delta - \frac{d \log P_{N,t}}{dt}. \quad (\text{A94})$$

Competitive capital income relative to output is $C_{K,t} \equiv R_t P_{N,t} K_t / Y_t$, and the pure profit share is

$$\Pi_t \equiv (1 - \zeta \chi) - C_{K,t}. \quad (\text{A95})$$

Enterprise value and Tobin's Q are

$$V_t = \nu_t (1 - \zeta \chi) Y_t, \quad Q_t \equiv \frac{V_t}{P_{N,t} K_t}. \quad (\text{A96})$$

Equations (A92)–(A96) hold at any feasible state $(N_t, K_t, \theta_t, \rho_t)$ supported by the equilibrium conditions of Appendix A.6.2.1; they do not require $d\theta_t/dt = 0$.

Anticipating the transitional dynamics analysis, another expression for Q_t can be obtained that will prove useful for the computation of transitional dynamics. Substituting (A92) into the definition of Q_t in (A96) and using $L_{Y,t} = 1 - L_{E,t}$ and $n_t = e^{\theta_t} N_t$,

$$Q_t = \frac{V_t}{P_{N,t} K_t} = \frac{\nu_t (1 - \zeta \chi) Y_t}{\frac{\zeta \chi}{L_{Y,t}} Y_t \cdot \frac{L_{E,t}^{1-\alpha}}{n_t} K_t} = \frac{(1 - \zeta \chi) \nu_t L_{Y,t} n_t}{\zeta \chi L_{E,t}^{1-\alpha} K_t}, \quad (\text{A97})$$

where the aggregate output Y_t cancels exactly. The free-entry first-order condition that holds at any state is

$$\frac{L_{E,t}^{1-\alpha}}{1 - L_{E,t}} = \frac{1 - \zeta \chi}{\zeta \chi} (1 - \gamma) x^\psi \nu_t e^{\omega \theta_t}, \quad \psi \equiv 1 - \omega + \omega \rho, \quad (\text{A98})$$

which on the BGP reduces to (A104) below. Therefore:

$$Q_t = \frac{N_t e^{(1-\omega)\theta_t}}{K_t (1 - \gamma) x^\psi}. \quad (\text{A99})$$

Specialization to a balanced growth path. On a BGP, L_E , γ , x , and ρ are constant, $g_{N,t} = g_N = \bar{\xi} L_E^{\alpha\sigma}$, and $d\theta_t/dt = 0$. We drop time subscripts and write $g = (1 - \zeta)g_N$. The measured stock satisfies

$$K_t = \frac{L_E^\alpha n_t}{g_N + \delta}, \quad (\text{A100})$$

so

$$\frac{P_N K}{Y} = \frac{\zeta \chi L_E}{(1 - L_E)(g_N + \delta)}. \quad (\text{A101})$$

The user cost (A94) simplifies because $P_{N,t}$ falls at rate ζg_N while wages grow at rate g :

$$R = \eta + \delta + g_N. \quad (\text{A102})$$

Labor income is $W L_Y = \zeta \chi Y$ and capital income is $(1 - \zeta \chi) Y$. Competitive capital income relative to output is

$$C_K = \frac{(\eta + \delta + g_N) \zeta \chi L_E}{(1 - L_E)(g_N + \delta)}. \quad (\text{A103})$$

Using the BGP free-entry condition (A51), which implies

$$\frac{\zeta \chi L_E}{(1 - \zeta \chi)(1 - L_E)} = (1 - \gamma) \frac{\delta + \omega g_N}{\eta + \delta + \omega g_N}, \quad (\text{A104})$$

and substituting (A102) into (A103):

$$C_K = (1 - \zeta \chi)(1 - \gamma) \mathcal{A}, \quad \mathcal{A} \equiv \frac{(\eta + \delta + g_N)(\delta + \omega g_N)}{(\eta + \delta + \omega g_N)(g_N + \delta)}. \quad (\text{A105})$$

The pure profit share (A95) becomes

$$\Pi = (1 - \zeta\chi) - C_K = (1 - \zeta\chi) \left[\gamma + (1 - \gamma)(1 - \mathcal{A}) \right]. \quad (\text{A106})$$

Expanding $1 - \mathcal{A}$ on the BGP yields

$$1 - \mathcal{A} = \frac{\eta(1 - \omega) g_N}{(g_N + \delta)(\eta + \delta + \omega g_N)}. \quad (\text{A107})$$

Substituting (A107) into (A106) gives

$$\Pi = (1 - \zeta\chi) \left[\gamma + (1 - \gamma) \frac{\eta(1 - \omega) g_N}{(g_N + \delta)(\eta + \delta + \omega g_N)} \right]. \quad (\text{A108})$$

The first term (γ) captures rents accruing to managers from the hold-up friction. The second term captures markup rents ($\omega < 1$), which accrue to entrepreneurs.

On the BGP, (A96) and (A101) imply

$$Q = \frac{\nu(1 - \zeta\chi)(g_N + \delta)}{\zeta\chi L_E} \frac{1 - L_E}{\eta + \delta + \omega g_N}, \quad (\text{A109})$$

where we used $\nu = 1/(\eta + \delta + \omega g_N)$ from (A52). Substituting (A104) to eliminate $(1 - L_E)/L_E$:

$$Q = \frac{1}{1 - \gamma} \cdot \frac{g_N + \delta}{\omega g_N + \delta}. \quad (\text{A110})$$

When $\omega = 1$ (no markups), the second factor equals one and $Q = 1/(1 - \gamma)$: Tobin's Q is elevated solely by the hold-up friction. When $\omega < 1$, the second factor exceeds one, reflecting additional rents from markups. Equation (A110) is the BGP specialization of the state-space identity (A99): substituting the BGP relations (A100) and (A87) into (A99) returns (A110).

Proof of Result 3, part (ii). When $\omega = 1$, $\Pi = (1 - \zeta\chi)\gamma$ and $Q = 1/(1 - \gamma)$, both strictly increasing in ρ since γ is increasing in ρ .

For general ω , write $Q(\rho; \omega) = F_1(\rho) F_2(\rho; \omega)$, where

$$F_1 \equiv \frac{1}{1 - \gamma}, \quad F_2 \equiv \frac{g_N + \delta}{\omega g_N + \delta}. \quad (\text{A111})$$

Since γ is increasing in ρ independently of ω , $d \log F_1 / d\rho > 0$. Moreover, $F_2(\rho; 1) = 1$ for all ρ , so $d \log F_2 / d\rho|_{\omega=1} = 0$. Therefore $d \log Q / d\rho|_{\omega=1} = d \log F_1 / d\rho > 0$. Since $d \log Q / d\rho$ is continuous in ω , there exists $\omega^* < 1$ such that $d \log Q / d\rho > 0$ for all $\omega \in (\omega^*, 1]$ and all $\rho \in [0, 1]$.

The same argument applies to Π . At $\omega = 1$, $d\Pi/d\rho = (1 - \zeta\chi) d\gamma/d\rho > 0$. By continuity in ω , Π is increasing in ρ for ω sufficiently close to 1. ■

A.5.3.3 Concentration

We derive the Herfindahl–Hirschman index of firm sales on the BGP. A firm born at date t_e that survives to date t produces x varieties, each with intangible input $n(m) = x^{-(1-\rho)} n_{t_e}$. Its total revenue is proportional to its total profits $\pi_{t_e}(t) = A_t x^{1-(1-\rho)\omega} n_{t_e}^\omega$. Since all firms face the same markup, a firm's sales share equals its share of aggregate profits:

$$s_{t_e}(t) = \frac{x^{1-(1-\rho)\omega} n_{t_e}^\omega}{N_t^\omega}. \quad (\text{A112})$$

On the BGP, $n_{t_e} = \Xi_{t_e}$ and $\Xi_{t_e}/N_{t_e} = e^\theta$ is constant. Using $N_{t_e}/N_t = e^{-g_N(t-t_e)}$:

$$s_{t_e}(t) = c e^{-\omega g_N(t-t_e)}, \quad c \equiv x^{1-(1-\rho)\omega} e^{\omega\theta}. \quad (\text{A113})$$

The term c is the sales share of the youngest cohort. Summing shares across all surviving firms and using the fact that L_E^α firms are born per unit of time, each surviving with probability $e^{-\delta(t-t_e)}$:

$$1 = \int_{-\infty}^t L_E^\alpha e^{-\delta(t-t_e)} s_{t_e}(t) dt_e = \frac{L_E^\alpha c}{\delta + \omega g_N}. \quad (\text{A114})$$

Solving: $c = (\delta + \omega g_N)/L_E^\alpha$. The Herfindahl–Hirschman index is:

$$H = \int_{-\infty}^t L_E^\alpha e^{-\delta(t-t_e)} s_{t_e}(t)^2 dt_e = \frac{L_E^\alpha c^2}{\delta + 2\omega g_N} = \frac{(\delta + \omega g_N)^2}{L_E^\alpha (\delta + 2\omega g_N)}, \quad (\text{A115})$$

where $g_N = \bar{\xi} L_E^{\alpha\sigma}$ is the BGP growth rate of intangible capital and $\bar{\xi} \equiv \xi(\gamma x^\rho)^\mu$ is the spillover rate.

Proof of Result 3, part (iii). When $\omega = 1$, the index (A115) reduces to

$$H = \frac{(\delta + g_N)^2}{L_E^\alpha (\delta + 2g_N)}. \quad (\text{A116})$$

Since $g_N = \bar{\xi} L_E^{\alpha\sigma}$ and $\bar{\xi} = \xi(\gamma x^\rho)^\mu$, this equals $h(L_E)$, where

$$h(L_E) \equiv \frac{(\delta + \bar{\xi} L_E^{\alpha\sigma})^2}{L_E^\alpha (\delta + 2\bar{\xi} L_E^{\alpha\sigma})}. \quad (\text{A117})$$

Direct computation shows $dh/dL_E < 0$: the denominator shrinks faster than the numerator as L_E falls. Since L_E is decreasing in ρ , H is increasing in ρ . When $\mu > 0$, $\bar{\xi}$ also increases with ρ , reinforcing the effect.

For general ω , the same structure applies: L_E^α enters the denominator, and the function $(\delta + \omega z)^2/(\delta + 2\omega z)$ is increasing in z for $z \geq 0$. At $\omega = 1$, $dH/d\rho > 0$ as shown above. By continuity in ω , H is increasing in ρ for ω sufficiently close to 1. ■

A.6 Appendix to Section 5

This appendix collects the implementation details behind the numerical illustration of Section 5: the procedure used to select parameter values, the algorithm used to solve the model globally and along the transition, and the

cohort accounting used to construct the aggregate transition-path moments.

A.6.1 Parameter selection procedure

This subsection details the procedure used in Section 5 to assign parameter values to the 1990 and 2020 balanced growth paths (BGPs). The key principle is *cross-BGP identification*: given the four data inputs $(\gamma_{1990}, \gamma_{2020}, x_{1990}, d)$ where $d \equiv \log(x_{2020}/x_{1990})$, together with the 1990 output growth rate g_{1990} , the five free internal parameters $\{\rho_{1990}, \rho_{2020}, \lambda, \kappa, \xi\}$ are recovered in closed form (up to a scalar root-finding step for ξ). The auxiliary parameter μ is not separately identified from BGP moments and is fixed at $\mu = 0$ in the baseline.

A.6.1.1 Data targets

This subsection describes how each of the five empirical inputs $(\gamma_{1990}, \gamma_{2020}, x_{1990}, d, g_{1990})$ is constructed and where in the paper the underlying time series can be inspected.

Managerial equity share, γ_{1990} and γ_{2020} . The model has only two equity-holding groups, managers and entrepreneurs, with no outsider equity. The managerial share is $\gamma = m/(m + e)$, where m and e are the equity stakes of managers and entrepreneurs respectively. The closest direct measure in the data is the equity-based compensation series of Eisfeldt et al. (2021), who report the dollar value of reserved shares (RS) for employee equity-based compensation scaled by aggregate market capitalization. Their Table 1, Panel B reports decade averages of RS/MktCap for the manufacturing sector:

Decade	1980–89	1990–99	2000–09	2010–19
RS / MktCap (%)	4	7	12	10

Their measure differs from the model’s γ because their denominator includes outsider equity:

$$\text{RS/MktCap} = \frac{m}{m + e + o}, \quad \gamma = \frac{m}{m + e}, \quad (\text{A118})$$

where o denotes outsider equity. Assuming all equity-based compensation accrues to managers in the model, the bridge between the two measures runs through the share of equity held by insiders,

$$\gamma = \frac{\text{RS/MktCap}}{s^{\text{insider}}}, \quad s^{\text{insider}} \equiv \frac{m + e}{m + e + o}. \quad (\text{A119})$$

We proxy s^{insider} using the post-IPO insider equity share collected from SEC IPO filings; Appendix A.1.3 documents the construction in detail. Decade averages of the post-IPO insider share, visible in Figure A-1, are approximately 46% over 1996–99 and 33% over 2010–19. Combining decade averages of RS/MktCap with decade averages of s^{insider} gives

$$\gamma_{1990} = \frac{0.07}{0.46} \approx 0.15, \quad \gamma_{2020} = \frac{0.10}{0.33} \approx 0.30, \quad (\text{A120})$$

and $\Delta \log \gamma = \log(0.30/0.15) \approx 0.69$. We anchor “1990” to the 1990s decade rather than the 1980s because

the IPO insider data starts in 1996. Newly-public firms typically have higher insider ownership than seasoned firms, so this proxy likely overstates s^{insider} at the listed-firm level and hence understates the rescaled γ ; our γ targets should be read as a lower bound on the model object.

Average firm scope, x_{1990} and d . The model's x_t is the scope at which firms deploy intangible capital across distinct varieties. We use the average number of establishments per firm in the NETS dataset, restricted to firms successfully matched to non-financial Compustat. The construction of the establishment-per-firm variable from raw NETS records (including the splice for the structural breaks in 2011 and 2015) is described in Appendix A.1.1; the Compustat–NETS matching protocol is described in Appendix A.1.2. Summary statistics for the underlying industry-year panel are reported in Table A-2, and time series of average scope are visible in Figures A-3–A-4 (which display log CBSAs per firm; the establishments-per-firm series follows a similar trend). Averaging establishments per firm over the windows 1988–1992 and 2014–2018 gives $x_{1990} \approx 5.0$ and $x_{2020} \approx 7.5$, so that $d = \log(x_{2020}/x_{1990}) \approx 0.40$.

Output growth, g_{1990} . We measure output growth as the real growth rate of value added in the non-financial corporate business (NFCB) sector, from National Income and Product Accounts (NIPA) Table 1.14 (FRED series B455RX1Q027SBEA). Construction details are in Appendix A.1.4, and the full annual time series is visible in panel (a) of Figure A-2. The 1990 calibration target $g_{1990} = 0.027$ is the average annualized log growth rate over the five-year window 1988–1992.

A.6.1.2 BGP equations

The model features eleven structural parameters, $\{\eta, \zeta, \chi, \delta, \rho, \lambda, \kappa, \alpha, \xi, \sigma, \mu\}$, with derived reduced-form parameter $\omega \equiv (1 - \zeta)\chi / (1 - \zeta\chi)$. In any BGP, the four equations relevant for parameter selection are

$$\text{Optimal scope: } x = \left(\frac{1 - \omega(1 - \rho)}{\kappa(1 - \omega(1 - \rho) + \lambda)} \right)^{1/\lambda}, \quad (\text{A121})$$

$$\text{Managerial share: } \gamma = \kappa x^\lambda, \quad (\text{A122})$$

$$\text{Free entry: } \frac{\zeta}{1 - \zeta} \frac{L_E}{1 - L_E} = \frac{1 - \gamma}{\omega} \frac{\delta + \omega \bar{\xi} L_E^{\alpha\sigma}}{\eta + \delta + \omega \bar{\xi} L_E^{\alpha\sigma}}, \quad (\text{A123})$$

$$\text{Output growth: } g = (1 - \zeta) \xi (\gamma x^\rho)^\mu L_E^{\alpha\sigma}, \quad (\text{A124})$$

where $\bar{\xi} \equiv \xi (\gamma x^\rho)^\mu$ is the spillover rate. Equations (A121)–(A122) are the BGP versions of the time-varying scope and managerial-share functions $x_t = \mathbf{x}(\rho_t)$ and $\gamma_t = \kappa x_t^\lambda$ in Appendix A.6.2.1; we restate them here in BGP form because they are needed in closed form below. Combining (A121) and (A122) yields the inverted relation we use repeatedly,

$$1 - \omega(1 - \rho) = \frac{\gamma \lambda}{1 - \gamma}. \quad (\text{A125})$$

Six of the eleven parameters are set externally as in Panel A of Table 3; ω is derived from (ζ, χ) . The five internal parameters are $\{\rho_{1990}, \rho_{2020}, \lambda, \kappa, \xi\}$, with μ left as a sensitivity parameter.

A.6.1.3 Cross-BGP identification of $(\lambda, \rho_{1990}, \rho_{2020}, \kappa)$

The structural parameters λ and κ are held constant across the two BGPs; only ρ shifts. Let $\Delta \log \gamma \equiv \log(\gamma_{2020}/\gamma_{1990})$.

Identification of λ . Since $\gamma = \kappa x^\lambda$ holds at both BGPs with common (κ, λ) , taking logs and differencing gives

$$\hat{\lambda} = \frac{\Delta \log \gamma}{d}. \quad (\text{A126})$$

Identification of ρ_t . Inverting (A125) and substituting $\hat{\lambda}$ yields

$$\hat{\rho}_t = 1 - \frac{1}{\omega} \left(1 - \frac{\hat{\lambda} \gamma_t}{1 - \gamma_t} \right), \quad t \in \{1990, 2020\}. \quad (\text{A127})$$

Identification of κ . From $\gamma_{1990} = \kappa x_{1990}^\lambda$,

$$\hat{\kappa} = \gamma_{1990} x_{1990}^{-\hat{\lambda}}. \quad (\text{A128})$$

We do *not* impose the normalization $x_{1990} = 1$: the level of x_{1990} enters $\hat{\kappa}$ explicitly. By construction the model exactly matches $\{\gamma_{1990}, \gamma_{2020}, x_{1990}, x_{2020}\}$, so none of these moments serves as an over-identification check.

A.6.1.4 Identification of ξ at $\mu = 0$

With $(\lambda, \kappa, \rho_{1990})$ in hand, fixing $\mu = 0$ collapses the spillover to $\bar{\xi} = \xi$. The 1990 free-entry condition (A123) and the output-growth equation (A124) together imply $\xi L_{E,1990}^{\alpha\sigma} = g_{1990}/(1 - \zeta)$. Substituting into (A123) eliminates ξ and leaves a scalar equation in $L_{E,1990}$,

$$\frac{\zeta}{1 - \zeta} \frac{L_{E,1990}}{1 - L_{E,1990}} = \frac{1 - \gamma_{1990}}{\omega} \frac{\delta + \omega g_{1990}/(1 - \zeta)}{\eta + \delta + \omega g_{1990}/(1 - \zeta)}, \quad (\text{A129})$$

whose right-hand side is a single constant. Because the left-hand side is strictly increasing in $L_{E,1990} \in (0, 1)$, $\hat{L}_{E,1990}$ is the unique root in $(0, 1)$ and is recovered by a one-dimensional bracketing solver. The spillover scale follows residually,

$$\hat{\xi} = \frac{g_{1990}/(1 - \zeta)}{\hat{L}_{E,1990}^{\alpha\sigma}}. \quad (\text{A130})$$

At the calibrated values $(\gamma_{1990}, g_{1990}) = (0.15, 0.027)$ and $(\eta, \delta, \zeta, \omega, \alpha, \sigma) = (0.02, 0.10, 0.70, 0.857, 0.5, 1)$, this delivers $\hat{L}_{E,1990} \approx 0.276$ and $\hat{\xi} \approx 0.171$.

A.6.1.5 Feasibility constraints

The parameter restrictions $\lambda \geq 1$ and $\rho_t \in [0, 1]$ translate into joint restrictions on the data inputs $(\gamma_{1990}, \gamma_{2020}, d, \omega)$. Combining (A125) with the bounds on ρ implies a feasibility window for $\hat{\lambda}$,

$$\max\left(1, \frac{(1 - \omega)(1 - \gamma_{1990})}{\gamma_{1990}}\right) \leq \hat{\lambda} \leq \frac{1 - \gamma_{2020}}{\gamma_{2020}}. \quad (\text{A131})$$

Equivalently, given $(\gamma_{1990}, \gamma_{2020}, \omega)$, the implied d must lie in

$$\frac{\Delta \log \gamma}{(1 - \gamma_{2020})/\gamma_{2020}} \leq d \leq \frac{\Delta \log \gamma}{\max(1, (1 - \omega)(1 - \gamma_{1990})/\gamma_{1990})}. \quad (\text{A132})$$

At the calibration values $\omega = 0.857$, $\gamma_{1990} = 0.15$, $\gamma_{2020} = 0.30$ used in Section 5, (A131) gives $\hat{\lambda} \in [1.00, 2.33]$ and (A132) gives $d \in [0.30, 0.69]$. The selected target $d = 0.40$ sits in the interior.

A.6.1.6 The auxiliary parameter μ

The parameter μ governs the strength of the manager-mediated knowledge spillover, $\bar{\xi} = \xi(\gamma x^\rho)^\mu$. From the cross-BGP identification of Sections A.6.1.3–A.6.1.4, μ is not separately identified from BGP moments at a single date: it enters the 1990 free-entry condition (A123) only through the product $\xi(\gamma_{1990} x_{1990}^{\rho_{1990}})^\mu$, and the same monotone scalar problem in $L_{E,1990}$ pins down $L_{E,1990}$ regardless of the value of μ . The implication is that the calibrated values of $(\hat{\lambda}, \hat{\rho}_{1990}, \hat{\rho}_{2020}, \hat{\kappa})$ and the 1990 model implications for $g, Q, V/Y, \Pi, \text{HHI}$ are all independent of μ . The value of $\hat{\xi}$ that fits 1990 growth scales as

$$\hat{\xi}(\mu) = \frac{\bar{\xi}_{1990}}{(\gamma_{1990} x_{1990}^{\rho_{1990}})^\mu}, \quad \bar{\xi}_{1990} \equiv \frac{g_{1990}}{(1 - \zeta) \hat{L}_{E,1990}^{\alpha\sigma}}, \quad (\text{A133})$$

with $\bar{\xi}_{1990}$ pinned down by the 1990 BGP.

The 2020 BGP *predictions* for non-targeted moments do depend on μ , since the spillover rate scales as

$$\frac{\bar{\xi}_{2020}}{\bar{\xi}_{1990}} = \left(\frac{\gamma_{2020} x_{2020}^{\rho_{2020}}}{\gamma_{1990} x_{1990}^{\rho_{1990}}} \right)^\mu. \quad (\text{A134})$$

At the calibrated values, the ratio inside parentheses is approximately 6.0, so $\bar{\xi}_{2020}/\bar{\xi}_{1990}$ ranges from 1 (at $\mu = 0$) to roughly 6 (at $\mu = 1$). The 2020 prediction for output growth is the most sensitive: since $g \propto \bar{\xi}$ holding L_E fixed, g_{2020} inherits the strong μ -dependence directly. Asset-pricing ratios ($Q, V/Y$) and concentration (HHI) depend on μ only through the joint pair $(L_{E,2020}, g_{N,2020})$, with substantial cancellation; their sensitivity to μ is much milder than that of g .

We set $\mu = 0$ as the baseline. This choice is consistent with the empirical 1990–2020 slowdown in g : amplifying the spillover ($\mu > 0$) makes the model overpredict 2020 growth more, not less. It also avoids tying the calibration to a parameter that BGP moments do not constrain. Predictions across a grid $\mu \in [0, 0.5]$ are reported in Section 5.

A.6.2 Numerical solution method

Outside of balanced growth, we solve the model globally using finite-difference methods on a grid of the sufficient state $\theta_t \equiv \log(\Xi_t/N_t)$. We then use the global solution to construct the transitional dynamics of the economy under a smooth, deterministic path of ρ_t . Throughout this appendix, we write $\nu(\theta, t)$ for the price-to-earnings ratio and $L_E(\theta, t)$ for entrepreneurial effort; in balanced growth, both reduce to time-invariant policy functions of θ alone.

A.6.2.1 The dynamic system in time-drift form

The equilibrium characterization of Section 3.3 reduces, when ρ is allowed to vary over calendar time, to a one-state, one-value-function dynamic system in (θ, t) . Let ρ_t be a known deterministic path; optimal scope x_t and the managerial share γ_t at any date are pinned down from ρ_t alone, through the closed-form expressions

$$x_t = \mathbf{x}(\rho_t) \equiv \left(\frac{1 - \omega(1 - \rho_t)}{\kappa(1 - \omega(1 - \rho_t) + \lambda)} \right)^{1/\lambda}, \quad \gamma_t = \kappa x_t^\lambda. \quad (\text{A135})$$

The spillover rate is

$$\bar{\xi}_t \equiv \xi(\gamma_t x_t^{\rho_t})^\mu. \quad (\text{A136})$$

With $(x_t, \gamma_t, \bar{\xi}_t)$ in hand, the price-to-earnings ratio $\nu(\theta, t)$ and entrepreneurial effort $L_E(\theta, t)$ are characterized by the following system (the BGP form is (58) and (55), augmented here with calendar-time drifts and time-varying parameters):

$$0 = 1 - \left(\eta + s(\theta, t) \right) \nu(\theta, t) + \partial_t \nu(\theta, t) + g_\theta(\theta, t) \partial_\theta \nu(\theta, t), \quad (\text{A137})$$

$$0 = \frac{L_E(\theta, t)^{1-\alpha}}{1 - L_E(\theta, t)} - \Xi(\theta, t), \quad (\text{A138})$$

where the auxiliary functions are

$$s(\theta, t) = x_t^{1-\omega+\rho_t\omega} L_E(\theta, t)^\alpha e^{\omega\theta}, \quad (\text{A139})$$

$$g_\theta(\theta, t) = \frac{1}{\omega} \left(\delta + \omega \bar{\xi}_t L_E(\theta, t)^{\alpha\sigma} - s(\theta, t) \right), \quad (\text{A140})$$

$$\Xi(\theta, t) = \frac{1 - \zeta\chi}{\zeta\chi} (1 - \gamma_t) x_t^{1-\omega+\rho_t\omega} \nu(\theta, t) e^{\omega\theta}. \quad (\text{A141})$$

We added the time drift $\partial_t \nu$ to the BGP-form ODE (58) because we will solve the HJB-like equation (A137) backward in calendar time, starting from a terminal condition at a date T at which ρ_t has settled at its long-run value (so that ν has converged to the corresponding BGP policy function). Equation (A138) is implicit in L_E . Given $\Xi \in (0, +\infty)$, the left-hand side of (A138) is strictly increasing in $L_E \in (0, 1)$ from 0 to $+\infty$, so a unique root exists; for $\alpha = 1/2$ in particular it admits the closed form

$$L_E = \frac{(1 + 2\Xi^2) - \sqrt{1 + 4\Xi^2}}{2\Xi^2} \in (0, 1). \quad (\text{A142})$$

For general α , L_E is recovered by a one-dimensional bracketing solver applied to (A138).

A.6.2.2 Discretization

We compute discrete approximations of the functions $(\nu, L_E, x, \gamma, \bar{\xi})$ at $N_\theta \times N_t$ points of the state-space. Denote these points by (θ_i, t_j) , with $1 \leq i \leq N_\theta$ and $1 \leq j \leq N_t$. The points along the θ -axis are equally spaced by Δ_θ and along the t -axis by Δ_t . Let $\nu_{i,j} \equiv \nu(\theta_i, t_j)$ and $\nu_{:,j}$ the $N_\theta \times 1$ column vector $(\nu_{i,j})_{i=1}^{N_\theta}$

representing the date- t_j iterate; similar notation is used for the other functions. We obtain the global solution by initializing $(\nu_{\cdot, N_t}, L_{E, \cdot, N_t}, x_{\cdot, N_t}, \gamma_{\cdot, N_t}, \bar{\xi}_{\cdot, N_t})$ at the long-run BGP values associated with $\rho_{N_t} \equiv \rho(t_{N_t})$, and then solving backward iteratively in calendar time.

At each time step $j < N_t$, given the date- $(j+1)$ iterates, we make the following three updates successively:

- (a) Update $\nu_{\cdot, j}$ using the date- $(j+1)$ value and policy functions, plus an upwind finite-difference approximation to $\partial_\theta \nu$;
- (b) Update x_j , γ_j , and $\bar{\xi}_j$ from ρ_j via the closed-form expressions (A135)–(A136);
- (c) Update $L_{E, \cdot, j}$ from $\nu_{\cdot, j}$, x_j , and γ_j using (A138).

Update (b) is immediate and does not depend on the spatial grid. Updates (a) and (c) are described next.

Update (a): Computation of $\nu_{\cdot, j}$. We define the forward and backward difference approximations of $\partial_\theta \nu$ as

$$\partial_\theta^{(b)} \nu_{i, j} \equiv \frac{\nu_{i, j} - \nu_{i-1, j}}{\Delta_\theta}, \quad \partial_\theta^{(f)} \nu_{i, j} \equiv \frac{\nu_{i+1, j} - \nu_{i, j}}{\Delta_\theta}. \quad (\text{A143})$$

The finite-difference approximation of $\partial_\theta \nu(\theta_i, t_j)$ with an upwinding scheme is

$$\partial_\theta \nu(\theta_i, t_j) \equiv \mathbb{I}_{\{g_{\theta, i, j+1} < 0\}} \partial_\theta^{(b)} \nu_{i, j} + \mathbb{I}_{\{g_{\theta, i, j+1} \geq 0\}} \partial_\theta^{(f)} \nu_{i, j}, \quad (\text{A144})$$

where $g_{\theta, i, j+1}$ is the drift of θ_t at (θ_i, t_{j+1}) ,

$$g_{\theta, i, j+1} = \frac{1}{\omega} \left(\delta + \omega \bar{\xi}_{j+1} L_{E, i, j+1}^{\alpha\sigma} - x_{j+1}^{1-\omega+\rho_{j+1}\omega} L_{E, i, j+1}^\alpha e^{\omega\theta_i} \right). \quad (\text{A145})$$

Substituting these finite differences into the discretized form of (A137), with backward Euler for $\partial_t \nu$, and using the date- $(j+1)$ iterates $(L_{E, \cdot, j+1}, x_{j+1}, \gamma_{j+1}, \bar{\xi}_{j+1})$ to evaluate the coefficients, leads to the following N_θ -dimensional linear system, expressed in matrix form:

$$\left(\left(\eta + \frac{1}{\Delta_t} \right) I_{N_\theta} + \text{diag}(s_{\cdot, j+1}) \right) \nu_{\cdot, j} - \mathcal{L}_{j+1} \nu_{\cdot, j} = \frac{1}{\Delta_t} \nu_{\cdot, j+1} + \mathbf{1}_{N_\theta}, \quad (\text{A146})$$

where

$$s_{\cdot, j+1} = x_{j+1}^{1-\omega+\rho_{j+1}\omega} L_{E, \cdot, j+1}^\alpha e^{\omega\theta}, \quad (\text{A147})$$

with θ denoting the $N_\theta \times 1$ vector of grid points and the powers and exponentials applied element-wise. The operator $\mathcal{L}_{j+1}$ is an $N_\theta \times N_\theta$ tri-diagonal matrix encoding the upwind first-order difference operator $g_\theta \partial_\theta$. Its diagonal and off-diagonal entries are

$$\begin{aligned} \mathcal{L}_{i, i, j+1} &= -\frac{|g_{\theta, i, j+1}|}{\Delta_\theta}, \\ \mathcal{L}_{i, i-1, j+1} &= -\frac{\min(0, g_{\theta, i, j+1})}{\Delta_\theta}, \\ \mathcal{L}_{i, i+1, j+1} &= \frac{\max(0, g_{\theta, i, j+1})}{\Delta_\theta}. \end{aligned} \quad (\text{A148})$$

Equation (A146) can equivalently be written as $B_{j+1}^{(\nu)} \nu_{\cdot,j} = z_{j+1}^{(\nu)}$, with

$$\begin{aligned} B_{j+1}^{(\nu)} &= \left(\eta + \frac{1}{\Delta_t} \right) I_{N_\theta} + \text{diag}(s_{\cdot,j+1}) - \mathcal{L}_{j+1}, \\ z_{j+1}^{(\nu)} &= \frac{1}{\Delta_t} \nu_{\cdot,j+1} + \mathbf{1}_{N_\theta}, \end{aligned} \quad (\text{A149})$$

and the date- j iterate is $\nu_{\cdot,j} = \left(B_{j+1}^{(\nu)} \right)^{-1} z_{j+1}^{(\nu)}$. In constructing $\mathcal{L}_{j+1}$ we impose the boundary conditions

$$\begin{aligned} \mathcal{L}_{1,1,j+1} &= -\frac{\min(0, g_{\theta,1,j+1})}{\Delta_\theta} - \frac{|g_{\theta,1,j+1}|}{\Delta_\theta} = -\mathcal{L}_{1,2,j+1}, \\ \mathcal{L}_{N_\theta,N_\theta,j+1} &= \frac{\max(0, g_{\theta,N_\theta,j+1})}{\Delta_\theta} - \frac{|g_{\theta,N_\theta,j+1}|}{\Delta_\theta} = -\mathcal{L}_{N_\theta,N_\theta-1,j+1}. \end{aligned} \quad (\text{A150})$$

If there were “shadow” grid points at $i = 0$ and $i = N_\theta + 1$, these conditions would impose $\nu_{0,j} = \nu_{1,j}$ and $\nu_{N_\theta+1,j} = \nu_{N_\theta,j}$, equivalently $\partial_\theta^{(b)} \nu_1 = 0$ and $\partial_\theta^{(f)} \nu_{N_\theta} = 0$, which is the discrete analogue of imposing $\partial_\theta \nu(\underline{\theta}) = 0$ and $\partial_\theta \nu(\bar{\theta}) = 0$ on the continuous-time problem. The normalizations also guarantee that each row of $\mathcal{L}_{j+1}$ sums to zero, ensuring that $\mathcal{L}_{j+1}$ represents a (transposed) Markov-intensity matrix.

Update (c): Computation of $L_{E,\cdot,j}$. We update $L_{E,\cdot,j}$ by solving (A138) pointwise on the θ -grid, with $\Xi_{\cdot,j}$ assembled from $\nu_{\cdot,j}$, x_j , and γ_j via (A141). For the symmetric Cobb-Douglas case $\alpha = 1/2$ used in our calibration (Table 3), $L_{E,i,j}$ is given in closed form by (A142), evaluated at $\Xi = \Xi_{i,j}$ and clipped to $[0, 1]$:

$$L_{E,i,j} = \min \left(1, \max \left(0, \frac{(1 + 2\Xi_{i,j}^2) - \sqrt{1 + 4\Xi_{i,j}^2}}{2\Xi_{i,j}^2} \right) \right). \quad (\text{A151})$$

For general $\alpha \in (0, 1)$, $L_{E,i,j}$ is recovered by a one-dimensional bracketing solver applied to (A138) on $(0, 1)$, using the value at the previous grid point as a warm start.

A.6.2.3 Initialization and bounds on the state space

We initialize the date- N_t iterates $(\nu_{\cdot,N_t}, L_{E,\cdot,N_t})$ at the BGP values associated with the long-run ρ_{LR} . Concretely, $\nu_{\cdot,N_t}^*$ is obtained by solving the BGP-form ODE (58) on the same θ -grid through value-function iteration—equivalent to the same finite-difference scheme described above, run with constant ρ_t and $\partial_t \nu = 0$ until convergence; we use the same algorithm because, for ρ_t constant in calendar time, the calendar-time-augmented HJB (A137) reduces to the BGP ODE (58). Once $\nu_{\cdot,N_t}$ is computed, $L_{E,\cdot,N_t}$ follows from the closed-form update (c).

Bounds on the θ -grid are set as $\underline{\theta} = \theta_{\rho=0}^* - \theta_{\text{pad}}$ and $\bar{\theta} = \theta_{\rho=1}^* + \theta_{\text{pad}}$, where $\theta_{\rho=0}^*$ and $\theta_{\rho=1}^*$ are the BGP values of θ corresponding to the extreme admissible values of ρ in the model, and $\theta_{\text{pad}} > 0$ is a small padding. Bounds on the calendar-time grid are $t_1 = 0$ and $t_{N_t} = T$, where T is chosen large enough that ρ_t has settled at ρ_{LR} for at least $T - \Delta_\rho$ years and $\nu(\theta, T)$ is indistinguishable from $\nu_{LR}^*(\theta)$, where Δ_ρ denotes the length of the window over which ρ_t increases. In our baseline implementation we use $\Delta_\rho = 10$ years for the window over which ρ_t rises from ρ_{1990} to ρ_{2020} , $T = 100$ years for the calendar horizon, $\Delta_t = 1/12$ years (monthly grid),

and Δ_θ chosen so that $N_\theta \approx 200$.

A.6.2.4 Transitional dynamics under a smooth path of ρ_t

Section 5 computes the transitional dynamics of the economy following a smooth, deterministic shift in ρ_t from ρ_{1990} to ρ_{2020} . We proceed in four steps.

Step 1: BGP endpoints. We solve for the BGP at the initial value ρ_{1990} and at the long-run value ρ_{2020} ; this gives the policy functions $\nu_{1990}^*(\theta)$ and $\nu_{2020}^*(\theta)$, and the BGP values θ_{1990}^* and θ_{2020}^* , by the BGP-iteration procedure described in Section A.6.2.3.

Step 2: Path of ρ_t . We specify a deterministic path ρ_t over $[0, T]$. Let Δ_ρ denote the length of the window over which ρ_t increases. The economy is in the initial BGP for $t \leq 0$; at $t = 0$ the path of ρ_t begins to rise toward its long-run value over a window of length Δ_ρ . We use a cubic smoothstep specified in equation (A153) of Appendix A.6.3.1; both ρ_t and its time derivative are continuous everywhere.

Step 3: Backward solve. We solve the global system of Sections A.6.2.1–A.6.2.2 backward in calendar time on $[0, T]$, using $\nu(\theta, T) = \nu_{2020}^*(\theta)$ as the terminal condition; this delivers the policy functions $\nu(\theta, t)$, $L_E(\theta, t)$, x_t , γ_t , and ξ_t on the (θ, t) -grid.

Step 4: Forward simulation. We forward-simulate the path of θ_t starting from $\theta(0) = \theta_{1990}^*$. At each step $j \geq 1$, we interpolate the policy functions $L_E(\cdot, t_j)$, $\nu(\cdot, t_j)$, and the drift $g_\theta(\cdot, t_j)$ at $\theta = \theta(t_j)$ from the grid solution, evolve the state by one step using

$$\theta(t_{j+1}) = \theta(t_j) + g_\theta(\theta(t_j), t_j) \Delta_t, \quad (\text{A152})$$

and continue until $j = N_t$. Aggregate observables along the transition path—including output growth $g_t = (1 - \zeta) \xi (\gamma_t x_t^{\rho_t})^\mu L_E(\theta(t), t)^{\alpha\sigma}$, the price-to-output ratio $V_t/Y_t = (1 - \zeta\chi) \nu(\theta(t), t)$, Tobin's $Q_t = V_t / (P_{N,t} K_t)$, the pure profit share Π_t from the user-cost formula (A186), and the firm-sales Herfindahl index H_t from (A193)—are constructed from the simulated path as in Appendices A.6.3.6–A.6.3.7. The level of intangible capital N_t and aggregate output Y_t are recovered along the transition by integrating the drifts of $\log N_t$ and $\log Y_t$ forward from their initial values; both are normalized to one in the initial BGP.

Sensitivity to μ . The procedure above takes μ as a fixed parameter throughout the backward solve and the forward simulation. To trace the sensitivity of transitional dynamics to μ , we repeat Steps 1 through 4 for each value of μ on the same grid used in Table 3, $\mu \in \{0, 0.10, 0.25, 0.50\}$. The two BGPs (ρ_{1990} and ρ_{2020}) and the time path ρ_t are the same across runs; what changes is the calibrated value of $\hat{\xi}(\mu)$ (Appendix A.6.1.6) and, through it, the spillover rate $\bar{\xi}_t$ along the transition.

A.6.3 Transitional dynamics of aggregate quantities

This subsection records the cohort-accounting equations used to construct the aggregate transition-path moments reported in Section 5 and Figure 5: the shock profile for ρ_t ; the average scope $\hat{x}_t$ and average managerial share $\hat{\gamma}_t$ across all incumbent firms; the firm-sales Herfindahl index H_t ; the measured intangible capital stock K_t and replacement cost $P_{N,t}$; aggregate enterprise value V_t and the price-to-output ratio V_t/Y_t ; Tobin's Q_t ;

and the pure profit share Π_t .

The starting point is that the object x_t used in Appendix A.6.2 is the *marginal* scope of **new entrants**: at date t , a firm created at t chooses $x_t = \mathbf{x}(\rho_t)$ from (A135). Incumbent firms keep the scope chosen at entry (scope is irreversible), so the cross-sectional distribution of scope, the managerial share, and the implied aggregate quantities all evolve along the transition only through entry and exit. Throughout this appendix we take the policy functions $\nu(\theta_t, t)$ and $L_E(\theta_t, t)$ and the simulated path (θ_t, N_t, Y_t) from the procedure in Appendix A.6.2.4 as given.

A.6.3.1 Shock profile for ρ_t

The transition figures in the main text and below trace the response of the economy to an unanticipated, deterministic increase in replicability from ρ_{1990} to ρ_{2020} over a window of length $\Delta_\rho > 0$, with all other parameters held fixed. We model the path of ρ_t as a cubic *smoothstep*:

$$\rho_t = \rho_{1990} + (\rho_{2020} - \rho_{1990}) S\left(\frac{t}{\Delta_\rho}\right), \quad S(u) = \begin{cases} 0 & u \leq 0, \\ u^2(3-2u) & u \in [0, 1], \\ 1 & u \geq 1. \end{cases} \quad (\text{A153})$$

Both ρ_t and its time derivative are continuous everywhere; $\rho_t = \rho_{1990}$ for $t \leq 0$, $\rho_t = \rho_{2020}$ for $t \geq \Delta_\rho$, and the slope $\dot{\rho}_t$ is zero at the two endpoints $t = 0$ and $t = \Delta_\rho$. The economy is at the initial BGP at ρ_{1990} for $t \leq 0$.

A.6.3.2 Variety mass, firm mass, and average scope and managerial share

Recall from Section 3.3 that J_t is the mass of active firms and

$$X_t \equiv \int_0^{J_t} e^{-\delta(t-t_e(j))} x_{t_e(j)} dj \quad (\text{A154})$$

is the total measure of actively supplied intermediate varieties. Each incumbent firm j entered at $t_e(j)$ with scope $x_{t_e(j)} = \mathbf{x}(\rho_{t_e(j)})$ and survives with survival probability $e^{-\delta(t-t_e(j))}$ in the continuum representation. Summing scopes across surviving firms yields X_t .

Define the **average firm scope** (establishments-per-firm analogue in the calibration) by

$$\hat{x}_t \equiv \frac{X_t}{J_t}. \quad (\text{A155})$$

On a BGP with constant ρ , all firms share the same scope x , so $\hat{x}_t = x$ and the average coincides with the marginal scope of entrants. Along a transition in which only ρ_t (and hence x_t) for *new* cohorts changes, $\hat{x}_t$ rises gradually toward $\mathbf{x}(\rho_{LR})$ as high-scope entrants replace exiting low-scope incumbents.

The same cohort accounting applies to the managerial share $\gamma_t = \kappa x_t^\lambda$: incumbents retain the share fixed at

entry, $\gamma_{t_e(j)} = \kappa x_{t_e(j)}^\lambda$. Define the **average managerial share** as the equal-weighted mean across active firms,

$$\hat{\gamma}_t \equiv \frac{G_t}{J_t}, \quad G_t \equiv \int_0^{J_t} e^{-\delta(t-t_e(j))} \gamma_{t_e(j)} dj = \int_{-\infty}^t e^{-\delta(t-s)} \gamma(\mathbf{x}(\rho_s); \lambda) E_s ds. \quad (\text{A156})$$

On a BGP with constant ρ , $\hat{\gamma}_t = \gamma$. Along a transition $\hat{\gamma}_t$ rises gradually toward $\gamma(\mathbf{x}(\rho_{LR}); \lambda)$ as high-share entrants replace exiting low-share incumbents.

A.6.3.3 Law of motion

Let $E_t \equiv L_{E,t}^\alpha L_{M,t}^{1-\alpha}$ denote the flow of new firms. With $L_{M,t} = 1$,

$$dJ_t = (E_t - \delta J_t) dt = (L_{E,t}^\alpha - \delta J_t) dt. \quad (\text{A157})$$

A firm born at t_e contributes x_{t_e} varieties and a managerial share $\gamma_{t_e} = \kappa x_{t_e}^\lambda$ while alive. The surviving contribution at t is $e^{-\delta(t-t_e)} x_{t_e}$ varieties and $e^{-\delta(t-t_e)} \gamma_{t_e}$ managerial-share units per firm, so the variety stock and the cohort-weighted share stock obey

$$dX_t = (x_t L_{E,t}^\alpha - \delta X_t) dt, \quad (\text{A158})$$

$$dG_t = (\gamma_t L_{E,t}^\alpha - \delta G_t) dt, \quad (\text{A159})$$

where $x_t = \mathbf{x}(\rho_t)$ and $\gamma_t = \kappa x_t^\lambda$ are the scope and managerial share of firms **entering** at t . In integrated form with $E_s \equiv L_{E,s}^\alpha$,

$$X_t = \int_{-\infty}^t e^{-\delta(t-s)} \mathbf{x}(\rho_s) E_s ds, \quad J_t = \int_{-\infty}^t e^{-\delta(t-s)} E_s ds, \quad (\text{A160})$$

and analogously for G_t in (A156), so $\hat{x}_t = X_t/J_t$ and $\hat{\gamma}_t = G_t/J_t$ are survival-weighted averages of past entrant scopes and managerial shares, with weights proportional to entry flows E_s .

Discrete-time implementation. On a grid with step Δ_t , we update

$$J_{t+\Delta_t} = e^{-\delta\Delta_t} J_t + L_{E,t}^\alpha \Delta_t, \quad (\text{A161})$$

$$X_{t+\Delta_t} = e^{-\delta\Delta_t} X_t + x_t L_{E,t}^\alpha \Delta_t, \quad (\text{A162})$$

$$G_{t+\Delta_t} = e^{-\delta\Delta_t} G_t + \gamma_t L_{E,t}^\alpha \Delta_t, \quad (\text{A163})$$

$$\hat{x}_t = X_t/J_t, \quad \hat{\gamma}_t = G_t/J_t. \quad (\text{A164})$$

At $t = 0$, the economy is in the initial BGP at ρ_{1990} . We initialize

$$J_0 = \frac{L_{E,0}^\alpha}{\delta}, \quad X_0 = J_0 \mathbf{x}(\rho_{1990}), \quad G_0 = J_0 \gamma(\mathbf{x}(\rho_{1990}); \lambda), \quad (\text{A165})$$

which is the unique BGP triple satisfying $E_0 = \delta J_0$, $\hat{x}_0 = x_0$, and $\hat{\gamma}_0 = \gamma_0$ when $L_{E,0}$, x_0 , and γ_0 are the BGP values at ρ_{1990} .

A.6.3.4 Marginal versus average scope and managerial share in the figures

Panels (b) and (c) of Figure 5 report the cohort-averaged objects $\hat{x}_t$ and $\hat{\gamma}_t$, not the marginal entrant scope and managerial share $x_t = \mathbf{x}(\rho_t)$ and $\gamma_t = \kappa x_t^\lambda$. Because $\mathbf{x}(\cdot)$ and $\gamma(\mathbf{x}(\cdot); \lambda)$ are nonlinear in ρ , neither $\hat{x}_t$ nor $\hat{\gamma}_t$ equals the corresponding evaluation at any single “average” $\bar{\rho}_t$ across incumbents. The X_t/J_t and G_t/J_t definitions are the objects consistent with (A154)–(A156) and with the calibration target for average establishments per firm in Appendix A.6.1.

A.6.3.5 Aggregate enterprise value and V_t/Y_t

Firm-level enterprise value is (31):

$$v_t(n, x) = \nu_t A_t x^{1-\omega+\omega\rho} n^\omega, \quad (\text{A166})$$

where ν_t is the price–earnings ratio implied by the NPV of earnings discounted at $r_{t+s} + \delta$ (Appendix to Section 3 and Appendix A.6.2). A firm born at t_e keeps scope $x_{t_e} = \mathbf{x}(\rho_{t_e})$ and entry intangible Ξ_{t_e} , so its calendar- t profits are $\Pi_{t_e}(t) = A_t x_{t_e}^{1-\omega+\omega\rho_{t_e}} \Xi_{t_e}^\omega$ and $\Pi_{t_e}(t+s)/\Pi_{t_e}(t) = A_{t+s}/A_t$ for all t_e .

When the economy-wide discount rate shifts, all firms are valued using the current multiple $\nu(\theta_t, t)$:

$$v_{t_e}(t) = \nu(\theta_t, t) \Pi_{t_e}(t) = \nu(\theta_t, t) A_t x_{t_e}^{1-\omega+\omega\rho_{t_e}} \Xi_{t_e}^\omega. \quad (\text{A167})$$

Cohorts differ in *earnings levels* (low scope for old firms, high scope for new entrants), not in the NPV formula per dollar of current earnings. Summing over surviving firms,

$$V_t = \nu(\theta_t, t) A_t \int_{-\infty}^t e^{-\delta(t-s)} \mathbf{x}(\rho_s)^{1-\omega+\omega\rho_s} \Xi_s^\omega E_s ds = \nu(\theta_t, t) A_t S_t, \quad (\text{A168})$$

where $S_t \equiv N_t^\omega$ is the intangible-index stock (distinct from measured TFP Z_t in Section 4),

$$N_t^\omega = \int_{-\infty}^t e^{-\delta(t-s)} \mathbf{x}(\rho_s)^{1-\omega+\omega\rho_s} \Xi_s^\omega E_s ds \equiv S_t, \quad (\text{A169})$$

with law of motion

$$dS_t = \mathbf{x}(\rho_t)^{1-\omega+\omega\rho_t} \Xi_t^\omega L_{E,t}^\alpha dt - \delta S_t dt. \quad (\text{A170})$$

Using $A_t = (1 - \zeta\chi) Y_t N_t^{-\omega}$,

$$V_t = \nu(\theta_t, t) (1 - \zeta\chi) Y_t, \quad \frac{V_t}{Y_t} = (1 - \zeta\chi) \nu(\theta_t, t). \quad (\text{A171})$$

On a BGP, $\nu(\theta_t, t)$ is constant and Y_t grows, so V_t grows with output; V_t/Y_t is constant. Along a transition, a MIT shock to ρ can jump $\nu(\theta_t, t)$ at $t = 0^+$ even if the earnings stock S_t is predetermined.

Discrete-time implementation. With $\Xi_t = e^{\theta_t} N_t$, $E_t = L_{E,t}^\alpha$, and output Y_t from the simulated path ($N(0) = 1$),

$$S_{t+\Delta t} = e^{-\delta\Delta t} S_t + \mathbf{x}(\rho_t)^{1-\omega+\omega\rho_t} \Xi_t^\omega L_{E,t}^\alpha \Delta t, \quad (\text{A172})$$

$$V_t = \nu(\theta_t, t) (1 - \zeta\chi) Y_t, \quad (\text{A173})$$

$$\frac{V_t}{Y_t} = (1 - \zeta\chi) \nu(\theta_t, t). \quad (\text{A174})$$

At $t = 0$ on the initial BGP, S_0 inherits the predetermined BGP stock; $V_0 = \nu(\theta_{0+}, 0^+) (1 - \zeta\chi) Y_0$ reprices all firms at the post-shock discount rate.

A.6.3.6 Measured capital stock K_t and replacement cost $P_{N,t}$

National accounts capitalize past intangible investment at replacement cost. Section 4.3 records the perpetual-inventory equation (63):

$$\frac{dK_t}{K_t} = \left(\frac{I_t/P_{N,t}}{K_t} - \delta \right) dt, \quad I_t = W_t L_{E,t}, \quad P_{N,t} = \frac{W_t L_{E,t}^{1-\alpha}}{n_t}. \quad (\text{A175})$$

Here I_t is total spending on firm creation (entrepreneurial labor at the wage), and $P_{N,t}$ is the output cost of creating one unit of *non-deployed* intangible capital. Because $L_{E,t}$ units of entrepreneurial labor produce $E_t = L_{E,t}^\alpha$ firms, each with intangible stock n_t ,

$$\frac{I_t}{P_{N,t}} = L_{E,t}^\alpha n_t, \quad (\text{A176})$$

which yields the equivalent form (63).

Replacement cost along the transition. Along the transition path solved in Appendix A.6.2, the marginal entrant at date t has intangible stock $n_t = \Xi_t = e^{\theta_t} N_t$ (Section 3.3). Labor market clearing implies $W_t = \zeta\chi Y_t / L_{Y,t}$, so

$$P_{N,t} = \frac{W_t L_{E,t}^{1-\alpha}}{n_t} = \frac{\zeta\chi}{L_{Y,t}} Y_t \frac{L_{E,t}^{1-\alpha}}{e^{\theta_t} N_t} = \frac{\zeta\chi L_{Y,t}^{\zeta-1} L_{E,t}^{1-\alpha}}{e^{\theta_t} N_t^\zeta}. \quad (\text{A177})$$

Given $(L_{E,t}, \theta_t, N_t, Y_t)$ from the forward simulation, $P_{N,t}$ is known in closed form. On a BGP with constant $(L_{E,t}, \theta_t)$, wages grow at rate $g = (1 - \zeta)g_N$ while n_t grows at rate g_N , so $P_{N,t}$ falls at rate ζg_N ; since K_t grows at rate g_N , the replacement-cost value of capital $P_{N,t} K_t$ grows at rate g along with V_t and Y_t , and Tobin's Q_t is constant. Along a transition, changes in $L_{E,t}$ and θ_t can move $P_{N,t}$ independently of this scale effect.

Cohort representation of K_t . Firms keep the intangible stock n_{t_e} chosen at entry. The measured stock is the undepreciated sum of past creation costs in *physical* units of intangible capital:

$$K_t = \int_{-\infty}^t e^{-\delta(t-s)} L_{E,s}^\alpha n_s ds, \quad (\text{A178})$$

where n_s is the stock of a cohort born at s (equal to Ξ_s at entry). Unlike the productive index N_t , K_t does **not** weight entrants by scope: it records units of intangible capital created, not varieties deployed.

Discrete-time implementation. On the grid with step Δ_t ,

$$K_{t+\Delta_t} = e^{-\delta\Delta_t} K_t + L_{E,t}^\alpha e^{\theta_t} N_t \Delta_t, \quad (\text{A179})$$

$$P_{N,t} = \frac{\zeta\chi}{L_{Y,t}} Y_t \frac{L_{E,t}^{1-\alpha}}{e^{\theta_t} N_t}. \quad (\text{A180})$$

At $t = 0$, the economy is on the initial BGP at ρ_{1990} . The **stock** K_t is predetermined and does not jump on impact:

$$K_0 = \frac{L_{E,0}^\alpha e^{\theta_0} N_0}{\delta + g_{N,0}}, \quad (\text{A181})$$

with $N_0 = 1$ by normalization at the shock date. On the pre-shock BGP segment ($t < 0$), $K_t = K_0 e^{g_{N,0}t}$ while $N_t = e^{g_{N,0}t}$. After the shock, K_t evolves forward from the same K_0 via (A179); only the investment flow $L_{E,t}^\alpha n_t$ and the replacement cost $P_{N,t}$ may jump. Measured TFP along the path is

$$Z_t = \frac{Y_t}{K_t^{1-\zeta} L_{Y,t}^\zeta} = \left(\frac{N_t}{K_t} \right)^{1-\zeta}, \quad (\text{A182})$$

using the simulated path of N_t (Appendix A.6.2) and K_t from (A179). Its growth rate follows directly:

$$\frac{d \log Z_t}{dt} = (1 - \zeta) \left(\frac{d \log N_t}{dt} - \frac{d \log K_t}{dt} \right). \quad (\text{A183})$$

On a BGP, N_t and K_t grow at the same rate (g_N), so $d \log Z_t / dt = 0$ even though output grows. Along a transition, N_t and K_t generally diverge because N_t weights entrants by scope while K_t does not; measured TFP growth can therefore be positive or negative even when true technology is unchanged.

Tobin's Q . Aggregate Tobin's Q is enterprise value divided by the replacement cost of measured intangible capital (Appendix A.5.3):

$$Q_t = \frac{V_t}{P_{N,t} K_t}, \quad V_t = (1 - \zeta\chi) \nu(\theta_t, t) Y_t, \quad (\text{A184})$$

with $P_{N,t}$ from (A180) and K_t from (A179). On a BGP, this ratio reduces to the closed form (A110). When there are deterministic changes in ρ_t , Q_t becomes:

$$Q_t = \frac{N_t e^{(1-\omega)\theta_t}}{K_t (1 - \gamma_t) x_t^{\psi_t}}, \quad \psi_t \equiv 1 - \omega + \omega \rho_t. \quad (\text{A185})$$

In particular, this expression implies that when ρ_t changes smoothly, Tobin's Q will not jump, consistent with the results reported in Figure 5.

Pure profit share. The economic profit share in Table 3 is capital income minus competitive returns to measured intangible capital (Appendix A.5.3). Along the transition,

$$\Pi_t = (1 - \zeta\chi) - R_t \frac{P_{N,t} K_t}{Y_t}, \quad R_t = r_t + \delta - \frac{d \log P_{N,t}}{dt}, \quad (\text{A186})$$

where $r_t = \eta + g_t$ is the real interest rate ($g_t = d \log Y_t / dt$) and $P_{N,t}$, K_t , and Y_t are from (A180)–(A179). On a BGP with constant $(L_{E,t}, \theta_t)$, $d \log P_{N,t} / dt = -\zeta g_N$, $R_t = \eta + \delta + g_N$, and (A186) is constant along the path.

A.6.3.7 Concentration along the transition path

We extend the BGP derivation in Appendix A.5.3.3 to allow for a time-varying replicability path ρ_t , with the corresponding entry scope $x_t = x(\rho_t)$ and entrant intangible Ξ_t . A firm born at date t_e retains scope x_{t_e} and entry intangible $n_{t_e} = \Xi_{t_e}$ for life. From (30), its calendar- t profit is

$$\pi_{t_e}(t) = A_t x_{t_e}^{1-\omega+\omega\rho_{t_e}} \Xi_{t_e}^\omega, \quad (\text{A187})$$

where $A_t = (1 - \zeta\chi) Y_t N_t^{-\omega}$ is the common variable-profit shifter. Because the markup $1/\chi$ is uniform across firms and varieties, sales shares equal profit shares. Aggregate profits are $A_t N_t^\omega$, so the date- t sales share of a single firm born at t_e is

$$s_{t_e}(t) = \frac{\psi_{t_e}}{N_t^\omega}, \quad \psi_{t_e} \equiv x_{t_e}^{1-\omega+\omega\rho_{t_e}} \Xi_{t_e}^\omega, \quad (\text{A188})$$

where ψ_{t_e} is a cohort-specific “intangible-services weight” fixed at entry. Aggregating with entry flow $E_{t_e} = L_{E,t_e}^\alpha$ and survival $e^{-\delta(t-t_e)}$, the firm-sales Herfindahl–Hirschman index is

$$H_t = \int_{-\infty}^t L_{E,t_e}^\alpha e^{-\delta(t-t_e)} s_{t_e}(t)^2 dt_e = \frac{T_t}{N_t^{2\omega}}, \quad (\text{A189})$$

where T_t is the survival-weighted stock of squared cohort weights,

$$T_t \equiv \int_{-\infty}^t L_{E,t_e}^\alpha e^{-\delta(t-t_e)} \psi_{t_e}^2 dt_e, \quad dT_t = \left(\psi_t^2 L_{E,t}^\alpha - \delta T_t \right) dt. \quad (\text{A190})$$

Equivalently, normalizing by $N_t^{2\omega}$ and using $\psi_t/N_t^\omega = x_t^{1-\omega+\omega\rho_t} e^{\omega\theta_t} \equiv c_t$, H_t itself satisfies the linear ODE

$$\frac{dH_t}{dt} = L_{E,t}^\alpha c_t^2 - \left(\delta + 2\omega g_{N,t} \right) H_t, \quad c_t \equiv x_t^{1-\omega+\omega\rho_t} e^{\omega\theta_t}, \quad (\text{A191})$$

with $g_{N,t} = d \log N_t / dt$. The driver $L_{E,t}^\alpha c_t^2$ is the rate at which new high-share entrants are added to the cross-section; the decay rate $\delta + 2\omega g_{N,t}$ combines firm exit and the dilution of incumbent shares as the productive index N_t grows.

Two consistency checks follow. First, on a BGP L_E , x , ρ , θ , and g_N are constant, so (A191) has a unique steady state

$$H = \frac{L_E^\alpha c^2}{\delta + 2\omega g_N} = \frac{(\delta + \omega g_N)^2}{L_E^\alpha (\delta + 2\omega g_N)}, \quad (\text{A192})$$

where the second equality uses $c = (\delta + \omega g_N) / L_E^\alpha$ from the cohort normalization

$$\int_{-\infty}^t L_E^\alpha e^{-\delta(t-t_e)} s_{t_e}(t) dt_e = 1.$$

This recovers (A115). Second, although ψ_t , T_t , and $N_t^{2\omega}$ all grow without bound on a BGP, the ratio $H_t = T_t / N_t^{2\omega}$ is dimensionless and remains bounded in $[0, 1]$ along any path.

Discrete-time implementation. On the grid $\{t_j\}$ with step Δt and $N_0 = 1$, with $g_{N,t} = (s_t - \delta) / \omega$ from the simulated path (where $s_t = \delta + \omega g_{N,t}$ is the BGP-form intangible-growth shifter from Appendix A.6.2), we

update

$$H_{t+\Delta_t} = H_t + \Delta_t \left(L_{E,t}^\alpha c_t^2 - (\delta + 2\omega g_{N,t}) H_t \right), \quad c_t = x_t^{1-\omega+\omega\rho_t} e^{\omega\theta_t}. \quad (\text{A193})$$

The stock is initialized at its pre-shock BGP value (A192), evaluated at the date-0 quantities, and is predetermined at $t = 0$: only the entry flow $L_{E,t}^\alpha$ and the cohort weight c_t may jump on impact.

Relation to other aggregate objects. The same entry–exit bookkeeping extends to other cohort-based aggregates ($\hat{x}_t, \hat{\gamma}_t, K_t$, and H_t). Output growth, $\nu(\theta_t, t)$, and $L_{E,t}$ are computed along (θ_t, t) as in Appendix A.6.2; $\hat{x}_t$ requires (J_t, X_t) ; $S_t = N_t^\omega$ is tracked as a cross-check on the productive index; H_t is integrated from (A193).

A.7 Appendix to Section 6

A.7.1 Static allocation equivalence

The social planner’s static allocation of labor and intangible capital across varieties is identical to the decentralized equilibrium allocation. Consider the planner’s problem of choosing labor $l_t(m)$ and intangible capital $n_t(m)$ for each variety, subject to the replicability constraint (4), and the aggregate resource constraints $\int l_t(m) dm \leq L_{Y,t}$ and $c_t(m) \leq y_t(m)$. The objective is aggregate output:

$$C_t = \left(\int_0^{X_t} c_t(m)^\chi dm \right)^{1/\chi}.$$

The production technology is

$$y_t(m) = n_t(m)^{1-\zeta} l_t(m)^\zeta.$$

The first-order conditions of this problem yield shadow prices (price equals marginal cost) and a shadow wage:

$$p_t(m) = mc_t(m),$$

$$W_t^{FB} = \zeta \left(\frac{N_t}{L_{Y,t}} \right)^{1-\zeta}.$$

The optimal allocation of labor across varieties is:

$$l_t(m) = \left(\frac{n_t(m)}{N_t} \right)^\omega L_{Y,t},$$

The optimal allocation of intangibles within each firm is symmetric:

$$n_t(m) = x_t(j)^{-(1-\rho)} n_t(j).$$

Aggregate output satisfies

$$C_t = N_t^{1-\zeta} L_{Y,t}^\zeta.$$

In the decentralized equilibrium, firms charge a constant markup $1/\chi$ over marginal cost. Prices and the

wage satisfy

$$p_t(m) = mc_t(m)/\chi,$$

$$W_t^{DE} = \zeta\chi (N_t/L_{Y,t})^{1-\zeta}.$$

However, the labor allocation, the intangible allocation, and aggregate output are all unchanged. The markup reduces wages by the factor χ but raises per-variety profits from $(1 - \zeta)$ to $(1 - \zeta\chi)$ times $C_t(n_t(m)/N_t)^\omega$. With a representative household that receives both wage and profit income, this redistribution does not affect the real allocation or the level of consumption.

Dynamic problem. Because the static allocation is the same, the planner's dynamic problem inherits the same capital index N_t and the same laws of motion for N_t and Ξ_t as the decentralized equilibrium. Aggregate output is again

$$Y_t = N_t^{1-\zeta} L_{Y,t}^\zeta.$$

The parameter ω is defined by

$$\omega = \frac{(1 - \zeta)\chi}{1 - \zeta\chi}.$$

It enters both problems identically: it is a technological parameter governing the curvature of the CES aggregator, not a consequence of market power. The planner's problem therefore reduces to the choice of entrepreneurial effort $L_{E,t}$ and scope x_t , subject to the same aggregate constraints as in Appendix A.4. This holds for the unconstrained first-best, the second-best with a tax on intangible investment, and the decentralized equilibrium.

A.7.2 First-best planning problem

Statement. The unconstrained first-best chooses scope at each date. Using the static aggregation of Section A.7.1, and defining

$$\theta_t \equiv \log(\Xi_t/N_t)$$

as in Appendix A.4, the problem can be stated as:

$$W(N_t, \theta_t) = \max_{\{x_{t+s}, L_{E,t+s}, L_{Y,t+s}, Y_{t+s}\}} \int_0^{+\infty} e^{-\eta s} \log(Y_{t+s}) ds$$

$$\text{s.t.} \quad Y_{t+s} \leq N_{t+s}^{1-\zeta} L_{Y,t+s}^\zeta,$$

$$L_{Y,t+s} + L_{E,t+s} \leq 1,$$

$$\frac{dN_{t+s}}{N_{t+s}} = \frac{1}{\omega} \left(x_{t+s}^{1-\omega+\rho\omega} L_{E,t+s}^\alpha e^{\omega\theta_{t+s}} - \delta \right) ds,$$

$$d\theta_{t+s} = \frac{1}{\omega} \left(\delta + \omega \xi \left(\gamma(x_{t+s}; \lambda) x_{t+s}^\rho \right)^\mu L_{E,t+s}^{\alpha\sigma} - x_{t+s}^{1-\omega+\rho\omega} L_{E,t+s}^\alpha e^{\omega\theta_{t+s}} \right) ds.$$

The states are (N_t, θ_t) . In recursive form, the Bellman equation is

$$\eta W(N_t, \theta_t) = \max_{\{x_t, L_{E,t}, L_{Y,t}, Y_t\}} \log(Y_t) + \partial_N W(N_t, \theta_t) \frac{dN_t}{dt} + \partial_\theta W(N_t, \theta_t) \frac{d\theta_t}{dt}$$

$$\begin{aligned}
\text{s.t.} \quad & Y_t \leq N_t^{1-\zeta} L_{Y,t}^\zeta, \\
& L_{Y,t} + L_{E,t} \leq 1, \\
& \frac{dN_t}{N_t} = \frac{1}{\omega} \left(x_t^{1-\omega+\rho\omega} L_{E,t}^\alpha e^{\omega\theta_t} - \delta \right) dt, \\
& d\theta_t = \frac{1}{\omega} \left(\delta + \omega \bar{\xi}_t L_{E,t}^{\alpha\sigma} - x_t^{1-\omega+\rho\omega} L_{E,t}^\alpha e^{\omega\theta_t} \right) dt,
\end{aligned}$$

where

$$\bar{\xi}_t = \xi \left(\gamma(x_t; \lambda) x_t^\rho \right)^\mu.$$

Derivation of the HJB. Define the normalized continuation value w^{FB} by

$$W(N_t, \theta_t) = \frac{1-\zeta}{\eta} \left(\log N_t + w^{FB}(\theta_t) \right).$$

The derivatives are

$$\begin{aligned}
\partial_N W &= \frac{1-\zeta}{\eta N_t}, \\
\partial_\theta W &= \frac{1-\zeta}{\eta} (w^{FB})'(\theta_t).
\end{aligned}$$

At the optimum, the resource constraint for output binds, so

$$Y_t = N_t^{1-\zeta} L_{Y,t}^\zeta,$$

and therefore

$$\log(Y_t) = (1-\zeta) \log N_t + \zeta \log(L_{Y,t}).$$

The law of motion for N_t implies

$$\frac{1}{N_t} \frac{dN_t}{dt} = \frac{1}{\omega} \left(x_t^{1-\omega+\rho\omega} L_{E,t}^\alpha e^{\omega\theta_t} - \delta \right).$$

Substitute these expressions into the Bellman equation and multiply through by η . The terms $(1-\zeta) \log N_t$ on the left from ηW cancel the same terms on the right from $\log(Y_t)$ and from

$$\partial_N W \cdot \frac{dN_t}{dt} = \frac{1-\zeta}{\eta} \frac{1}{N_t} \frac{dN_t}{dt}.$$

What remains is an equation in $(w^{FB}(\theta_t), (w^{FB})'(\theta_t), L_{E,t}, x_t, \theta_t)$ alone: no $\log N_t$ appears. If w^{FB} depended on N_t , the HJB would contain the additional term $((1-\zeta)/\eta) \partial_N w^{FB} \cdot (dN_t/dt)$. Along the aggregate law for N_t , this term is proportional to $N_t \partial_N w^{FB}$. For the residual Bellman equation to hold uniformly in N_t given θ_t , we need

$$\partial_N w^{FB} = 0.$$

Thus w^{FB} depends on θ_t only. Rearranging the equation, the reduced HJB is

$$w^{FB}(\theta_t) = \max_{\{x_t, L_{E,t}\}} \frac{\zeta}{1-\zeta} \log(1 - L_{E,t}) + \frac{1}{\eta\omega} \left(x_t^{1-\omega+\rho\omega} L_{E,t}^\alpha e^{\omega\theta_t} - \delta \right) \\ + \frac{1}{\eta\omega} \left(\delta + \omega \bar{\xi}_t L_{E,t}^{\alpha\sigma} - x_t^{1-\omega+\rho\omega} L_{E,t}^\alpha e^{\omega\theta_t} \right) (w^{FB})'(\theta_t).$$

First-order conditions and envelope condition. The optimality conditions for the reduced HJB are as follows.

Differentiating with respect to $L_{E,t}$ and multiplying through by $L_{E,t}$:

$$\frac{\zeta}{1-\zeta} \frac{L_{E,t}}{1-L_{E,t}} = \frac{\alpha}{\eta\omega} x_t^{1-\omega+\rho\omega} L_{E,t}^\alpha e^{\omega\theta_t} \left(1 - (w^{FB})'(\theta_t) \right) + \frac{\alpha\sigma}{\eta} \bar{\xi}_t L_{E,t}^{\alpha\sigma} (w^{FB})'(\theta_t).$$

Differentiating with respect to x_t :

$$0 = \frac{1-\omega+\rho\omega}{x_t} \left(1 - (w^{FB})'(\theta_t) \right) x_t^{1-\omega+\rho\omega} L_{E,t}^\alpha e^{\omega\theta_t} + \omega \frac{d}{dx_t} [\xi(\gamma(x_t; \lambda) x_t^\rho)^\mu] L_{E,t}^{\alpha\sigma} (w^{FB})'(\theta_t).$$

The envelope condition for $(w^{FB})'(\theta_t)$ is obtained by differentiating the HJB with respect to θ_t :

$$(w^{FB})'(\theta_t) = \frac{1}{\eta} x_t^{1-\omega+\rho\omega} L_{E,t}^\alpha e^{\omega\theta_t} \left(1 - (w^{FB})'(\theta_t) \right) \\ + \frac{1}{\eta\omega} \left(\delta + \omega \bar{\xi}_t L_{E,t}^{\alpha\sigma} - x_t^{1-\omega+\rho\omega} L_{E,t}^\alpha e^{\omega\theta_t} \right) (w^{FB})''(\theta_t).$$

Economic interpretation of $(w^{FB})'(\theta_t)$. Define

$$f(\theta_t) \equiv \frac{1}{\eta} \left(1 - (w^{FB})'(\theta_t) \right).$$

Substituting into the envelope condition and rearranging:

$$\eta f(\theta_t) = 1 - x_t^{1-\omega+\rho\omega} L_{E,t}^\alpha e^{\omega\theta_t} f(\theta_t) + \frac{1}{\omega} \left(\delta + \omega \bar{\xi}_t L_{E,t}^{\alpha\sigma} - x_t^{1-\omega+\rho\omega} L_{E,t}^\alpha e^{\omega\theta_t} \right) f'(\theta_t).$$

We now show that $f(\theta_t)$ equals the price-to-earnings ratio $\nu(\theta_t)$ that would prevail under the prices implementing the planner's allocation. By the same argument as in Appendix A.4, the price-to-earnings ratio under any allocation satisfying the aggregate constraints is

$$\nu_t = \int_0^{+\infty} e^{-\int_0^s (r_{t+u} + \delta) du} \frac{A_{t+s}}{A_t} ds,$$

where

$$r_t dt \equiv \frac{dC_t}{C_t} + \eta dt, \\ A_t \equiv (1-\zeta) C_t N_t^{-\omega}.$$

Since $C_t = Y_t = N_t^{1-\zeta} L_{Y,t}^\zeta$, this simplifies to

$$\nu_t = N_t^\omega \int_0^{+\infty} e^{-(\eta+\delta)s} N_{t+s}^{-\omega} ds.$$

Because θ_t is a sufficient state for the dynamics of the economy, we can write $\nu_t = \nu(\theta_t)$. The same manipulation as in Appendix A.4 (defining $h(N_t) = N_t^{-\omega} \nu(\theta_t)$ and differentiating its ODE) shows that $\nu(\theta_t)$ satisfies

$$\eta \nu(\theta_t) = 1 - x_t^{1-\omega+\rho\omega} L_{E,t}^\alpha e^{\omega\theta_t} \nu(\theta_t) + \frac{1}{\omega} \left(\delta + \omega \bar{\xi}_t L_{E,t}^{\alpha\sigma} - x_t^{1-\omega+\rho\omega} L_{E,t}^\alpha e^{\omega\theta_t} \right) \nu'(\theta_t).$$

This is the same ODE as the one satisfied by $f(\theta_t)$. Under the same boundary conditions (transversality), the solution is unique, so

$$f(\theta_t) = \nu(\theta_t),$$

or equivalently,

$$(w^{FB})'(\theta_t) = 1 - \eta \nu(\theta_t).$$

Optimal scope is infinite. Recall that $(w^{FB})'(\theta_t) = 1 - \eta \nu(\theta_t)$. Since $\nu(\theta_t) > 0$ and $\eta \nu(\theta_t) < 1$ (the price–earnings ratio satisfies $\nu(\theta_t) < 1/(\eta + \delta)$, because current earnings are discounted at rate $\eta + \delta$ and diluted by the growth of the capital index), we have $(w^{FB})'(\theta_t) \in (0, 1)$. In the first-order condition for x_t , the first term is then strictly positive—its sign is that of $1 - (w^{FB})'(\theta_t) = \eta \nu(\theta_t) > 0$ —and the second is nonnegative, strictly positive when $\mu > 0$ because the spillover rate $\xi(\gamma(x_t; \lambda) x_t^\rho)^\mu$ is then strictly increasing in x_t . The right-hand side is therefore strictly positive for any finite x_t , so the first-order condition cannot hold. Hence the planner always benefits from wider scope, and

$$x^{FB} \rightarrow \infty.$$

BGP entry condition at fixed scope. We now specialize to a balanced growth path with constant scope x and constant $\theta_t = \theta$. Write

$$\bar{\xi} = \xi(\gamma x^\rho)^\mu, \quad \gamma \equiv \gamma(x; \lambda).$$

On the BGP, the stationarity condition $d\theta_t = 0$ implies that the drift coefficient multiplying $\nu'(\theta)$ in the ODE vanishes. The ODE for ν therefore reduces to

$$\nu = \frac{1}{\eta + \delta + \omega \bar{\xi} L_E^{\alpha\sigma}}.$$

The stationarity condition $d\theta_t = 0$ requires

$$x^{1-\omega+\rho\omega} L_E^\alpha e^{\omega\theta} = \delta + \omega \bar{\xi} L_E^{\alpha\sigma}.$$

Using these two conditions together with $(w^{FB})'(\theta) = 1 - \eta \nu$, the FOC for L_E becomes

$$\frac{\zeta}{1-\zeta} \frac{L_E}{1-L_E} = \frac{\alpha}{\omega} (\delta + \omega \bar{\xi} L_E^{\alpha\sigma}) \nu + \frac{\alpha\sigma}{\eta} \bar{\xi} L_E^{\alpha\sigma} (1 - \eta \nu)$$

$$\begin{aligned}
&= \frac{\alpha}{\omega} \frac{\delta + \omega \bar{\xi} L_E^{\alpha\sigma}}{\eta + \delta + \omega \bar{\xi} L_E^{\alpha\sigma}} + \frac{\alpha\sigma}{\eta} \bar{\xi} L_E^{\alpha\sigma} \frac{\delta + \omega \bar{\xi} L_E^{\alpha\sigma}}{\eta + \delta + \omega \bar{\xi} L_E^{\alpha\sigma}} \\
&= \frac{\alpha}{\omega} \frac{\delta + \omega \bar{\xi} L_E^{\alpha\sigma}}{\eta + \delta + \omega \bar{\xi} L_E^{\alpha\sigma}} \left(1 + \frac{\sigma}{\eta} \omega \bar{\xi} L_E^{\alpha\sigma} \right).
\end{aligned}$$

This establishes

$$\frac{\zeta}{1-\zeta} \frac{L_E^{FB}}{1-L_E^{FB}} = \frac{\alpha}{\omega} \frac{\delta + \omega \bar{\xi} L_E^{\alpha\sigma}}{\eta + \delta + \omega \bar{\xi} L_E^{\alpha\sigma}} \left(1 + \frac{\sigma}{\eta} \omega \bar{\xi} L_E^{\alpha\sigma} \right). \quad (\text{A194})$$

Proof of Result 4. Compare the first-best entry condition (A194):

$$\frac{\zeta}{1-\zeta} \frac{L_E}{1-L_E} = \underbrace{\alpha}_{(ii)} \cdot \frac{1}{\omega} \frac{\delta + \omega \bar{\xi} L_E^{\alpha\sigma}}{\eta + \delta + \omega \bar{\xi} L_E^{\alpha\sigma}} \cdot \underbrace{\left(1 + \frac{\sigma}{\eta} \omega \bar{\xi} L_E^{\alpha\sigma} \right)}_{(i)}$$

with the decentralized BGP entry condition (60):

$$\frac{\zeta}{1-\zeta} \frac{L_E}{1-L_E} = \underbrace{(1-\gamma)}_{(iii)} \cdot \frac{1}{\omega} \frac{\delta + \omega \bar{\xi} L_E^{\alpha\sigma}}{\eta + \delta + \omega \bar{\xi} L_E^{\alpha\sigma}}.$$

The common factor $(1/\omega) \cdot (\delta + \omega \bar{\xi} L_E^{\alpha\sigma}) / (\eta + \delta + \omega \bar{\xi} L_E^{\alpha\sigma})$ is the BGP value of a marginal unit of capital. Three differences remain:

- (i) The factor $(1 + \frac{\sigma}{\eta} \omega \bar{\xi} L_E^{\alpha\sigma})$ captures the dynamic value of entry: each new firm raises Ξ_t for future entrants. This term is absent in the DE.
- (ii) The factor α in the FB reflects the social marginal product of entry labor (α times the private marginal product when $\alpha < 1$).
- (iii) The factor $(1 - \gamma)$ in the DE reflects the hold-up friction. In the FB, the planner retains the full value of each new firm.

■

A.7.3 Welfare in the decentralized equilibrium

The same derivation as in Section A.7.2 applies to any allocation in which $L_{E,t}$ and x_t are functions of θ_t alone. In the decentralized equilibrium, optimal scope satisfies

$$x_t = x,$$

where x solves (36), and $L_{E,t}$ is determined by (55). The normalized welfare function

$$W(N_t, \theta_t) = \frac{1-\zeta}{\eta} \left(\log N_t + w(\theta_t) \right)$$

satisfies the reduced HJB

$$w(\theta_t) = \frac{\zeta}{1-\zeta} \log(1 - L_{E,t}) + \frac{1}{\eta\omega} \left(x^{1-\omega+\rho\omega} L_{E,t}^\alpha e^{\omega\theta_t} - \delta \right) + \frac{1}{\eta\omega} \left(\delta + \omega \bar{\xi} L_{E,t}^{\alpha\sigma} - x^{1-\omega+\rho\omega} L_{E,t}^\alpha e^{\omega\theta_t} \right) w'(\theta_t), \quad (\text{A195})$$

where

$$\bar{\xi} = \xi(\gamma x^\rho)^\mu.$$

In the second-best with a tax on intangible investment, $L_{E,t}$ satisfies a modified free-entry condition and x_t is determined by the tax rate. The same functional form (A195) applies; only the mappings from θ_t to $(L_{E,t}, x_t)$ differ.

A.7.4 Second-best planning problem with a tax on intangible investment

Decentralized equilibrium with tax. Fix a tax path $\{\tau_t\}$. A competitive equilibrium with this policy is an allocation and prices such that: (a) firms optimize and markets clear as in Section 3; (b) free entry satisfies (78); (c) the manager's incentive constraint binds at the scaled outside option (77); (d) the price-to-earnings ratio $\nu_t = \nu(\theta_t)$ solves the same ODE (A47) as in Appendix A.4, evaluated at the policy-induced scope. Aggregate constraints are unchanged: $Y_t = N_t^{1-\zeta} L_{Y,t}^\zeta$, $L_{Y,t} + L_{E,t} \leq 1$, and the laws of motion for N_t , Ξ_t , θ_t from Appendix A.4.

Optimal scope as a function of τ . Given τ , the entrepreneur chooses scope to maximize the retained share of enterprise value. Under the taxed IC (77), the manager's share of enterprise value is $(1 - \tau) \gamma(\tilde{x}; \lambda)$, because the outside option is scaled by $(1 - \tau)$. The entrepreneur's retained fraction of firm value is therefore $1 - (1 - \tau) \gamma(\tilde{x}; \lambda)$, and optimal scope solves

$$\hat{x}(\tau) \in \arg \max_{\tilde{x} > 1} \left(1 - (1 - \tau) \gamma(\tilde{x}; \lambda) \right) \tilde{x}^{1-\omega+\rho\omega}. \quad (\text{A196})$$

The first-order condition is

$$(1 - \omega + \rho\omega) \left(1 - (1 - \tau) \gamma(\hat{x}; \lambda) \right) = (1 - \tau) \gamma'(\hat{x}; \lambda) \hat{x}. \quad (\text{A197})$$

Under the functional form $\gamma(\tilde{x}; \lambda) = \kappa \tilde{x}^\lambda$, this simplifies to

$$1 - (1 - \tau) \gamma(\hat{x}; \lambda) = \frac{(1 - \tau) \lambda \gamma(\hat{x}; \lambda)}{1 - \omega(1 - \rho)}, \quad (\text{A198})$$

or equivalently,

$$(1 - \tau) \gamma(\hat{x}(\tau); \lambda) = \frac{1 - \omega(1 - \rho)}{\lambda + 1 - \omega(1 - \rho)}.$$

A higher τ requires a higher $\gamma(\hat{x})$ and therefore a wider $\hat{x}(\tau)$.

Effective entrepreneurial wedge. The IC-adjusted residual share retained by the entrepreneur, before the entry tax, is independent of τ —as τ rises the entrepreneur widens scope so as to hold the manager's claim

$(1 - \tau)\gamma(\hat{x}(\tau); \lambda)$ fixed—and equals

$$1 - (1 - \tau)\gamma(\hat{x}(\tau); \lambda) = \frac{\lambda}{\lambda + 1 - \omega(1 - \rho)}.$$

After the entrepreneur's entry payoff is taxed at the same rate τ , the composite wedge that enters the free-entry condition is

$$1 - \varepsilon(\tau) \equiv (1 - \tau) \left(1 - (1 - \tau)\gamma(\hat{x}(\tau); \lambda) \right) = \frac{\lambda(1 - \tau)}{\lambda + 1 - \omega(1 - \rho)}. \quad (\text{A199})$$

When $\tau = 0$, we recover $1 - \varepsilon(0) = 1 - \gamma(x^{DE}; \lambda)$. The free-entry condition under the tax becomes

$$\frac{\zeta}{1 - \zeta} \frac{L_{E,t}^{1-\alpha}}{1 - L_{E,t}} = \frac{1 - \zeta\chi}{\zeta\chi} \left(1 - \varepsilon(\tau_t) \right) \hat{x}(\tau_t)^{1-\omega+\rho\omega} \nu(\theta_t) e^{\omega\theta_t}. \quad (\text{A200})$$

On the BGP, this simplifies to

$$\frac{\zeta}{1 - \zeta} \frac{L_E}{1 - L_E} = \frac{1 - \varepsilon(\tau)}{\omega} \frac{\delta + \omega \bar{\xi}(\tau) L_E^{\alpha\sigma}}{\eta + \delta + \omega \bar{\xi}(\tau) L_E^{\alpha\sigma}}, \quad (\text{A201})$$

where

$$\bar{\xi}(\tau) = \xi\left(\gamma(\hat{x}(\tau); \lambda) \hat{x}(\tau)^\rho\right)^\mu.$$

Statement of the optimal policy problem. The planner chooses a tax path $\{\tau_t\}$ to maximize

$$\max_{\{\tau_t\}} \int_0^\infty e^{-\eta t} \log Y_t dt \quad (\text{A202})$$

subject to the aggregate constraints of the model, the entry condition (A200), the price-to-earnings ODE (A47) evaluated at $\hat{x}(\tau_t)$, and the feasibility restriction

$$\tau_t < \frac{\lambda}{\lambda + 1 - \omega(1 - \rho)}.$$

The planner cannot choose $(L_{E,t}, x_t)$ directly. Scope is pinned down by the coalition's incentive problem through (A196), and entry is determined by free entry (A200). The tax is the only instrument.

On the BGP, θ_t , L_E , ν , and x are all constant. BGP welfare is

$$W(\tau) = \frac{\zeta}{\eta} \log(1 - L_E(\tau)) + \frac{g(\tau)}{\eta^2}, \quad (\text{A203})$$

where

$$g(\tau) = (1 - \zeta) \bar{\xi}(\tau) L_E(\tau)^{\alpha\sigma}$$

and $L_E(\tau)$ solves (A201). The BGP planning problem reduces to $\max_\tau W(\tau)$.

Proof of Result 5. Set $\sigma = 0$. Then $L_E^{\alpha\sigma} = 1$, so

$$g(\tau) = (1 - \zeta) \bar{\xi}(\tau),$$

which depends on τ only through scope. The BGP entry condition (A201) becomes

$$\underbrace{\frac{\zeta}{1-\zeta} \frac{L_E}{1-L_E}}_{\equiv \ell(L_E)} = \underbrace{\frac{1-\varepsilon(\tau)}{\omega} \frac{\delta + \omega \bar{\xi}(\tau)}{\eta + \delta + \omega \bar{\xi}(\tau)}}_{\equiv R(\tau)}.$$

The right-hand side $R(\tau)$ does not depend on L_E when $\sigma = 0$. Write

$$\mathcal{Q}(\tau) \equiv \frac{\delta + \omega \bar{\xi}(\tau)}{\eta + \delta + \omega \bar{\xi}(\tau)},$$

so that

$$R(\tau) = \frac{1-\varepsilon(\tau)}{\omega} \mathcal{Q}(\tau).$$

Implicit differentiation. Since R is independent of L_E , we have

$$\ell'(L_E) L'_E(\tau) = R'(\tau), \quad \ell'(L_E) = \frac{\zeta}{(1-\zeta)(1-L_E)^2} > 0,$$

so $L'_E(\tau)$ has the same sign as $R'(\tau)$. Differentiating R :

$$R'(\tau) = \frac{1}{\omega} [(1-\varepsilon)'(\tau) \mathcal{Q}(\tau) + (1-\varepsilon(\tau)) \mathcal{Q}'(\tau)].$$

From (A199),

$$(1-\varepsilon)'(\tau) = -\frac{\lambda}{\lambda + 1 - \omega(1-\rho)}.$$

At $\tau = 0$,

$$1 - \varepsilon(0) = \frac{\lambda}{\lambda + 1 - \omega(1-\rho)} = 1 - \gamma,$$

so

$$(1-\varepsilon)'(0) = -(1-\gamma).$$

Therefore

$$R'(0) = \frac{1-\gamma}{\omega} \left[-\mathcal{Q}_0 + \frac{\omega \eta \bar{\xi}'(0)}{(\eta + \delta + \omega \bar{\xi}_0)^2} \right],$$

where $\mathcal{Q}_0 \equiv \mathcal{Q}(0)$, $\bar{\xi}_0 \equiv \bar{\xi}(0)$, and we used $\mathcal{Q}'(\tau) = \omega \eta \bar{\xi}'(\tau) / (\eta + \delta + \omega \bar{\xi}(\tau))^2$.

Signing $W'(0)$. Differentiate BGP welfare (A203):

$$W'(0) = -\frac{\zeta}{\eta(1-L_E)} L'_E(0) + \frac{1-\zeta}{\eta^2} \bar{\xi}'(0).$$

Substituting $L'_E(0) = R'(0) / \ell'(L_E) = (1-\zeta)(1-L_E)^2 R'(0) / \zeta$:

$$W'(0) = -\frac{(1-\zeta)(1-L_E)}{\eta} R'(0) + \frac{1-\zeta}{\eta^2} \bar{\xi}'(0).$$

Substituting the expression for $R'(0)$ and using the BGP entry condition

$$\frac{(1-\gamma)(1-L_E)}{\omega} \mathcal{Q}_0 = \frac{\zeta L_E}{1-\zeta},$$

we obtain

$$W'(0) = \frac{\zeta L_E}{\eta} + \frac{(1-\zeta)\bar{\xi}'(0)}{\eta} \left(\frac{1}{\eta} - \frac{(1-\gamma)(1-L_E)\eta}{(\eta+\delta+\omega\bar{\xi}_0)^2} \right).$$

The first term is strictly positive. The second term is non-negative because: (a) $\bar{\xi}'(0) \geq 0$, with equality when $\mu = 0$; and (b) the parenthetical factor is strictly positive, since $(1-\gamma)(1-L_E) < 1$ and $(\eta+\delta+\omega\bar{\xi}_0)^2 > \eta^2$ (using $\delta > 0$), so

$$\frac{(1-\gamma)(1-L_E)\eta}{(\eta+\delta+\omega\bar{\xi}_0)^2} < \frac{\eta}{\eta^2} = \frac{1}{\eta}.$$

Therefore

$$W'(0) > 0.$$

A small tax strictly raises welfare. To conclude that $\tau^{SB} > 0$, observe that $W(\tau)$ extends continuously to the closed interval $[0, \tau_{\max}]$ with $\tau_{\max} \equiv \lambda/(\lambda+1-\omega(1-\rho))$: at $\tau = \tau_{\max}$, the scope condition (A198) pins down $\gamma(\hat{x}(\tau_{\max}); \lambda) = 1$, so $\hat{x}(\tau_{\max}) = \kappa^{-1/\lambda}$ is finite, and $1-\varepsilon(\tau)$, $\bar{\xi}(\tau)$, and $L_E(\tau)$ from (A201) all admit finite, positive limits. By Weierstrass, W attains a maximum on $[0, \tau_{\max}]$, and since $W'(0) > 0$ that maximum is not at $\tau = 0$. Therefore $\tau^{SB} > 0$.

Note that $L'_E(0)$ itself may be positive or negative depending on parameters. When $\mu = 0$, $\bar{\xi}'(0) = 0$ and $R'(0) < 0$, so $L'_E(0) < 0$: the tax unambiguously reduces entry. When $\mu > 0$, the positive spillover effect in $R'(0)$ could in principle outweigh the negative wedge effect, making $L'_E(0) > 0$. The welfare result $W'(0) > 0$ holds regardless, because the direct growth gain from wider scope and the consumption gain from the entry condition adjustment combine to dominate any adverse effect. ■

Proof of Result 6. We now allow $\sigma > 0$. The BGP entry condition (A201) becomes

$$\underbrace{\frac{\zeta}{1-\zeta} \frac{L_E}{1-L_E}}_{\ell(L_E)} = \underbrace{\frac{1-\varepsilon(\tau)}{\omega} \frac{\delta+\omega\bar{\xi}(\tau)L_E^{\alpha\sigma}}{\eta+\delta+\omega\bar{\xi}(\tau)L_E^{\alpha\sigma}}}_{R(\tau, L_E)}.$$

Unlike the $\sigma = 0$ case, R now depends on L_E through $L_E^{\alpha\sigma}$. Write

$$z(\tau, L_E) \equiv \bar{\xi}(\tau) L_E^{\alpha\sigma}, \quad \mathcal{Q}(z) \equiv \frac{\delta+\omega z}{\eta+\delta+\omega z},$$

so that

$$R(\tau, L_E) = \frac{1-\varepsilon(\tau)}{\omega} \mathcal{Q}(z(\tau, L_E)).$$

Implicit differentiation. Total differentiation of the entry condition gives

$$L'_E(\tau) = \frac{\partial R / \partial \tau}{\ell'(L_E) - \partial R / \partial L_E},$$

where

$$\begin{aligned}\ell'(L_E) &= \frac{\zeta}{(1-\zeta)(1-L_E)^2}, \\ \frac{\partial R}{\partial \tau} &= \frac{1}{\omega} [(1-\varepsilon)'(\tau) \mathcal{Q}(z) + (1-\varepsilon(\tau)) \mathcal{Q}'(z) L_E^{\alpha\sigma} \bar{\xi}'(\tau)], \\ \frac{\partial R}{\partial L_E} &= \frac{1-\varepsilon(\tau)}{\omega} \mathcal{Q}'(z) \alpha\sigma \bar{\xi}(\tau) L_E^{\alpha\sigma-1},\end{aligned}$$

with

$$\mathcal{Q}'(z) = \frac{\omega \eta}{(\eta + \delta + \omega z)^2} > 0.$$

Define the denominator

$$\Delta \equiv \ell'(L_E) - \frac{\partial R}{\partial L_E} \Big|_{\tau=0}.$$

At the unique BGP, ℓ crosses R from below as a function of L_E , so $\ell'(L_E) > \partial R / \partial L_E$ and

$$\Delta > 0.$$

Welfare derivative. Differentiate (A203) and decompose $g'(\tau)$ into scope and entry channels as in (84):

$$W'(\tau) = \underbrace{\frac{(1-\zeta) L_E^{\alpha\sigma} \bar{\xi}'(\tau)}{\eta^2}}_{\text{scope channel}} + \underbrace{\left(-\frac{\zeta}{\eta(1-L_E)} + \frac{(1-\zeta) \alpha\sigma L_E^{\alpha\sigma-1} \bar{\xi}(\tau)}{\eta^2} \right)}_{\equiv \mathcal{M}(\tau)} L_E'(\tau).$$

The coefficient $\mathcal{M}(\tau)$ collects the two effects of L_E on welfare: the consumption cost (negative) and the dynamic spillover benefit (positive, proportional to σ). Evaluating at $\tau = 0$ and substituting $L_E'(0) = (\partial R / \partial \tau|_{\tau=0}) / \Delta$:

$$W'(0) = \frac{(1-\zeta) L_E^{\alpha\sigma} \bar{\xi}'(0)}{\eta^2} + \mathcal{M}(0) \frac{\partial R / \partial \tau|_{\tau=0}}{\Delta}. \quad (\text{A204})$$

The first term is the direct growth gain from wider scope. It is non-negative ($\bar{\xi}'(0) \geq 0$), strictly positive when $\mu > 0$. The second term is the welfare effect of the induced change in L_E . Its sign depends on two objects: $\mathcal{M}(0)$ and $\partial R / \partial \tau|_{\tau=0}$.

Sign of $\partial R / \partial \tau$ at $\tau = 0$. From the expressions above, at $\tau = 0$:

$$\frac{\partial R}{\partial \tau} \Big|_{\tau=0} = \frac{1}{\omega} [-(1-\gamma) \mathcal{Q}(z_0) + (1-\gamma) \mathcal{Q}'(z_0) L_E^{\alpha\sigma} \bar{\xi}'(0)],$$

where $z_0 \equiv \bar{\xi}(0) L_E^{\alpha\sigma}$ and we used $(1-\varepsilon)'(0) = -(1-\gamma)$. The first term is strictly negative; the second is non-negative. When $\mu = 0$, $\bar{\xi}'(0) = 0$ and $\partial R / \partial \tau < 0$. When $\mu > 0$, the sign is ambiguous.

Sign of $\mathcal{M}(0)$. The coefficient is

$$\mathcal{M}(0) = -\frac{\zeta}{\eta(1-L_E)} + \frac{(1-\zeta)\alpha\sigma L_E^{\alpha\sigma-1}\bar{\xi}(0)}{\eta^2}.$$

The first term is the marginal consumption cost of entry (negative). The second is the marginal growth benefit of entry through dynamic spillovers (positive, proportional to σ). When $\sigma = 0$, $\mathcal{M}(0) < 0$. As σ increases, the second term grows and $\mathcal{M}(0)$ can become positive. Define σ^{**} as the value at which $\mathcal{M}(0) = 0$:

$$\sigma^{**} : \frac{\zeta}{\eta(1-L_E)} = \frac{(1-\zeta)\alpha\sigma^{**} L_E^{\alpha\sigma^{**}-1}\bar{\xi}(0)}{\eta^2}.$$

For $\sigma < \sigma^{**}$, the consumption cost dominates ($\mathcal{M}(0) < 0$), and the tax-induced reduction in L_E raises welfare through this channel. For $\sigma > \sigma^{**}$, the dynamic spillover benefit dominates ($\mathcal{M}(0) > 0$), and reducing L_E lowers welfare.

Characterization of σ^* . The threshold σ^* in Result 6 is defined by

$$W'(0; \sigma^*) = 0,$$

with $W'(0)$ given by (A204). This is a scalar equation in σ (all other parameters fixed), where the BGP allocation ($L_E, \bar{\xi}, \gamma, x$) itself depends on σ through (A201) at $\tau = 0$.

At $\sigma = 0$, Result 5 established $W'(0) > 0$. As $\sigma \rightarrow \infty$, the dynamic spillover benefit of entry becomes arbitrarily large: $\mathcal{M}(0) \rightarrow +\infty$. Since $\partial R / \partial \tau|_{\tau=0}$ is bounded and generically negative (the wedge effect on R dominates for moderate μ), the second term in (A204) becomes arbitrarily negative. Since the first term is bounded (for finite μ), $W'(0) < 0$ for σ sufficiently large. By the intermediate value theorem, there exists a $\sigma^* > 0$ such that $W'(0; \sigma^*) = 0$.

For $\sigma < \sigma^*$, $W'(0) > 0$ and the optimal tax is positive. For $\sigma > \sigma^*$, $W'(0) < 0$ and the optimal policy is a subsidy. This establishes Result 6. ■

A.7.5 Optimal IP protection

Proof of Result 7. We treat λ as a parameter of the DE with $\tau = 0$. On the BGP, from (59), the growth rate is

$$g(\lambda) = (1-\zeta)\bar{\xi}(\lambda)L_E(\lambda)^{\alpha\sigma},$$

where

$$\bar{\xi}(\lambda) = \xi\left(\gamma(x(\lambda); \lambda) x(\lambda)^\rho\right)^\mu,$$

and $x(\lambda)$ is the DE optimal scope solving (36) at the given λ . BGP welfare is

$$W(\lambda) = \frac{\zeta}{\eta} \log(1-L_E(\lambda)) + \frac{g(\lambda)}{\eta^2}.$$

Differentiating:

$$\frac{dW}{d\lambda} = -\frac{\zeta}{\eta(1-L_E)} \frac{dL_E}{d\lambda} + \frac{1}{\eta^2} \frac{dg}{d\lambda}. \quad (\text{A205})$$

Using the decomposition (85):

$$\frac{dg}{d\lambda} = g\mu \frac{d}{d\lambda} \log(\gamma x^\rho) + g\alpha\sigma \frac{d \log L_E}{d\lambda}.$$

Sign of $d \log(\gamma x^\rho)/d\lambda$. Under the functional form $\gamma(x; \lambda) = \kappa x^\lambda$ and optimal scope from (36):

$$\gamma x^\rho = \kappa x^{\lambda+\rho}, \quad x = \left(\frac{1}{\kappa} \frac{1 - \omega(1-\rho)}{1 - \omega(1-\rho) + \lambda} \right)^{1/\lambda}.$$

Therefore

$$\log(\gamma x^\rho) = \log \kappa + (\lambda + \rho) \log x = \log \kappa + \frac{\lambda + \rho}{\lambda} \left(\log \frac{1 - \omega(1-\rho)}{1 - \omega(1-\rho) + \lambda} - \log \kappa \right).$$

Differentiating with respect to λ . Write $\phi \equiv 1 - \omega(1 - \rho) > 0$ for brevity:

$$\log x = \frac{1}{\lambda} \log \frac{\phi}{\kappa(\phi + \lambda)}.$$

Then

$$\frac{d \log x}{d\lambda} = -\frac{1}{\lambda^2} \log \frac{\phi}{\kappa(\phi + \lambda)} - \frac{1}{\lambda(\phi + \lambda)}.$$

The first term is negative (since $x > 1$ implies $\log x > 0$). The second is negative. So $d \log x/d\lambda < 0$: weaker IP protection reduces scope. Since $\gamma = \kappa x^\lambda$:

$$\frac{d \log \gamma}{d\lambda} = \log x + \lambda \frac{d \log x}{d\lambda}.$$

And

$$\frac{d \log(\gamma x^\rho)}{d\lambda} = \frac{d \log \gamma}{d\lambda} + \rho \frac{d \log x}{d\lambda} = \log x + (\lambda + \rho) \frac{d \log x}{d\lambda}.$$

Substituting:

$$\frac{d \log(\gamma x^\rho)}{d\lambda} = \log x + (\lambda + \rho) \left(-\frac{\log x}{\lambda} - \frac{1}{\lambda(\phi + \lambda)} \right) = -\frac{\rho}{\lambda} \log x - \frac{\lambda + \rho}{\lambda(\phi + \lambda)}.$$

Both terms are strictly negative (since $x > 1$). Therefore

$$\frac{d \log(\gamma x^\rho)}{d\lambda} < 0.$$

Weaker IP protection unambiguously reduces the spillover rate when $\mu > 0$.

Sign of $dL_E/d\lambda$. From the BGP entry condition (60), L_E depends on λ through γ and $\bar{\xi}$. When $\mu = 0$, $\bar{\xi} = \xi$ is independent of λ , and $dL_E/d\lambda > 0$ follows from $d\gamma/d\lambda < 0$ (weaker IP protection reduces the managerial share, raising the private return to entry). When $\mu > 0$, the decline in $\bar{\xi}$ partially offsets this, but $dL_E/d\lambda > 0$

continues to hold under the maintained assumption that the direct effect through $(1 - \gamma)$ dominates.

Part (i): $\mu = 0$. When $\mu = 0$, $\bar{\xi}(\lambda) = \xi$ for all λ . The growth rate is $g = (1 - \zeta)\xi L_E^{\alpha\sigma}$, which depends on λ only through L_E . Since $dL_E/d\lambda > 0$ and $\alpha\sigma > 0$, $dg/d\lambda > 0$. The welfare derivative (A205) becomes

$$\frac{dW}{d\lambda} = \left(-\frac{\zeta}{\eta(1 - L_E)} + \frac{(1 - \zeta)\alpha\sigma\xi L_E^{\alpha\sigma-1}}{\eta^2} \right) \frac{dL_E}{d\lambda}.$$

The sign depends on whether the parenthetical term is positive. With $\sigma = 0$, the second term vanishes and $dW/d\lambda < 0$: the planner prefers stronger IP protection. With $\sigma > 0$, the second term is positive and grows with σ . The parenthetical term is positive whenever the marginal growth benefit of entry exceeds its consumption cost, which holds when L_E is below the socially optimal level. Under the maintained assumption that $L_E(\lambda)$ is below the consumption-growth optimum for all feasible λ , $dW/d\lambda > 0$ for all λ . This establishes part (i).

Part (ii): $\mu > 0$. When $\mu > 0$, as $\lambda \rightarrow +\infty$, scope collapses to $x \rightarrow 1$ and $\gamma \rightarrow \kappa$. The spillover rate $\bar{\xi} = \xi(\kappa \cdot 1^\rho)^\mu = \xi\kappa^\mu$ converges to a finite lower bound. Meanwhile L_E converges to a finite limit determined by the entry condition with $\gamma = \kappa$. So $g(\lambda)$ and $W(\lambda)$ converge to finite limits as $\lambda \rightarrow \infty$.

As $\lambda \rightarrow 0^+$, scope $x \rightarrow \infty$ and $\gamma \rightarrow 1$. The managerial share absorbs all of enterprise value. Entry collapses ($L_E \rightarrow 0$), consumption $(1 - L_E) \rightarrow 1$ but growth $g \rightarrow 0$ (since $L_E^{\alpha\sigma} \rightarrow 0$). Welfare $W(\lambda) \rightarrow -\infty$ (from the g/η^2 term).

For the existence of an interior maximum, it suffices that $dW/d\lambda > 0$ at some λ_0 and $dW/d\lambda < 0$ at some $\lambda_1 > \lambda_0$. The first condition holds near $\lambda = 0$ (the entry channel dominates when L_E is close to zero). The second holds for λ large enough if μ is large enough: the scope-spillover term $g\mu d \log(\gamma x^\rho)/d\lambda$ is negative and grows in magnitude with μ , eventually dominating the positive entry term. By the intermediate value theorem, an interior λ^W with $dW/d\lambda = 0$ exists.

$\lambda^W < \hat{\lambda}$. Suppose $\mu > 0$ is large enough that a growth-maximizing $\hat{\lambda}$ exists (i.e., $dg/d\lambda$ changes sign). At $\hat{\lambda}$, $dg/d\lambda = 0$ by definition. The welfare derivative (A205) at $\hat{\lambda}$ reduces to

$$\left. \frac{dW}{d\lambda} \right|_{\hat{\lambda}} = -\frac{\zeta}{\eta(1 - L_E)} \left. \frac{dL_E}{d\lambda} \right|_{\hat{\lambda}}.$$

Since $dL_E/d\lambda > 0$, this is strictly negative. Welfare is declining at $\hat{\lambda}$, so the welfare maximum must occur at $\lambda^W < \hat{\lambda}$. ■

	Count	Mean best name score	Mean years matched
<i>By confidence tier</i>			
High	13,393	98.0	12.3
Medium	6,560	83.8	6.2
Low	426	75.3	4.9
<i>By matching pass (modal)</i>			
Pass 1 (ZIP + SIC2)	8,138		
Pass 2 (ZIP only)	3,792		
Pass 3 (SIC2 only)	7,931		
Pass 4 (No blocking)	518		
Total matched GVKEYs	20,379	93.0	10.2
Total Compustat GVKEYs	47,119		

Table A-1: Compustat–DNB Matching Summary

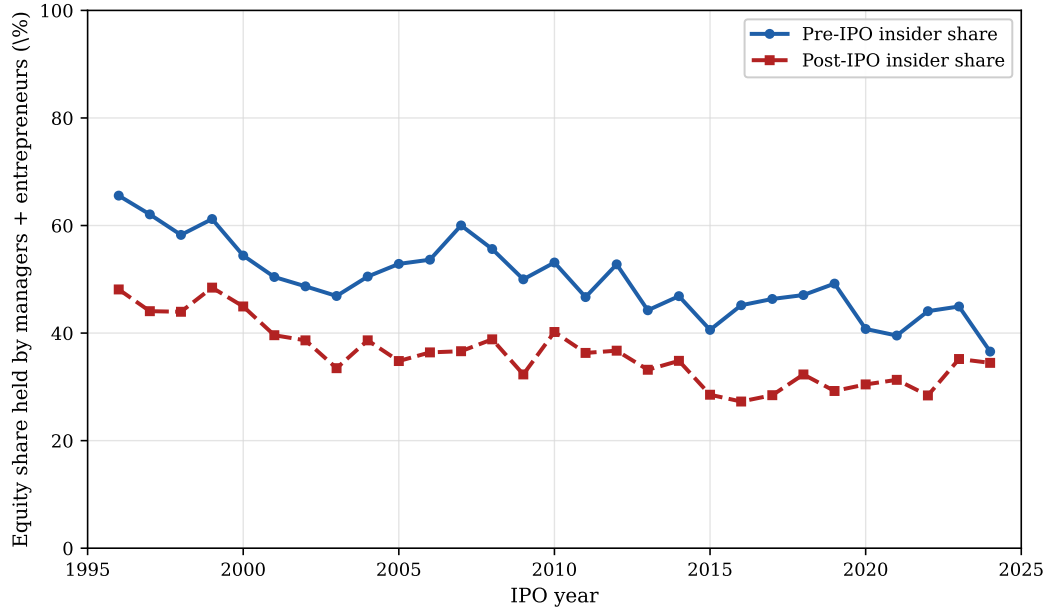


Figure A-1: Equal-weighted yearly mean of insider equity share at IPO, 1996–2023. The insider share is the share of common stock beneficially held by the firm’s executive officers and directors as a group (collectively, “insiders”). Solid blue line: pre-IPO snapshot (insider share before issuance of new shares to the public). Dashed red line: post-IPO snapshot (insider share after issuance). The post-IPO snapshot is the proxy for s^{insider} used in Section 5. Sample: U.S. public firms that completed an IPO on Nasdaq or the NYSE between 1996 and 2023, excluding ADRs, natural resource limited partnerships and trusts, closed-end funds, REITs, SPACs, banks and S&Ls (SIC 6011–6099), unit offers, penny stocks (offer price below \$5), and spinoffs. Sample-selection criteria follow [Ritter and Welch \(2002\)](#). Ownership data for 1996–2016 are from an earlier dataset constructed by Andrea Eisfeldt; ownership data for 2017–2023 are constructed by the authors from IPO prospectuses scraped from SEC EDGAR.

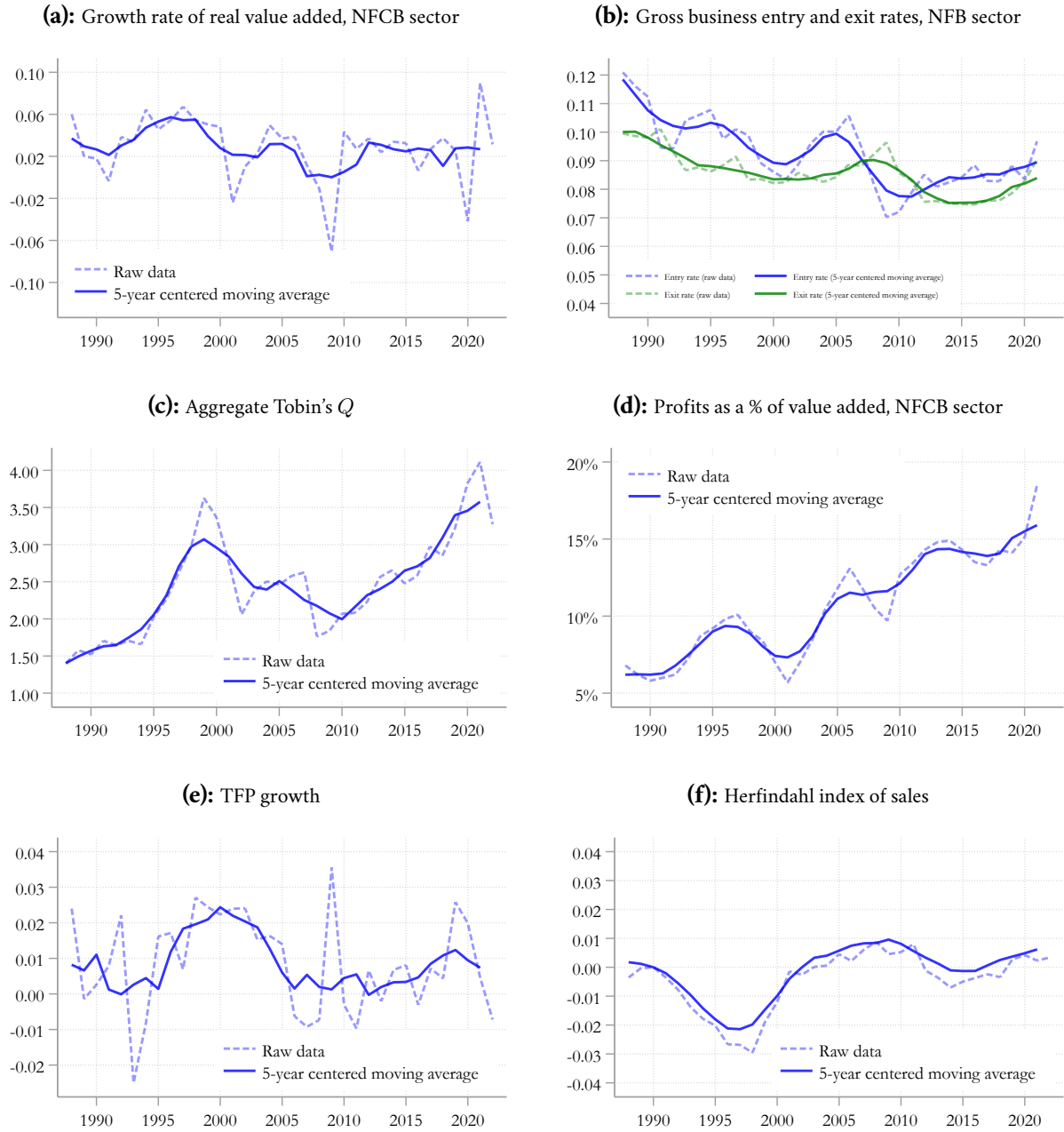


Figure A-2: Other aggregate moments used in the numerical illustration of Section 5. The top left panel shows the growth rate of real value added for the non-financial corporate business (NFCB) sector. The top right panel shows the gross business entry and exit rates for the non-financial business (NFB) sector. The middle left panel shows aggregate Tobin's Q for the NFCB sector, with the denominator defined as the sum of estimates of the physical and intangible capital stocks. The middle right panel shows the ratio of profits to value added for the NFCB sector. The bottom left panel shows total factor productivity growth for the business sector, from [Fernald \(2014\)](#). The bottom right panel shows the average Herfindahl index of sales across NAICS-3 sectors in Compustat. Data sources are reported in Appendix A.1.4.

	<i>N</i>	Mean	SD	Min	P25	P75	Max
<i>Panel A: Entry rates</i>							
Firm entry rate	8,545	0.0828	0.0547	0.0000	0.0502	0.1074	1.3810
Establishment entry rate	8,545	0.1032	0.0476	0.0000	0.0702	0.1270	1.4138
<i>Panel B: Scope measures (levels)</i>							
CBSA/firm	8,545	3.158	1.596	0.750	2.176	3.674	23.360
ZIP/firm	8,545	4.749	3.684	1.000	2.917	5.222	67.809
Establishment/firm	8,545	5.557	4.460	1.333	3.466	5.917	81.539
<i>Panel C: Scope measures (logs)</i>							
log(CBSA/firm)	8,545	1.0606	0.4026	−0.2877	0.7775	1.3012	3.1510
log(ZIP/firm)	8,545	1.4163	0.4791	0.0000	1.0704	1.6528	4.2167
log(Estab/firm)	8,545	1.5715	0.4743	0.2877	1.2430	1.7778	4.4011
<i>Panel D: Five-year differences</i>							
Δ_5 firm entry rate	8,545	−0.0048	0.0531	−0.7519	−0.0229	0.0118	0.8218
Δ_5 establishment entry rate	8,545	−0.0049	0.0414	−1.3733	−0.0177	0.0073	1.3160
Δ_5 log(CBSA/firm)	8,545	0.0363	0.1982	−1.7411	−0.0519	0.1259	1.8409
Δ_5 log(ZIP/firm)	8,545	0.0337	0.2303	−1.8928	−0.0681	0.1347	1.8824
Δ_5 log(Estab/firm)	8,545	0.0330	0.2330	−1.9386	−0.0673	0.1282	1.6094

Table A-2: Summary statistics, industry-year sample for entry–scope regressions. Entry rates are from the BDS mapped to SIC3; scope is from NETS (multi-establishment firms with two or more employees), adjusted for structural breaks. Firm births use $\max(0, \text{firms}_t - \text{firms}_{t-1} + \text{deaths}_t)$. See Appendix A.2 for additional details.

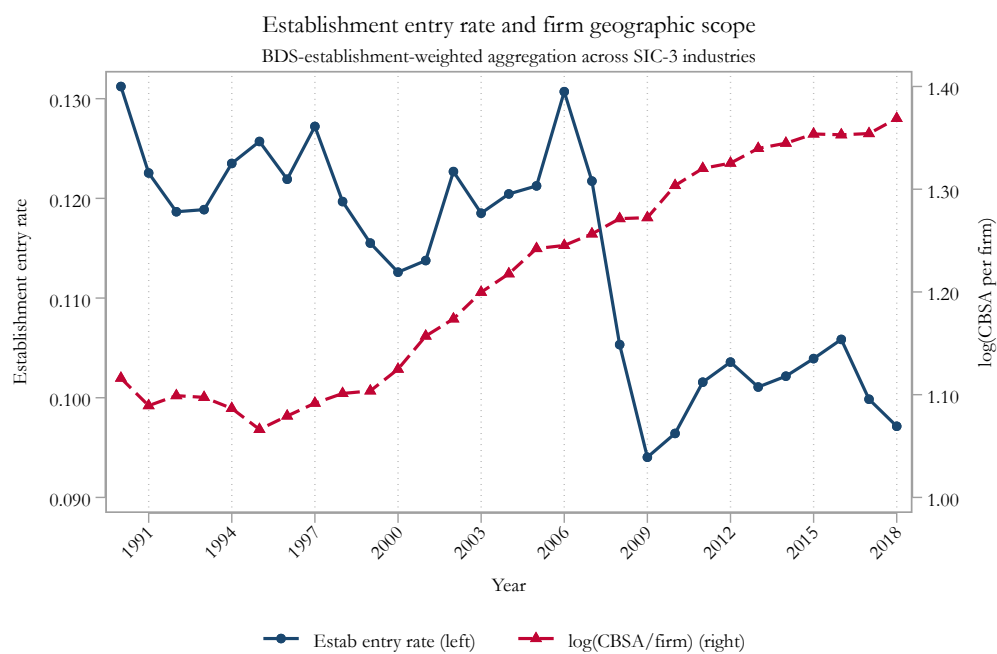


Figure A-3: Aggregate establishment entry rate and firm geographic scope (BDS establishment-weighted; log CBSAs per firm from NETS, multi-establishment firms with two or more employees), 1991–2018. Scope series adjusted for NETS breaks in 2011 and 2015. See replication code for construction.

Establishment entry rate and firm scope by SIC-1 division

BDS-establishment-weighted, adjusted NETS scope

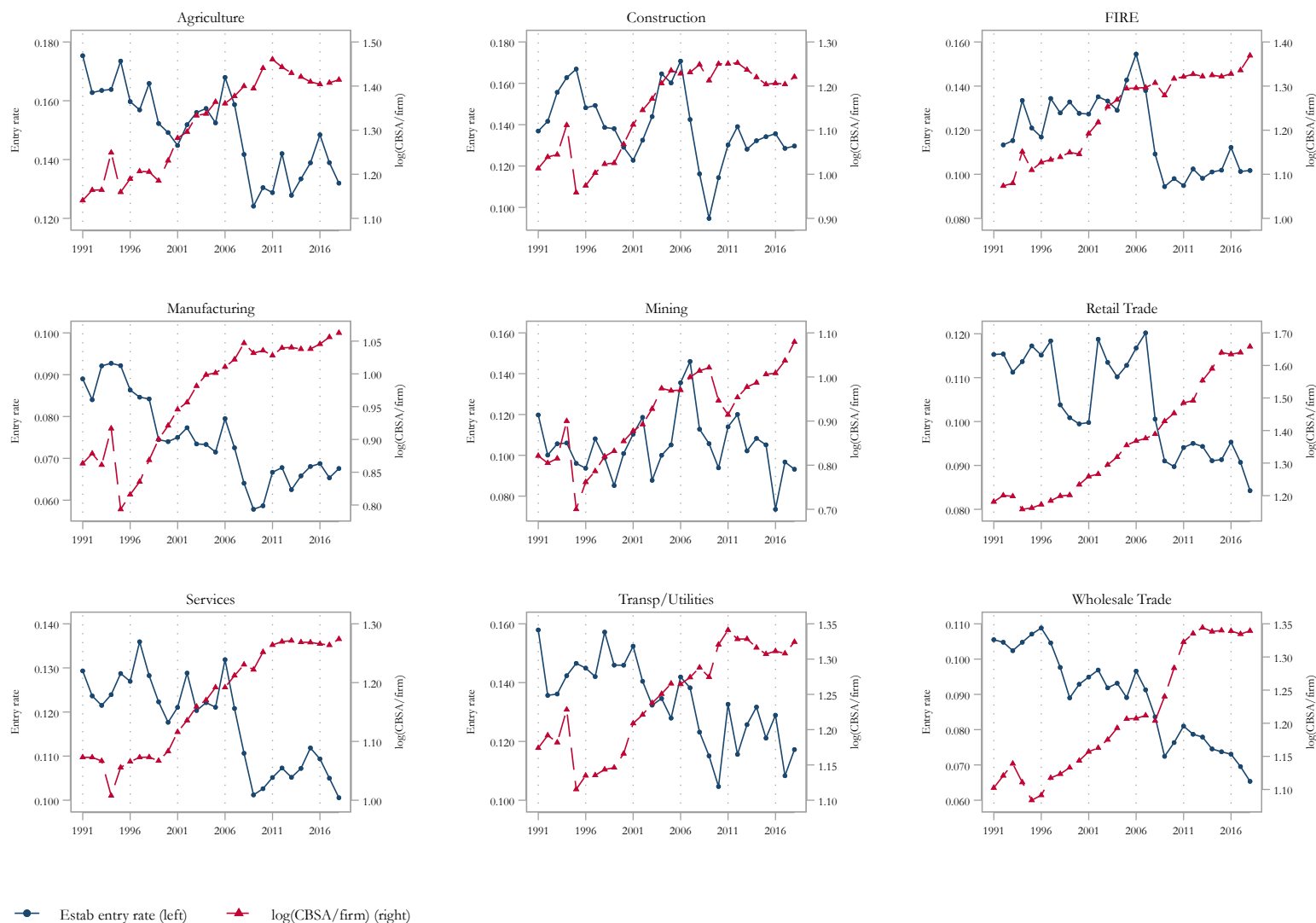


Figure A-4: Establishment entry rate and log CBSAs per firm by one-digit SIC division. Each subplot: establishment entry rate (left axis) and log scope (right axis) for one SIC-1 division.

	Five-year change in log scope					
	CBSA / firm		ZIP / firm		Establishment / firm	
	(1)	(2)	(3)	(4)	(5)	(6)
Δ_5 firm entry rate	−0.095 (0.075)	−0.090 (0.094)	−0.127 (0.088)	−0.078 (0.113)	−0.156* (0.083)	−0.127 (0.114)
Sample	All	Ex-FIRE	All	Ex-FIRE	All	Ex-FIRE
Observations	8,545	7,855	8,547	7,857	8,547	7,857
R^2	0.098	0.088	0.102	0.094	0.083	0.078

Table A-3: Five-year changes in firm scope and *firm* entry rates (SIC-3 panel). Same specification as Table 2, replacing the establishment entry rate with the firm entry rate. Firm births are imputed as in the “Variable definitions” subsection. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.